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Asset Allocation via Machine Learning and Factor Models:

An Empirical Evaluation of Portfolio Optimization Approaches

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Abstract

This thesis empirically investigates whether Machine Learning methods produce more accurate return forecasts than classical factor models and econometric approaches, and whether any statistical advantage translates into economically superior asset allocation strategies within a Mean-Variance Optimization (MVO) framework. The analysis covers a 35-year out-of-sample period (May 1989 – November 2024) across five US asset classes: equities (S&P 500), government bonds (7–10 year Treasuries), commodities (S&P GSCI), gold, and REITs.

The study compares four forecasting models, all tested inside a quarterly MVO framework. Two of these are based on theoretical models. The first, *OLS-Base*, uses a three-factor Fama-French framework and estimates factor premia using 37 macroeconomic predictors from Goyal et al. (2023). The second, *OLS-Lasso*, builds on this by adding L_1 penalization and a larger predictor set of 57 variables, composed of the original 37 GWZ variables and 20 engineered features (such as rolling momentum and volatility). The other two models are based entirely on data. Both the Elastic Net (*ENet*) and a three-hidden-layer neural network (*NN-3*) predict returns directly, using 60 predictor variables that combine macroeconomic variables, Fama-French factors and the same engineered features.

The unregularised *OLS-Base* model shows low OOS predictive power ($R^2_{\text{OOS}} = -87.3\%$) because its predictor variables are too correlated, and adding L_1 regularization to *OLS-Lasso* and including all predictors, we obtain a much more stable R^2_{OOS}, although it remains slightly negative (-1.7%). Both ML models achieve positive R^2_{OOS} (*ENet*: $+2.75\%$; *NN-3*: $+1.04\%$) and forecast return directions with statistically significant accuracy, in particular for equities and REITs. Among all models, *ENet* delivers the best risk-adjusted performance (Sharpe Ratio of 0.80). However, unconstrained MVO results in uninvestable portfolios with an annual turnover between 241% and 468%. Once realistic constraints are applied, the alpha falls to zero in all models, and none of them outperforms the passive 60/40 benchmark (SR = 0.69).

In summary, ML models demonstrate a genuine advantage in terms of forecasting, although this predictability is concentrated in equities and REITs. Moreover, the MVO behaves as an 'error maximizer': rather than leveraging regularised forecasts, it amplifies estimation noise in unstable and high-cost allocations. Although MVO is the main bottleneck, predictive signals remain modest and asset-class-specific, suggesting that translating statistical accuracy into economic value requires both better forecasts and more robust optimization tools.

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List of Abbreviations

1/N Equal-Weight (Naïve) Portfolio

Adam Adaptive Moment Estimation (optimizer, Kingma and Ba, 2014)

APT Arbitrage Pricing Theory

BH Buy-and-Hold passive benchmark

BN Batch Normalization

CAPM Capital Asset Pricing Model

CVaR Conditional Value at Risk

DM Diebold–Mariano (test)

EMH Efficient Market Hypothesis

ENet Elastic Net

FF Fama–French

FF-3 Fama–French Three-Factors

GARCH Generalized Autoregressive Conditional Heteroskedasticity

GFC Global Financial Crisis (2007–2009)

GWZ Goyal–Welch–Zafirov (predictor set, Goyal et al., 2023)

HAC Heteroskedasticity and Autocorrelation Consistent

HML High Minus Low (value factor of FF-3)

IS In-Sample

LASSO Least Absolute Shrinkage and Selection Operator (L_1 regularization)

LOCF Last Observation Carried Forward

LO Long-Only (unconstrained portfolio setup)

MDD Maximum Drawdown

Mkt-RF Excess Market Return ($R_M - R_f$, market factor of FF-3)

ML Machine Learning

MPT Modern Portfolio Theory

MSE Mean Squared Error

MVO Mean-Variance Optimization

NAV Net Asset Value

NN Neural Network (NN-3 refers to a 3-hidden-layer architecture)

OLS Ordinary Least Squares

OLS-Base Ordinary Least Squares Baseline Model

OLS-Lasso Ordinary Least Squares with LASSO Penalization (L_1)

OOS Out-of-Sample

RE Realistic (weight and turnover constrained portfolio setup)

REIT Real Estate Investment Trust

ReLU Rectified Linear Unit

RMSE Root Mean Squared Error

SD Standard Deviation

SLSQP Sequential Least Squares Programming

SMB Small Minus Big (size factor of FF-3)

SNR Signal-to-Noise Ratio

SR Sharpe Ratio

TF Technical Features (custom momentum and volatility indicators)

VaR Value at Risk

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Chapter 1

Introduction

This opening chapter establishes the foundational context for the study by discussing the multiple challenges of modern portfolio allocation. Specifically, it discusses the main challenges in forecasting financial returns, explaining the need for more advanced predictive models. Finally, the chapter outlines the research objectives and the overall organization of the thesis.

1.1 Context and Problem Statement

1.1.1 Strategic Asset Allocation: The Research Focus

The investment management process is typically organized into three levels: *Capital Allocation* (how much to invest in risky vs. risk-free assets), *Asset Allocation* (how to distribute capital across asset classes), and *Security Selection* (which individual instruments to hold within each class). This thesis operates at the second, intermediate level.

In particular, the study focuses on *strategic asset allocation*, and therefore aims to determine the optimal allocations within a multi-asset portfolio comprising equities, government bonds, property (REITs), and commodities, with a view to maximizing risk-adjusted returns and creating investable portfolios. The analysis deliberately excludes idiosyncratic risk by using general market indices rather than individual securities to construct the portfolios.

This scope is consistent with the classical result of Tobin (1958), who showed that the investment problem can be decomposed into two independent steps: first, identifying the optimal risky portfolio that maximizes the risk-return trade-off; second, blending it with the risk-free asset according to individual risk aversion. This thesis addresses only the first step.

1.1.2 Estimation Uncertainty and Return Predictability

The theoretical foundation for constructing the risky portfolio is represented by *Modern Portfolio Theory* (MPT), introduced by the foundational work of Markowitz (1952). The author formalized the problem of portfolio selection as a trade-off between expected return and risk. Under this framework, known as *Mean-Variance Optimization* (MVO), rational investors aim to minimize return variance for any given target return. In practice, it requires estimating two parameters: the vector of expected returns (μ) and the variance-covariance matrix (Σ).

It is well known that MVO is highly sensitive to input estimates. Researchers Chopra and Ziemba (1993) demonstrate that errors in expected returns (μ) affect portfolio performance much more than errors in risk estimates. This sensitivity gives rise to what Michaud (1989) terms "error maximization": the optimiser tends to overweight assets with overestimated returns and underestimated variances, producing concentrated and unstable portfolios that perform poorly out-of-sample.

To mitigate the estimation problem, researchers have often used various macroeconomic predictive variables (such as the *Dividend-Price Ratio*). However, as will be discussed in *Chapter 2*, Goyal and Welch (2008) have shown that such predictors suffer from marked structural instability: while they appear significant within the observed sample (*in-sample*), they often fail to generate robust forecasts on new data (*out-of-sample*). This evidence suggests the limitations of purely classical approaches and motivates the search for more sophisticated methodologies, such as those proposed in this work.

1.1.3 The Proposed Solution: Factor Models and Machine Learning

To overcome these limitations, this thesis proposes a framework that compares a structured, theory-driven approach with a purely *data-driven* one based on Machine Learning.

In the first phase, the Fama and French (1993) three-factor model is used to estimate returns. The β coefficients are estimated via regression, and the factor premia are then forecast using the macroeconomic predictors of Goyal, Welch, and Zafirov (2023) as explanatory variables. This two-step structure produces conditional expected returns that leverage both systematic factor exposures and forward-looking macroeconomic signals, with the covariance matrix estimated consistently within the same factor structure. However, applying standard OLS to a high-dimensional set of collinear predictors introduces severe overfitting. To address this within the same structural framework, an *OLS-Lasso* variant is introduced, which applies L_1 regularization to the factor forecasting step, shrinking irrelevant predictors to zero and producing more stable estimates of the factor premia.

In the second phase, the *a priori* factor structure is removed entirely to test flexible, data-driven approaches. Two distinct models are implemented on a comprehensive dataset comprising the Goyal–Welch predictors, Fama–French factors, and historical returns: an *Elastic Net*, which combines L_1 and L_2 penalties for regularized variable selection; and shallow *Neural Networks*, introduced to capture non-linearities and complex interactions that linear models cannot explain. Together, these strategies span two fundamentally different forecasting paradigms: an indirect *structural approach*, where macroeconomic predictors forecast the factor premia, and a direct *data-driven approach*, where the algorithms map predictors to asset excess returns without imposing any prior factor structure.

1.2 Research Objective

The primary objective of this thesis is to empirically evaluate whether the integration of *Machine Learning* techniques into the estimation of expected returns can generate more efficient *multi-asset* portfolios compared to traditional econometric approaches and market benchmarks. Specifically, this work aims to address the following research questions:

1. **Methodological Comparison:** Is it possible to improve *out-of-sample* performance relative to the simple historical mean by integrating macroeconomic predictors (Goyal & Welch) into a structured factor model (Fama–French), and does introducing L_1 regularization into the factor forecasting step further improve forecast stability?

2. **The Value Added of Machine Learning:** Can a *data-driven* approach based on *Elastic Net* and *Neural Networks* outperform the classical linear approach by capturing more complex relationships among predictors?

3. **Drivers of Predictability and Economic Value:** Which variables are the most significant drivers of out-of-sample returns? Furthermore, does this statistical predictability translate into tangible economic value compared to passive benchmarks?

4. **Portfolio Stability and the MVO Bottleneck:** How do the different methodologies affect turnover, and can the MVO framework effectively translate statistical signals into economically meaningful allocations? How does imposing realistic constraints impact performance and practical stability?

To answer these questions, the analysis compares the risk-adjusted performance of all active strategies, including the Fama–French structured model and its OLS-Lasso extension, as well as machine learning approaches (Elastic Net and neural networks), with a range of passive benchmarks (1/N, Buy-and-Hold and a classic 60/40 portfolio, with 60% in the S&P 500 and 40% in Treasury bonds). Performance is assessed using a range of metrics, including the Sharpe ratio, maximum drawdown, cumulative returns and Jensen's alpha, both in an unconstrained (Long-Only) context and in a realistic one (with weighting constraints). Detailed answers to these questions are provided in *Section 6.1*.

1.3 Thesis Structure

This thesis is organized as follows:

Chapter 2 establishes the theoretical foundations. It reviews Markowitz's *Mean-Variance Optimization* and the estimation risk problem, analyzes the literature on macroeconomic return predictability and the critique by Goyal et al. (2023), outlines the Fama–French three-factor model and its integration with macroeconomic predictors, and introduces the Machine Learning methodologies adopted, *OLS-Lasso*, *Elastic Net*, and *Neural Networks*, following Gu et al. (2020), together with the validation strategies used to prevent overfitting.

Chapter 3 presents the empirical dataset, describing the data sources for the selected asset classes (US equities, commodities, gold, REITs and Treasury bonds) and the macroeconomic predictors from Goyal et al. (2023), along with custom features calculated from the data. Finally, it describes the time-alignment pipeline designed to prevent look-ahead bias.

Chapter 4 describes the empirical framework, providing a detailed evaluation of the implementation of all forecasting models, from the two-stage Fama–French benchmark model and its OLS-Lasso extension through to Elastic Net and neural networks. It also formalises the portfolio optimization problem, addressing covariance estimation, realistic constraints and the walk-forward validation architecture.

Chapter 5 presents the empirical results in terms of predictive accuracy, measured using R^2_{OOS}, directional accuracy and the Diebold–Mariano test; and in terms of economic value, measured using the Sharpe ratio, maximum drawdown and Jensen's alpha, in both unconstrained and realistic settings. The chapter also examines the importance of predictors and concludes with robustness checks across different training window configurations.

Chapter 6 summarises the main findings of the empirical analysis, examines its methodological limitations, and offers suggestions for future work and possible extensions.

Chapter 2

Theoretical Framework and Literature

This chapter illustrates the theoretical framework required to understand the empirical choices made in this thesis. It begins with the mean-variance model, focusing on its sensitivity to input estimates and why this poses a practical problem. It then moves on to the predictive models used in this study, ranging from simple linear approaches to regularised and non-linear methods, explaining the rationale behind each step.

2.1 Portfolio Selection and Mean-Variance Optimization

2.1.1 The Mean-Variance Framework: Modern Portfolio Theory

Modern portfolio theory is rooted in the seminal work of Markowitz (1952), titled *Portfolio Selection*. Before this work, investment strategies largely evaluated securities in isolation, overlooking the crucial correlations between different assets. Markowitz shifted this paradigm by formally defining portfolio construction as a quantitative trade-off between expected return and return variance. According to his theory, assuming risk-averse and rational investors, they will prefer the portfolio with the lower variance for a given level of expected return.

Formally, consider an investment universe of N risky assets, with w an $N \times 1$ vector of portfolio weights, μ the $N \times 1$ vector of expected returns, and Σ the $N \times N$ variance-covariance matrix. The expected portfolio return and variance are:

$$\mathbb{E}(R_p) = w'\mu \qquad \text{and} \qquad \sigma_p^2 = w'\Sigma w \tag{2.1}$$

Following the "E-V rule" (Markowitz, 1952, p. 79), the optimal investor seeks the weight vector w that minimizes portfolio variance for a given target return $\bar{\mu}$. In matrix notation, this becomes the constrained quadratic minimization problem:

$$
\begin{aligned}
\min_{w} \quad & \frac{1}{2} w' \Sigma w \\
\text{s.t.} \quad & w'\mu = \bar{\mu} \\
& w'\mathbb{1} = 1
\end{aligned}
\tag{2.2}
$$

The $\frac{1}{2}$ term in the objective function is a mathematical convention used to simplify the first-order conditions (first derivative) without altering the problem's solution. In this minimization problem, the first constraint ($w'\mu = \bar{\mu}$) ensures that the portfolio achieves the desired target return, while the second constraint ($w'\mathbb{1} = 1$, where $\mathbb{1}$ is a vector of ones) ensures that the sum of the weights equals 100% of the invested capital (the budget constraint).

2.1.2 Estimation Risk and the "Error Maximization" Problem of MVO

While Markowitz's theoretical model assumes that the vector of expected returns μ and the covariance matrix Σ are known and certain parameters, in reality, the investor must base his decision on sample estimates derived from historical time series. Such estimates are necessarily exposed to *estimation risk*, as past data represent only one of many possible realizations of the stochastic process generating returns.

The practical implementation of MVO is complicated by the algorithm's extreme sensitivity to these input errors. Since the optimizer treats estimates as deterministic parameters, it tends to aggressively overweight assets with positive estimation errors and sell or underweight those with negative errors[1]. Michaud formalized this phenomenon by defining mean-variance optimizers as "estimation-error maximizers" (Michaud, 1989, p. 33): the algorithm does not maximize the investor's true utility, but rather the impact of the statistical errors contained in the data. Consequently, the optimized portfolios often exhibit extreme weights, poor diversification, and severe temporal instability.

As we mentioned earlier, Chopra and Ziemba (1993) established a clear hierarchy regarding the severity of forecasting errors. Their empirical results show that the negative impact on portfolio performance resulting from errors in the estimation of expected returns (μ) is an order of magnitude greater than that of errors in the estimation of variances (Σ). In particular, the authors note that, in a baseline scenario, the accuracy of mean estimates is approximately eleven times more important than the accuracy of variance estimates[2].

---

[1] Positive estimation errors lead to *overestimated* returns or *underestimated* risk, making assets appear more attractive than they truly are; the converse applies to negative errors.

[2] Chopra and Ziemba (1993, p. 10, Exhibit 5) show that for $\tau = 50$, expected return errors are nearly eleven

This evidence provides a strong theoretical justification for moving beyond the use of the simple historical mean (*Sample Mean*), which, given the low *signal-to-noise ratio*[3] of financial returns, produces excessively noisy and unreliable estimates of μ. Therefore, the primary objective of this work is to improve the estimation of the expected returns vector, the most critical parameter according to Chopra and Ziemba (1993), by leveraging the informational content of macroeconomic variables. To this end, linear regularization techniques such as *Elastic Net* will be employed to manage dimensionality, complemented by non-linear models such as *Neural Networks* to capture complex interactions among predictors. The explicit formulation of the structural constraints and the regularized covariance matrix will be formalized in the empirical framework.

2.2 Linear Regressions for Expected Returns

2.2.1 The Efficient Market Hypothesis and Return Predictability

The theoretical starting point for any analysis of stock return predictability is the *Efficient Market Hypothesis* (EMH) proposed by Fama (1970), which defines a market as ”efficient” if asset prices fully and instantaneously reflect all available information. Under this view, price evolution follows a *Random Walk*, making it impossible to systematically outperform the market using past information. Therefore, the best predictor of the future return $E_t[R_{t+1}]$ is simply the historical mean adjusted by a constant risk premium, and any deviation is considered unpredictable *white noise*.

However, from the 1980s onwards, empirical data began to challenge this view; indeed, Shiller (1981) demonstrated that the volatility of stock prices is too high to be explained exclusively by changes in dividend expectations, referring to this phenomenon as *excess volatility*[4]. In addition, Campbell and Shiller (1988) demonstrated that valuation ratios, particularly the *price-to-dividend ratio* (`d/p`), have significant predictive power for long-term returns, arguing that when prices are high relative to dividends, future returns tend to be low, and viceversa[5].

times more impactful than variance errors, a dominance that persists even as risk aversion increases.

[3]The *Signal-to-Noise Ratio* (SNR) measures how much of a return is driven by real economic fundamentals (the ”signal”) versus random fluctuations (the ”noise”). In the stock markets, monthly volatility is much higher than the average return, so the signal-to-noise ratio (SNR) is very low, making the historical average uninformative about future returns.

[4]Shiller (1981, p. 434) formally concludes: ”We have seen that measures of stock price volatility over the past century appear to be far too high—five to thirteen times too high—to be attributed to new information about future real dividends [...]”.

[5]As noted by the authors: ”The returns regressions show that stock returns are somewhat predictable. The lagged log dividend-price ratio has a positive effect on stock returns [...]” (Campbell and Shiller, 1988, p. 214).

This predictive power is best interpreted not as a violation of market efficiency, but as a manifestation of *Time-Varying Risk Premia*. Investor risk aversion fluctuates with the business cycle: during recessions it rises, driving up the required risk premium, pushing prices down and hence raising the d/p ratio[6]. Because the aggregate risk premium moves with macroeconomic conditions, utilizing macroeconomic indicators to estimate these variations represents a theoretically sound approach. From a modern asset pricing perspective, it is crucial to distinguish between two probability measures. The physical, or real-world, probability measure ($\mathbb{P}$) is the one that generates the observed data. Under this measure, asset prices are not required to be martingales, and risk premia exist. Conversely, the risk-neutral measure ($\mathbb{Q}$) is an artificial probability measure used for pricing, under which discounted asset prices are strictly martingales. The transition from $\mathbb{P}$ to $\mathbb{Q}$ is governed by the state price density (or stochastic discount factor), which shifts the probability weights. The fact that $\mathbb{P}$ and $\mathbb{Q}$ are formally distinct implies that returns contain a predictable component determined by risk compensation, in full accordance with modern no-arbitrage theories (Cochrane, 2005).

2.2.2 Predictive Regressions and the Critique by Goyal and Welch

The standard tool to assess whether a set of macroeconomic variables contains statistically significant information about future excess returns is the predictive regression:

$$r_{t+1} = \alpha + \beta' X_t + \varepsilon_{t+1} \tag{2.3}$$

- r_{t+1} represents the *excess return* (the logarithmic return of the risky asset minus the *risk-free* rate) calculated over the time horizon from t to $t+1$.
- $X_t \in \mathbb{R}^{M\times 1}$ is the vector of M predictive variables (or predictors) observed at the beginning of t. Note that X_t is known to the investor at the time of the portfolio decision.
- ε_{t+1} is the error term (*disturbance term*), assumed to have a zero mean and to be uncorrelated with the information available at time t.
- $\alpha \in \mathbb{R}$ is the scalar intercept and $\beta \in \mathbb{R}^{M\times 1}$ is the vector of coefficients to be estimated.

The economic content of the model rests on the vector of slope coefficients β. Under the null hypothesis of return unpredictability, $\beta = \mathbb{0}$ (where $\mathbb{0}$ denotes a null vector of dimension M) and the best forecast is simply the constant α, which approximates the historical mean. A statistically significant β, by contrast, suggests that the macroeconomic variables in X_t

[6] In particular, Campbell and Shiller (1988, p. 196) observe that "when discount rates are high, the dividend-price ratio is high". They found that ratio fluctuations are mainly caused by discount rates, not by future dividend growth (1988, pp. 214–216).

are able to track the variable component of the risk premium, allowing for the formulation of conditional expectations on the future return, $E_t[r_{t+1}] = \alpha + \beta' X_t$, that differ from the unconditional mean.

Although economic theory suggested numerous predictive variables for years, their empirical effectiveness had never been rigorously proven. Fundamental in this regard is the work of Goyal and Welch (2008), who conducted a systematic analysis on the ability of classic macroeconomic variables to predict the equity premium of the US stock market (using the S&P 500 as a benchmark).

The authors' critique is based on the methodological distinction between *in-sample* (IS) and *out-of-sample* (OOS) analysis. While IS analyses use the entire dataset (including future data relative to the forecasting moment) to estimate coefficients, Goyal and Welch simulate the behavior of a real-world investor who, to make forecasts, only has access to information prior to the forecasting moment, adopting an *expanding window* approach. In this scheme, the forecast for time $t + 1$ is generated using exclusively the data available up to time t. Subsequently, the dataset is updated by including the observation at $t + 1$, the model is re-estimated, and the forecast for $t + 2$ is generated.

To evaluate the predictive accuracy of the models, Goyal and Welch (2008, p. 1461) propose the Out-of-Sample R^2 statistic (R^2_{OOS}), defined as:

$$R^2_{OOS} = 1 - \frac{\sum_{t=1}^{T} (r_t - \hat{r}_{t,\text{model}})^2}{\sum_{t=1}^{T} (r_t - \bar{r}_{t,\text{historical}})^2} = 1 - \frac{MSE_{\text{model}}}{MSE_{\text{mean}}} \tag{2.4}$$

where $\hat{r}_{t,\text{model}}$ is the forecast based on the regression and $\bar{r}_{t,\text{historical}}$ is the simple arithmetic mean of past returns (that is the Historical Mean).

The interpretation is the following:

- If $R^2_{OOS} > 0$, the predictive model has a lower mean squared error (MSE) than the historical mean, thus offering additional value to the forecast.
- If $R^2_{OOS} < 0$, the model is less accurate than simply using the historical mean, meaning the predictors X_t add no useful information.

In their work, Goyal and Welch analyze a broad spectrum of variables, including the classic *Dividend-Price Ratio* (`d/p`), *Dividend-Yield* (`d/y`), short and long term interest rates (T-Bill, `tms`), inflation, and the *Book-to-Market Ratio*, arriving at an important conclusion: most of these variables largely fail in the OOS test. Many predictors show forecasting ability only in specific sub-periods (for example, during the oil shock of the 1970s)[7] to lose significance later.

The main cause of this failure is based on the evidence that "most models seem unstable or even spurious"[8] (Goyal and Welch, 2008, p. 1504). The estimated $\hat{\beta}$ coefficients are not stable over time but fluctuate significantly, sometimes even changing sign in response to exogenous shocks or changes in the economic regime. A static linear model, which assumes the input β coefficients as constant, fails to adapt to their variation, projecting obsolete statistical relationships into the future.

The validity of these conclusions has recently been confirmed and explored in greater depth in an update, in which Goyal, Welch and Zafirov (2023) re-examined the predictive performance of 29 new variables proposed in 26 articles published after 2008, together with the 17 original predictive variables drawn from the previous study. Using data updated to the end of 2021, the authors demonstrate that the new results also confirm the fundamental flaws of the single-variable linear approach. In particular, the authors find that about half of the newly proposed variables have no empirical significance even *in-sample* when using updated data and a consistent OLS framework[9]. Even more surprising is the fact that, for the variables that do show in-sample significance, about half fail to generate *out-of-sample* forecasts superior to the historical mean.

The 2023 study goes further by highlighting the phenomenon of performance degradation: many predictors that appeared robust in their original publication samples show a systematic decline in predictive power in the months following their discovery[10]. Only a few variables, primarily those related to interest rate spreads or specific technical indicators, demonstrate some degree of resilience in the extended sample (Goyal et al., 2023, Section V). However, their predictive power remains economically modest: the authors highlight that the highest monthly R^2_{OOS} among the newly proposed variables is just 1.26% (2023, p. 32). Furthermore, when evaluating the practical utility of these models, only about 16% of the investment strategies based on these predictors would have managed to exceed a passive buy-and-hold

[7] Goyal and Welch (2008) explicitly note that "For many models, any earlier apparent statistical significance was often based exclusively on years up to *and especially on* the years of the Oil Shock of 1973–1975" (p. 1456), further concluding that "if we exclude the Oil Shock, most models perform even worse" (p. 1504).

[8] A regression is *spurious* when it looks significant, but the relationship is not real, driven by a shared trend rather than a genuine economic link.

[9] In the introduction of their updated study, Goyal, Welch, and Zafirov (2023, p. 2) report that "about half of our variables have already lost their predictive power", highlighting that many newly proposed predictors fail to remain significant even in-sample when tested against the extended data through 2021.

[10] Goyal et al. (2023, p. 45) note that, among the new monthly variables, "16 variables showed a deterioration, 9 of which to such an extent that they have already lost their IS significance since publication". This deterioration suggests that many of the initial results may be the result of sample-specific noise or certain biases.

performance (2023, p. 64), confirming that the vast majority of these variables do not provide a reliable basis for profitable market timing.

This persistent difficulty in finding a single linear predictor reinforces the necessity of moving beyond simple models. The authors conclude that the equity premium remains remarkably difficult to predict (2023, Section VII), which provides the ultimate motivation for this thesis to adopt more sophisticated multivariate methodologies. By integrating regularization techniques like *Elastic Net* to manage high-dimensional noise and non-linear models like *Neural Networks* to capture complex interactions, this work attempts to address the instability and lack of robustness identified by Goyal et al.

2.3 Asset Pricing Models: The Factor Framework

2.3.1 The Fama–French Three-Factor Model

The central problem of asset pricing lies in identifying the determinants of expected returns. The *Capital Asset Pricing Model* (CAPM), developed independently by Sharpe (1964) and Lintner (1965), provides the classical answer: in equilibrium, the expected return on any asset constitutes a premium exclusively for its exposure to systematic risk, captured by a single market factor:

$$E[R_i] - R_f = \beta_i(E[R_m] - R_f) \tag{2.5}$$

However, extensive empirical research has demonstrated that market β alone is insufficient to explain the cross-section of expected returns. Fama and French (1992) showed that the relationship between β and average return is statistically weak when controlling for other variables, and identified two accounting variables that capture a large portion of the variation in average returns: market capitalization, *Size*, and the ratio of book equity to market equity, *Book-to-Market Equity* (1992, p. 429).

Building on this evidence, Fama and French (1993) formalized the *Three-Factor Model*. This framework postulates that an asset's excess return is a function of its sensitivity to three distinct systematic risk factors. In the section dedicated to *time-series regressions*, the model takes the following form for each asset i (1993, Section 4):

$$R_{it} - R_{ft} = a_i + b_i(R_{Mt} - R_{ft}) + s_i SMB_t + h_i HML_t + e_{it} \tag{2.6}$$

Where b_i, s_i, and h_i are the regression *slopes* measuring the exposure to the factors:

- **Market ($R_m - R_f$):** This is the market risk factor, representing the excess return of the value-weighted market portfolio over the *risk-free* rate.
- **SMB (*Small Minus Big*):** This is the size-related risk factor, constructed as the difference between the average returns of small-capitalization (*Small*) and large-capitalization (*Big*) stock portfolios.
- **HML (*High Minus Low*):** This is the value-related risk factor, calculated as the difference between the returns of stock portfolios with a high book-to-market ratio (*High*) and those with a low ratio (*Low*).

The model assumes that, if the three factors fully capture systematic risk, the intercept a_i should be statistically indistinguishable from zero[11]. In the context of this thesis, Equation (2.6) serves as a "structured" benchmark. Unlike Machine Learning algorithms, which explore the predictor space independently, the three-factor model imposes a precise economic *a priori* structure, assuming that the sole sources of risk are Market, Size, and Value.

The three-factor model is widely used, but it does not capture every market dynamic. Building on their earlier research, Fama and French (2015) eventually created a five-factor model by adding profitability and investment to the original equation. Furthermore, the academic literature has identified up to 400 potential factors that correlate with asset returns, a phenomenon often described as the "factor zoo" (Cochrane, 2011; Harvey et al., 2016). However, facing this huge range of variables makes it increasingly difficult to distinguish true risk factors from statistical noise. To maintain parsimony, this thesis relies on the classic three-factor model, using it as a solid and intuitive baseline to integrate macroeconomic predictors without overcrowding the factor space.

2.3.2 Conditioning Factor Returns on Macroeconomic Predictors

To construct the baseline portfolio, we link the macroeconomic predictors identified by Goyal et al. with the Fama and French factor framework, formalizing a two-step structural approach. Let y_{t+1} be the $N \times 1$ vector of asset excess returns realized at time $t + 1$, and f_{t+1} be the $K \times 1$ vector of systematic factor returns. The cross-sectional linear factor model dictates that:

$$y_{t+1} = \alpha + Bf_{t+1} + \varepsilon_{t+1} \tag{2.7}$$

[11] Formally, the null hypothesis $H_0 : \alpha_i = 0$ is tested via a *t-test*. An α significantly different from zero would indicate that the model is unable to fully explain the asset's return, leaving a portion of "excess return" (or anomaly) unjustified by the risk factors.

where α is the $N \times 1$ vector of structural pricing errors (Jensen's alphas), B is the $N \times K$ matrix of factor loadings defining the asset risk profile, and ε_{t+1} is the $N \times 1$ vector of idiosyncratic, diversifiable shocks. Assuming expectations are conditional on the information set $\mathcal{F}_t$ available at time t, the expected asset returns are intrinsically linked to expected factor premia:

$$\mathbb{E}_t[y_{t+1}] = \alpha + B \cdot \mathbb{E}_t[f_{t+1}] \tag{2.8}$$

where $\mathbb{E}_t[\cdot] \equiv \mathbb{E}[\cdot \mid \mathcal{F}_t]$ denotes the conditional expectation given all information available at time t. Rather than predicting y_{t+1} directly, the structural approach leverages an $M \times 1$ vector of contemporaneous macroeconomic variables, Z_t, to forecast the factor realizations:

$$f_{t+1} = CZ_t + u_{t+1} \tag{2.9}$$

where C is a $K \times M$ matrix of predictive slopes mapping macroeconomic states to risk premia, and u_{t+1} captures the unforecastable factor innovations. So, the expected factor premia are:

$$\mathbb{E}_t[f_{t+1}] = CZ_t \tag{2.10}$$

Substituting Equation (2.10) into Eq. (2.8) yields the conditional expected returns at time t:

$$\mathbb{E}_t[y_{t+1}] = \alpha + B \cdot (CZ_t) \tag{2.11}$$

The economic intuition behind this architecture is foundational: under the maintained assumption that an asset's risk profile (B) remains fundamentally stable, the market compensation for bearing that risk ($\mathbb{E}_t[f_{t+1}]$) fluctuates dynamically with the business cycle.

Furthermore, this factor structure provides a robust mathematical solution to the sample covariance instability highlighted by Chopra and Ziemba (1993). The same factor structure can also be used to derive the covariance matrix, $\Sigma_{y,t+1}$:

$$\Sigma_{y,t+1} = B\Sigma_f B^T + \Sigma_\varepsilon \tag{2.12}$$

where Σ_f is the $K \times K$ covariance matrix of the Fama–French factors and Σ_ε is the strictly diagonal covariance matrix of idiosyncratic residuals. In this framework we assume a constant Σ_f, although GARCH-type models could in principle capture time-varying second moments; however, given that errors in expected returns have a far greater impact on portfolio performance than errors in covariance matrix (Chopra and Ziemba, 1993), the modelling effort is concentrated on the return forecasting step, and Σ_f is kept constant to preserve parsimony.

Finally, this factor-based framework offers a major conceptual advantage: it is consistent with the *Arbitrage Pricing Theory* (APT) by Ross (1976). The APT states that, to prevent arbitrage opportunities, expected returns must be a linear function of risk exposures. Our Two-Step approach strictly respects this economic rule, while ML methods are purely statistical. They extract patterns directly from data without imposing any economic structure, meaning their forecasts are not guaranteed to respect standard no-arbitrage conditions.

2.4 Machine Learning in Asset Pricing

2.4.1 The General Forecasting Framework

Following Gu et al. (2020), forecasting excess returns is a conditional expectation problem. The model uses a vector of predictors X_t observed at time t. Note that X_t is a general predictor vector that may include the raw macroeconomic variables Z_t introduced in *Section 2.1*, as well as transformations of them (such as lagged returns or rolling volatilities) defined by the researcher prior to estimation. Given this information set, the excess return r_{t+1} is modeled as:

$$r_{t+1} = \mathbb{E}_t[r_{t+1}] + \varepsilon_{t+1} = g(X_t) + \varepsilon_{t+1}, \tag{2.13}$$

where $g(X_t)$ represents the conditional expectation of the return given the information at time t, and ε_{t+1} is an idiosyncratic error with $\mathbb{E}_t[\varepsilon_{t+1}] = 0$. In econometrics, it is usually assumed that $g(X_t)$ is a linear function (for example, $g(X_t) = \alpha + \beta' X_t$); however, ML algorithms allow $g(\cdot)$ to assume more complex, non-linear functional forms, so that it can capture relationships that elude classical linear models.

In line with the methodology described by Gu et al. (2020), the estimation of $g(\cdot)$ is achieved by minimizing an objective function based on a generic empirical loss, augmented by a penalty term that controls model complexity. As formalized in their general framework, the optimization problem for a single asset can be written as:

$$\hat{g} = \arg\min_{g \in \mathcal{G}} \left\{ \frac{1}{T} \sum_{t=1}^{T} \ell(r_{t+1}, g(X_t)) + \phi(g) \right\} \tag{2.14}$$

where the first term represents the empirical loss function $\ell(\cdot)$ measuring the forecasting error, and $\phi(g)$ is the regularization penalty[12]. The specific form of $\ell(\cdot)$ depends on the chosen algorithm; in this thesis, the Mean Squared Error (MSE) is adopted across all models.

The penalty term is fundamental in the context of *Machine Learning* to prevent *overfitting*. Given the low *signal-to-noise ratio* typical of financial returns, simply minimizing the error on past data would tend to produce models that memorize noise rather than learning the structural

[12] Equation (2.14) applies the panel setup in Gu et al. (2020, Eqs. (1)–(2), (4), (7)–(8)) to a single-asset time.

relationship[13]. The penalization $\phi(g)$ imposes parsimony on the forecasting function, reducing the variance of the estimates at the cost of introducing a slight bias, with the goal of improving out-of-sample predictive performance.

2.4.2 Linear Regularization: Elastic Net

The first approach adopted in this thesis is the *Elastic Net*, introduced by Zou and Hastie (2005) and applied to asset pricing by Gu et al. (2020). The Elastic Net combines the strengths of two classic regularization methods: Ridge (L_2) and LASSO (L_1) regressions[14]. In a predictive linear model where $g(X_t;\theta) = X_t'\theta$, the Elastic Net estimator is defined as the solution to the following optimization problem:

$$\hat{\theta}_{EN} = \arg\min_{\theta} \left\{ \frac{1}{2} \sum_{t=1}^{T} (y_{t+1} - X_t'\theta)^2 + \alpha \left((1-\rho)\, \frac{1}{2} \sum_{j=1}^{P} \theta_j^2 + \rho \sum_{j=1}^{P} |\theta_j| \right) \right\}, \tag{2.15}$$

where the term in parentheses represents the Elastic Net penalty, a convex combination of the Ridge and LASSO penalties.

The penalty function includes two fundamental hyperparameters:

- $\alpha > 0$: determines the strength of the regularization penalty; indeed, a higher value of α increases the weight of the penalty, which reduces the coefficients more decisively towards zero. This regularization simplifies the model by accepting slightly biased estimates in exchange for lower variance.
- $\rho \in [0, 1]$: controls the balance between the two regularizations L_2 (Ridge) and L_1 (LASSO). Specifically, when $\rho = 1$, the penalty becomes purely L_1 (LASSO), which helps the model by setting many coefficients to zero; when $\rho = 0$, the penalty is purely L_2 (Ridge), which does not set the coefficients to zero but reduces them continuously and manages multicollinearity well.

In this sense, α plays the same role as the regularization parameter in both LASSO ($\frac{1}{2}\sum(y_{t+1} - X_t'\theta)^2 + \alpha\sum_j |\theta_j|$) and Ridge ($\frac{1}{2}\sum(y_{t+1} - X_t'\theta)^2 + \alpha\sum_j \theta_j^2$); in the Elastic Net, the same α scales the entire combination of the two penalties, while ρ determines whether the model's behaviour is closer to LASSO or Ridge.

[13] In asset pricing, the *signal-to-noise ratio* measures the predictable component of returns relative to unpredictable market noise. Gu et al. (2020) emphasize that this ratio is notoriously low in stock return prediction, making regularization and out-of-sample validation particularly important (2020, Section 1.3).

[14] For the formal definition, see Zou and Hastie (2005, Section 2.1), specifically Eqs. (3)–(5), where the Elastic Net is introduced as a penalized least squares estimator combining Ridge and LASSO penalties (Least Absolute Shrinkage and Selection Operator).

A crucial methodological requirement when implementing penalized regressions like the Elastic Net is the pre-processing of the predictor variables. Because the L_1 and L_2 penalties shrink coefficients based on their absolute or squared magnitudes, they are highly sensitive to the scale of the input features. If variables are left unstandardized, a feature with a naturally large scale will have a correspondingly small coefficient, artificially shielding it from the regularization penalty. To ensure a fair and unbiased penalization, all predictors must be strictly standardized (to have zero mean and unit variance) prior to model training. The optimal calibration of (α, ρ) via a two-dimensional Grid Search on a dedicated Validation Set is formalized in *Chapter 4*.

2.4.3 Capturing Non-Linearities: Neural Networks

As shown by Gu et al. (2020), linear models, even when penalized via Elastic Net, significantly improve upon OLS regression but remain inferior to non-linear algorithms, particularly trees and neural networks, in predicting both individual stock returns and portfolio returns[15]. This suggests that, to capture complex interactions among predictors, it is useful to introduce a controlled form of non-linearity.

In this thesis, we focus on shallow *feed-forward* neural networks with a maximum of $L = 3$ hidden layers, in line with the NN1–NN3 architectures proposed by Gu et al. (2020), which were found to be the best-performing in their study[16]. The basic idea is to build a parametric map $g(X_t)$ that approximates the expected return as a composition of linear and non-linear transformations applied to the predictors X_t.

Generally, we consider a network with $L \leq 3$ hidden layers and denote by $h^{(0)} = X_t$ the input vector of predictors and by $h^{(l)}$ the vector of activations at hidden layer l. The output of layer l is defined recursively as:

$$h^{(l)} = f\big(W^{(l)} h^{(l-1)} + b^{(l)}\big), \tag{2.16}$$

where:

- $W^{(l)}$ is the weight matrix connecting layer $l-1$ to layer l;

[15] In particular, Gu et al. (2020) show in Tables 1 and 3 that the Elastic Net achieves a monthly R^2_{OOS} of approximately 0.11%, while neural networks (NN1–NN3) reach values between 0.33% and 0.40%. Section 2.4 of their study examines portfolios built on Neural Network return forecasts and provides further supporting evidence.

[16] Specifically, in Section 1.7 and Table 1 of their work, Gu, Kelly, and Xiu (2020) compare architectures with 1–5 hidden layers and find that peak performance is achieved with three layers; deeper models do not provide additional benefits in the context of monthly returns.

- $b^{(l)}$ is the vector of bias terms for layer l;
- $f(\cdot)$ is a non-linear activation function applied element-wise.

This is the compact vector form of the scalar recursion reported by Gu, Kelly, and Xiu in their equations (17)–(18), where the output of each neuron is given by a linear combination of the previous layer's outputs followed by an activation function.[17]

The output layer is a linear transformation of the last hidden layer. Unlike cross-sectional panels that aggregate thousands of equities into a single loss function, this thesis focuses on forecasting five distinct macroeconomic asset classes with fundamentally different risk-return profiles. Consequently, the architecture is instantiated as a *Single-Task* network. A dedicated and independent neural network is trained for each specific target asset i, modeling the scalar conditional expected excess return as:

$$g_i(X_t) = W_i^{(L+1)} h_i^{(L)} + b_i^{(L+1)}. \tag{2.17}$$

This separation allows the hidden layers of each network to learn asset-specific non-linear representations of the macroeconomic predictors, preventing the dominant volatility of equity indices from overshadowing the subtle signals driving fixed-income or alternative assets.

Following the implementation by the cited literature, we use the ReLU (*Rectified Linear Unit*) as the activation function, defined as $f(x) = \max(0, x)$[18]. This introduces non-linearity through a simple, computationally efficient design suited for shallow networks.

To control overfitting, the theoretical framework of Gu et al. (2020) relies on a combination of regularization techniques. Among these, two deserve a brief theoretical note. *Learning Rate Shrinkage* consists of adaptively reducing the step size as optimization approaches a minimum, stabilizing gradient descent and preventing the algorithm from oscillating around the optimal weights; this is typically embedded within modern optimizers such as Adam (Kingma and Ba, 2014). *Batch Normalization* standardizes the inputs of each hidden layer across every mini-batch, mitigating the so-called *internal covariate shift*, i.e., the continuous change in the distribution of layer inputs during training, which would otherwise slow convergence

[17] In Section 1.7 of their work, Gu et al. (2020) describe feed-forward neural networks by defining each neuron's output as a linear combination of the previous layer's outputs followed by a non-linear activation function (ReLU). The formulation provided in Eq. (2.16) is a compact matrix rewriting of the scalar notation used in their original study; weights are indexed by the destination layer, consistently with the implementation notation in *Appendix C*.

[18] The ReLU function is defined as the maximum between 0 and x, where x represents the pre-activation input to a given neuron, calculated as the linear combination of the previous layer's outputs plus a bias term. This function maps positive inputs to themselves while assigning zero to all negative values. Despite its straightforward form, it allows neural networks to capture intricate non-linearities more effectively than traditional saturating functions, such as the sigmoid or hyperbolic tangent.

and destabilize the optimization. The remaining regularization choices, including L_1 weight penalty, early stopping, and ensemble averaging, are described in full in *Chapter 4*, where their specific implementation and hyperparameter choices are formalized.

2.4.4 Validation and Hyperparameter Tuning

To avoid *look-ahead bias*[19] and overfitting, the training of ML models follows a rigorous procedure of sample splitting over time. Hyperparameters are not chosen arbitrarily, but optimized on a *Validation Set*, which is strictly distinct from the *Training Set* (used to estimate the model parameters) and the *Test Set* (used exclusively for the final out-of-sample evaluation)[20]. The structure of this temporal division follows the scheme:

Figure 2.1: Chronological sample splitting for time series data. The diagram illustrates a single iteration of the validation approach. The dataset is divided strictly sequentially to preserve temporal dependence and prevent look-ahead bias. *Source:* Author's elaboration.

To simulate a realistic investment process, the evaluation advances on a quarterly basis, recursively updating both the estimated parameters and the hyperparameter choices as new data become available. Two distinct windowing philosophies can be adopted for this recursive procedure. A *rolling window* maintains a fixed historical depth by dropping the oldest observations as new ones arrive, allowing the model to adapt quickly to structural breaks and regime shifts. An *expanding window*, by contrast, continuously accumulates all available historical data, reducing estimator variance at the cost of slower adaptation to recent market changes. The choice between the two involves a fundamental bias-variance trade-off, and the appropriate scheme depends on the model's theoretical requirements and the signal-to-noise characteristics of the target environment. The specific windowing choices adopted for each model in this thesis are formalized in *Chapter 4*.

[19]*Look-ahead bias* occurs when a model uses information that was not yet available at the time of the investment decision. In return forecasting, this leads to an overestimation of how well the model actually predicts; see Gu et al. (2020, Section 1.1).

[20]Gu et al. (2020, Section 1.1) explicitly describe the sample splitting scheme into three temporally separate sub-periods: one for estimating the models, one for the optimal choice of hyperparameters, and one for the purely out-of-sample evaluation.

Chapter 3

Data and Variable Selection

This chapter describes the source and creation of the dataset used in this thesis. It explains the selection of target variables and the set of predictor variables, which combine macroeconomic variables with technical characteristics derived from the data. It then outlines the pre-processing steps required to prepare the data for both linear models and ML methods.

3.1 Data Description

For the empirical analysis of this thesis, five asset classes were selected to create the typical investable universe of a US-based investor: US Equity, Fixed Income, Commodities, Gold, and Real Estate. In addition to the returns of the selected indices, the dataset includes traditional macroeconomic indicators, systematic risk factors, and technical indicators. The analyzed timeframe spans from May 1972 to December 2024, covering over half a century of financial history and including periods of significant market turbulence, such as the inflationary crisis of the 1970s, the dot-com bubble, the 2008 GFC, and the recent COVID-19 pandemic.

3.1.1 Selected Asset Classes

To ensure the representativeness and transparency of the results, each asset class is proxied by a specific market index. Monthly total returns for the following components were retrieved from the *FactSet* database (accessed February 2026):

- **US Equity (S&P 500 Index – *FactSet* Ticker: *SP50*)**: It represents the US stock market, as it comprises the 500 companies with the highest market capitalization listed in the United States. The index, in its *Total Return* version, is the leading indicator of the US *large-cap* market and is widely used in the literature.

- **Fixed Income (ICE BofA US Treasury 7-10 Year Index – *FactSet* Ticker: *MLG4O2)***: This *Total Return* index tracks the performance of US dollar-denominated government bonds (Treasuries) with a remaining maturity of between 7 and 10 years. The index covers the medium-to-long end of the yield curve and represents the asset class with low credit risk.

- **Commodities (S&P Goldman Sachs Commodity Index – *FactSet* Ticker: *SPGSCI*)**: The *S&P GSCI* is a *Total Return* index that serves as a diversified benchmark for the commodities market. It comprises futures contracts on 24 commodities across five sectors: energy (which accounts for the largest weighting), agricultural products, livestock, industrial metals and precious metals. It was included because it has historically shown a low correlation with traditional financial assets.

- **Gold (LBMA Gold Price – *FactSet* Ticker: *GOLD-FDS*)**: The spot price of gold (*Gold - Free Deliverable Standard*). Unlike other commodities, gold is regarded as an asset class in its own right, chosen for its role as the ultimate safe-haven asset.

- **Real Estate (FTSE NAREIT All Equity REITs Index – *FactSet* Ticker: *FNER*)**: This *Total Return* index comprises all US publicly traded real estate investment trusts (REITs) (with the exception of mortgage REITs) that own and manage physical real estate assets (offices, shopping centres, residential properties). It was included because it provides exposure to the real estate market, in which a large number of investors allocate a significant portion of their wealth.

Specifically, all returns were converted into *Excess Returns* by subtracting the Risk-Free rate (1-month T-Bill), making the target variables consistent with risk premium theory.

3.1.2 Macroeconomic and Financial Predictors

The models' predictive power depends on the quality and variety of the input data. To provide a comprehensive view of the financial environment, the predictors (Z_t) are divided into three main categories: Fama–French factors capture traditional risk premiums; macroeconomic and fundamental predictors provide indications of long-term equilibrium conditions; finally, technical features encode short-term price dynamics and volatility regimes.

Fama–French Factors (FF-3) In addition to representing a fundamental pillar in the theoretical framework of asset pricing, the Fama and French (1993) three-factor model, including the *Market Risk Premium* (Mkt-RF), *Small Minus Big* (SMB), and *High Minus Low* (HML), has been integrated into the predictive framework. The data, sourced from the Kenneth R. French Data Library[1], allow the models to condition forecasts on the prevailing performance of risk premiums related to market exposure, company size, and style (Value/Growth).

The three Fama–French factors are used differently depending on the model. In the OLS approach, they enter through a *two-stage procedure*: factor loadings (β) are estimated first for each asset class, and expected returns are then computed by combining these exposures with the estimated factor risk premia. In the ML models the three factors are used as lagged *state variables*, shifted by one month to serve as forward-looking predictive features. This captures broad market risk signals without introducing *look-ahead bias*.

Goyal, Welch, and Zafirov Predictors (GWZ) The core macroeconomic information used in the different specifications is based on the work originally developed by Goyal and Welch (2008) and recently expanded and subjected to rigorous critical revision by Goyal, Welch, and Zafirov (2023). As discussed in *Section 2.2.2*, the 2008 study consolidated 17 macroeconomic predictor variables, demonstrating their fragility *out-of-sample*, while the 2023 implementation introduced 29 additional variables, of which only 20 are available from mid-1973, for the prediction of the equity premium[2].

Specifically, the dataset includes 37 predictors starting from mid-1973, all processed to ensure stationarity and economic consistency. As the authors noted, to avoid the issues associated with raw price levels, the predictors are transformed into valuation ratios and spreads. For instance, dividends (`d12`) and earnings (`e12`) are calculated as 12-month rolling sums to remove seasonal fluctuations in corporate data. Similarly, bond market indicators like the *Term Spread* and *Default Spread* are constructed as differentials. This approach allows us to isolate the risk premiums associated with liquidity and default from the overall level of interest rates.

To ensure the predictive architecture is strictly *ex-ante*, all features are lagged by one month ($t - 1$) relative to the target returns (t). This temporal alignment is strictly required to preclude

[1] The data were extracted from the file *"Fama/French 3 Factors [Monthly]"*. Available at: `https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html` (French, 2024)

[2] The data were exported from the *"Monthly"* sheet of the official database provided by the authors. Accessible at: `https://docs.google.com/spreadsheets/d/1O_nkOkJPvq4eZgNl-1ys63PzhbnM3S2y/edit?gid=1060486139` (Goyal et al., 2024)

look-ahead bias, thereby guaranteeing that the forecasting algorithms are trained exclusively on information available at the decision point. By imposing this one-month lag, the model is forced to rely solely on historical data, as in a real-world setting, ensuring that the predictive signals are causal and the backtesting results are economically realistic.

To better understand the nature of these macroeconomic predictors, *Table 3.1* below lists the main predictors selected for this study on the basis of their empirical relevance. A comprehensive assessment of their specific predictive contributions and how these are calculated will be presented alongside the final results in *Section 5.1.4*.

Table 3.1: Selected GWZ Predictors and Their Economic Description

Code	**Variable Name**	**Construction and Economic Description**
Panel A: Price and Valuation Ratios		
`price`	Price Index Level	S&P 500 price index value.
`d12`	Dividends (12m sum)	12-month rolling sum of dividends paid on the S&P 500.
`e12`	Earnings (12m sum)	12-month rolling sum of S&P 500 earnings.
`d/p`	Dividend–Price Ratio	Log 12-month dividends minus log prices.
`e/p`	Earnings–Price Ratio	Log 12-month earnings minus log prices.
`d/e`	Dividend–Payout Ratio	Log 12-month dividends minus log earnings.
Panel B: Interest Rates and Credit Spreads		
`tbl`	Treasury Bill Rate	3-month Treasury Bill secondary market rate.
`lty`	Long-Term Bond Yield	Yield on long-term government bonds.
`AAA`	AAA Bond Yield	Yield on AAA-rated corporate bonds.
`tms`	Term Spread	Long-term government bond yield minus Treasury-bill rate.
`dfy`	Default Yield Spread	BAA-rated minus AAA-rated corporate bond yield.
`dfr`	Default Return Spread	Long-term corp. bond minus long-term gov. bond return.
`corpr`	Corporate Bond Return	Long-term corporate bond return.
Panel C: Macro and Market Condition Indicators		
`ygap`	Stock–Bond Yield Gap	Dividend-price ratio minus 10-year Treasury bond yield.
`lzrt`	Log Zero-Return Illiquidity	Log fraction of trading days with zero equity return.
`svar`	Stock Variance	Sum of squared daily returns on the S&P 500.
`rdsp`	Stock Return Dispersion	Cross-sectional SD of returns on 100 size/BM portfolios.
`dtoat`	Distance to ATH	Dow Jones index distance from its all-time historical high.

Note: All variables are lagged one month ($t-1$) relative to the target return (t) to prevent look-ahead bias. A complete variable-by-variable description of all 37 predictors, including construction details and original references, is provided in *Appendix A, Table A.1*. *Source:* Author's elaboration based on Goyal and Welch (2008) and Goyal et al. (2023)

Technical Features for Machine Learning In addition to the fundamental macroeconomic variables from Goyal et al., to capture signals derived from price evolution and short-term volatility, 20 additional technical *features* were created (4 for each of the 5 target asset classes), derived entirely from return time series. The inclusion of these indicators is based on empirical evidence provided by Neely et al. (2014), which suggests that technical variables capture information orthogonal to macroeconomic fundamentals: while economic predictors capture slow changes in the business cycle, price indicators identify regime shifts and market crashes[3]. Specifically, the following metrics were calculated:

- **12-Month Momentum** (`[Asset]_Mom_12m`): Sum of returns over the last 12 months. As demonstrated by Moskowitz, Ooi, and Pedersen (2012) with their work on *Time Series Momentum*, this metric can intercept persistent trends across all considered asset classes (stocks, bonds, and commodities).
- **1-Month Lagged Return** (`[Asset]_Ret_1m`): Return of the strictly preceding month, used to capture very short-term autocorrelation dynamics (continuation or *reversal* effects). Moskowitz et al. show that the return of the first preceding month plays a primary statistical role in triggering price persistence[4].
- **3-Month Moving Average** (`[Asset]_MA_3m`): Short-term moving average that captures short-term trend reversals over a quarterly horizon, in line with the technical indicators used by Neely et al. (2014).
- **6-Month Volatility** (`[Asset]_Vol_6m`): 6-month *rolling* standard deviation. Providing this metric as input allows for the timely mapping of transitions toward high risk-aversion regimes. As deducible from Neely et al. (2014), the predictive efficacy of technical indicators tends to maximize during phases of economic contraction and financial stress[5].

These technical variables were integrated for a precise purpose related to Machine Learning capabilities. The simultaneous inclusion of numerous, highly correlated predictors creates a dense, high-dimensional environment in which standard parametric models inherently struggle: as demonstrated in the empirical analysis, adding these features to an unregularized OLS regression severely degrades its *out-of-sample* performance, leading to coefficient instability and extreme *overfitting* driven by multicollinearity. Consequently, these custom features are primarily designed to be exploited by advanced Machine Learning algorithms, which possess the structural tools to handle such complexity through penalization and non-linear

[3] As highlighted by Neely et al. (2014, p. 1772): "Technical indicators better detect the typical decline in the actual equity risk premium near business-cycle peaks".

[4] Moskowitz et al. (2012, p. 228) document "significant time series momentum in equity index, currency, commodity, and bond futures [...] with persistence in returns for one to 12 months." Formally testing individual lags $k = 1, \dots, 60$, the authors further show that the coefficient of the first monthly lag ($k = 1$) exhibits, on average, one of the strongest and most statistically significant positive autocorrelation signals across all asset classes (2012, Section 3).

[5] In formalizing Moving Average (MA) rules, Neely et al. (2014, Section 2.1) specify the use of short-term averages of 1, 2, and 3 months as opposed to long-term averages. The authors further explicitly link the success of price indicators to inefficiencies generated by fire sales during crisis periods, emphasizing that "[...] technical indicators display enhanced predictive ability during recessions" (2014, p. 1773).

representations. The extent to which each model actually leverages these signals, and which specific features prove most informative, is examined empirically in the feature importance analysis of *Chapter 5*.

3.2 Dataset Construction and Pre-processing

The dataset preparation phase followed a rigorous protocol to avoid statistical biases and ensure the total reproducibility of the empirical results. The entire *pipeline* was structurally developed within a *Python* environment, and the data matrix consolidation process was organized into the following steps:

1. **Temporal Alignment**: All time series sourced from *FactSet*, French, and Goyal-Welch were synchronized on a monthly basis using the last available observation.

2. **Reconstruction of the S&P 500 Index**: Prior to 1988, data from *FactSet* for the S&P 500 Total Return Index is not available, but the time series has been extended back to 1973 using the Fama–French market proxy, reconstructed as Mkt = (Mkt-RF) + R_f, which captures the gross market return by adding the risk-free rate to the excess return factor.

3. **Handling of Missing Data and Burn-in Period**: The *Forward-Fill* (LOCF - Last Observation Carried Forward) technique was applied to fill any specific gaps in the price data. Subsequently, the generation of the 12-month Momentum created missing values for the first 12 rows of each time series; consequently, these initial observations (May 1972 – April 1973) were removed across all columns.

4. **Standardisation**: The features were standardised as required by regularization and deep learning algorithms. However, the mean and variance parameters were calculated exclusively on the *training* window and subsequently applied to the *validation* and *test* sets to avoid *look-ahead bias*.

The final dataset thus takes the form of a structured *panel* consisting of 608 monthly observations and 65 variables in total: 5 *excess returns* of the asset classes (the *target* variables) and 60 independent variables, composed of 3 Fama–French factors (FF-3), 37 macroeconomic predictors from Goyal et al. (GWZ), and 20 *technical features* (TF). This high-dimensional matrix is now ready to be processed by the *backtesting* architecture described in the following chapter.

Chapter 4

Model Architecture and Implementation

This chapter translates the theoretical framework of *Chapter 2* into an empirical pipeline, detailing the implementation of all forecasting models, from the OLS-Base and its Lasso extension to Elastic Net and Neural Networks, and formalizing the portfolio optimization engine, including covariance estimation, constraints, and walk-forward validation architecture. The empirical analysis and multi-asset backtesting are fully implemented in Python[1].

4.1 Linear Modeling and Regularization: Empirical Design

The architecture of the baseline model implements a Two-Step Predictive Regression framework. The empirical objective is to estimate the vector of expected returns $E_t[Y_{t+1}]$ and the expected covariance matrix $\Sigma_{Y,t+1}$ for a universe of N assets (in our case $N = 5$), utilizing a set of K contemporaneous risk factors and P lagged macroeconomic variables as predictors. Let us define the following vectors and matrices at time t:

- $y_{i,t} \in \mathbb{R}$: excess return of asset i at time t.
- $F_t \in \mathbb{R}^{K\times 1}$: vector of Fama–French factors (Mkt-RF, SMB, HML), with $K = 3$.
- $Z_t \in \mathbb{R}^{P\times 1}$: a vector of predictor variables, which includes the 37 macroeconomic variables from Goyal-Welch and the 20 custom technical characteristics, with $P = 57$.

The out-of-sample backtest is constructed by iterating all models over historical time windows, with the evaluation strictly beginning at index 180 ($t = 180$) to ensure a perfectly aligned

[1] See *Appendix C.1* for the pseudocode of all algorithms, and *Appendix C.2* for a summary of model architectures and hyperparameters. To ensure transparency and the complete reproducibility of the results, the source code, data processing pipelines, and interactive visualizations are hosted on *GitHub* at the following repository: `https://github.com/botta-andrea/Bachelor_Thesis_USI-Asset_Allocation_via_ML_Factor-Models`.

comparison across methodologies. For the linear models discussed in this section, both the Baseline OLS and the OLS-Lasso, the estimation relies on a *rolling* window of 120 months, whose rationale is discussed below.

4.1.1 The OLS-Base: Implementation of the Two-Step Framework

The factor-based framework established in *Chapter 2* is translated here into a computational pipeline that maps raw historical data into OOS portfolio parameters within a rolling window. Within each iteration, the algorithm executes four sequential steps.

Step A: Estimation of Risk Exposures (Factor Loadings). The algorithm first considers the structural risk model (Eq. 2.7) for each asset i over the active training window $[t - T_{train}, t]$:

$$Y_{i,\tau} = \alpha_i + B_i^T F_\tau + \epsilon_{i,\tau} \quad \text{for } \tau \in [t - T_{train}, t] \tag{4.1}$$

Within each rolling window, the true parameters $\alpha_i \in \mathbb{R}$ and $B_i \in \mathbb{R}^K$ are estimated by OLS, yielding the estimates $\hat{\alpha}_i \in \mathbb{R}$ and $\hat{B}_i \in \mathbb{R}^K$ for asset i. Aggregating across all N assets, the algorithm extracts the full estimated factor exposures matrix $\hat{B} \in \mathbb{R}^{N\times K}$ and the intercept vector $\hat{\alpha} \in \mathbb{R}^{N\times 1}$. The in-sample residuals $\hat{\epsilon}_{i,\tau} = Y_{i,\tau} - \hat{\alpha}_i - \hat{B}_i^T F_\tau$ are simultaneously extracted to populate the diagonal matrix of sample idiosyncratic variances, $\hat{V}_\varepsilon \in \mathbb{R}^{N\times N}$.

Step B: Forecasting Fama–French Factors (Predictive Step). To project the risk premia forward (Eq. 2.9), we specify a predictive model that projects the true factor returns onto the expanded set of one-period lagged macroeconomic predictors ($Z_{\tau-1}$):

$$F_{k,\tau} = c_k + C_k^T Z_{\tau-1} + u_{k,\tau} \quad \text{for } \tau \in [t - T_{train}, t] \tag{4.2}$$

The true coefficient vector $C_k \in \mathbb{R}^P$ and intercept c_k are estimated by OLS (or Lasso in the penalized variant), yielding the estimates $\hat{C}_k$ and $\hat{c}_k$. While Gu et al. (2020) rely on a Huber loss to limit the influence of severe outliers[2], our baseline minimizes the standard MSE. Diversified factor portfolios are less prone to extreme outliers than individual equities, making Huber truncation unnecessary and potentially harmful to forecasting power.

Once the full coefficient matrix $\hat{C} \in \mathbb{R}^{K\times M}$ and the intercept vector $\hat{c}$ are estimated, the conditional expected factor returns for the out-of-sample period $t + 1$ are computed by

[2]For a prediction error e, the MSE applies a strictly quadratic penalty: $L_{MSE}(e) = e^2$. The Huber loss introduces a threshold δ: $L_{Huber}(e) = \frac{1}{2}e^2$ for $|e| \leq \delta$, and $L_{Huber}(e) = \delta|e| - \frac{1}{2}\delta^2$ for $|e| > \delta$, so that extreme outliers receive a linear rather than quadratic penalty.

evaluating the estimated regression on the latest available row of predictors (Z_t):

$$\hat{\mathbb{E}}_t[F_{t+1}] = \hat{c} + \hat{C}Z_t \tag{4.3}$$

Step C & D: Parameter Recombination and Covariance Construction. Following the logic of Eq. (2.11), the macroeconomic factor forecasts from Step B are multiplied by the static factor exposures from Step A to compute the estimated expected asset returns:

$$\hat{\mathbb{E}}_t[Y_{t+1}] = \hat{\alpha} + \hat{B} \cdot \hat{\mathbb{E}}_t[F_{t+1}] \tag{4.4}$$

Concurrently, the empirical covariance matrix $\hat{\Sigma}_{Y,t+1}$ required by the optimizer is constructed using the factor-implied formulation from Eq. (2.12):

$$\hat{\Sigma}_{Y,t+1} = \hat{B}\hat{V}_f\hat{B}^T + \hat{V}_\varepsilon \tag{4.5}$$

where $\hat{V}_f \in \mathbb{R}^{K\times K}$ is the 120-month sample covariance matrix of the Fama–French factors.

Theoretical Validation: The "Oracle" Benchmark. To verify the pipeline works correctly, an *Oracle* test was run using perfect foresight: instead of forecasting the factors, the realized values F_{t+1} were fed directly into Step C, bypassing Step B entirely. The results confirm that the factors are strong structural drivers of returns, especially for the S&P 500, where the Out-of-Sample R^2 reaches $\sim$ 99%[3]. The Oracle test confirms that the factor exposures ($\hat{B}$) are well estimated for most asset classes, meaning their performance depends on how well Step B forecasts the macroeconomic factors. For some assets, in particular gold, the factor structure appears weaker, suggesting that the FF-3 alone may not fully capture their return dynamics.

Empirical Justification for the Rolling Window. For the implementation of OLS-Base, a 120-month moving window was chosen rather than an expanding one. Practically, using a rolling window can be seen as a nonparametric way to account for the time-variation of the parameters (B and C): while macroeconomic relationships and factor loadings evolve over time, it is reasonable to assume they remain stable within a shorter, fixed window. The choice of window length is therefore a trade-off: a shorter window adapts faster to structural breaks but uses fewer observations, increasing estimation variance; while a longer window produces statistically more stable estimates but may retain outdated economic relationships (such as the high-inflation environment of the 1970s). While an expanding window reduces overall estimator variance and slightly improves out-of-sample R^2, this rigidity does not translate into better forecasts for Mean-Variance Optimization (MVO), leading instead to large allocation

[3]Full out-of-sample metrics across all evaluated asset classes are reported in *Appendix B.2*, *Table A.3*.

errors. The rolling window effectively acts as a *forgetting mechanism*, discarding obsolete data to maintain alignment with current market conditions.

4.1.2 The Curse of Dimensionality in Standard OLS

While the Two-Step framework provides a solid structural baseline, relying on unpenalized OLS in Step B becomes highly problematic with large datasets due to the *Curse of Dimensionality*[4].

In standard OLS, the stability of the coefficient vector $\hat{\Lambda}$ depends on $(Z^T Z)^{-1}$. As the number of predictors (P) grows relative to training observations (T_{train}), the natural multicollinearity of macroeconomic indicators makes this matrix ill-conditioned, causing extreme sensitivity to sample noise. This ill-conditioning renders unpenalized OLS unreliable: the estimator overfits the training sample and loses all predictive power out-of-sample (Gu et al., 2020, Section 1.3)

To observe this within our framework, we compared the *Baseline OLS* (restricted to 37 Goyal-Welch predictors) against an *Alternative OLS* incorporating the full 57-feature set. The results clearly show the curse of dimensionality at work: instead of improving accuracy, adding 20 features causes the OOS R^2 to collapse, dropping below -100% for certain asset classes (mean $R^2_{OOS} = -289.9\%$)[5]. This statistical instability directly translates into erratic portfolio turnover and a severely degraded Sharpe Ratio. This evidence highlights a fundamental trade-off. Restricting OLS to 37 variables maintains basic stability but still yields a mean R^2_{OOS} of -87.34% (as shown in *Chapter 5*). This limitation provides the key justification for introducing regularization techniques such as Lasso and Elastic Net, specifically designed to extract predictive signals from high-dimensional, collinear spaces.

4.1.3 Penalized OLS-Lasso to Address Overfitting

To overcome the curse of dimensionality and safely extract signals from the expanded set of 57 predictors, we replace the standard unpenalized OLS in Step B with a LASSO predictive regression. Incorporating this full feature space of 57 predictors is essential to permit a direct and rigorous comparison with the advanced Machine Learning models presented later.

Implementation of the L_1 Penalty in Step B. In our Two-Step framework, Lasso regularization is applied during the Step B predictive regression. To resolve the ill-conditioning of

[4]As discussed by Hastie et al. (2009, p. 22), the feature space becomes increasingly sparse as dimensions grow, forcing models to extrapolate over large distances and inflating the variance of unpenalized estimates.

[5]This aggregate figure is reported for diagnostic purposes only; the Alternative OLS is not used in the portfolio analysis and does not appear in *Table 5.1*. A brief note documenting the asset-level breakdown of this diagnostic is provided in *Appendix B.1*.

the high-dimensional covariance matrix, an L_1 penalty is added to the objective function:

$$\hat{\lambda}_k = \arg\min_{\lambda_k} \left(\frac{1}{T} \sum_{t=1}^{T} \left(F_{k,t} - \lambda_k^T Z_{t-1} \right)^2 + \alpha \sum_{j=1}^{P} |\lambda_{k,j}| \right) \tag{4.6}$$

where $\lambda_k \in \mathbb{R}^P$ is the coefficient vector estimated separately for each FF factor $k \in \{1, \ldots, K\}$, collectively aggregated into the penalized matrix $\hat{\Lambda} = [\hat{\lambda}_1, \ldots, \hat{\lambda}_K]^\top$. The geometric properties of the L_1 norm allow Lasso to shrink the coefficients of noisy or redundant predictors exactly to zero. This built-in variable selection naturally reduces dimensionality, preventing the model from overfitting to sample noise while isolating only the most robust macroeconomic signals.

Walk-Forward Hyperparameter Tuning. The efficacy of this regularization depends on the penalty parameter (α), which we optimize at each quarterly rebalancing via a walk-forward grid search. To ensure a fair structural comparison, the Lasso predictive regression maintains 120-month inner training window as the Baseline OLS to estimate its core coefficients. Penalized regressions require a dedicated tuning phase, so the historical block is expanded to 180 months to include a 60-month validation set. Using a strict 120-month window would have left only 60 observations to estimate 57 predictors, critically compromising the degrees of freedom. At each iteration, standard scaling is applied to both predictors (Z) and targets (F) to ensure uniform penalization across variables. The optimal α is selected from a log-linear grid of 50 values ranging from 0.01 to 1.0, choosing the value that minimizes the validation Mean Squared Error (MSE). The model is then refitted on the full 180-month window to produce the out-of-sample factor forecasts ($E_t[F_{t+1}]$).

Predictor Sparsity and Stability relative to the OLS-Base. Lasso acts as a dynamic filter that safely incorporates the full feature space, overcoming the multicollinearity that collapses the unpenalized OLS-Base. Observations from the algorithm's execution confirm this behavior: the optimized penalty restricts the 57 available predictors to a global average of just 4.2 active variables per window, with median α values of 0.22, 0.17, and 0.20 for the Mkt-RF, SMB, and HML targets, respectively[6]. This extreme parsimony structurally validates the methodology: the Step B regression effectively neutralizes multicollinear noise, stabilizing the expected return forecasts before passing them to the portfolio optimizer.

[6]Full distributions of the selected hyperparameters are shown in *Figure A.1* in *Appendix C.3*.

4.2 Machine Learning Forecasting Models

4.2.1 Elastic Net Implementation

Transitioning from the sequential Two-Step framework to a direct forecasting approach, the Elastic Net (ENet) bypasses the factor regression phase to forecast asset returns directly from the expanded predictor set, which now includes both the 57 Goyal-Welch and custom features and the 3 Fama–French factors as inputs. In this high-dimensional setting, a pure L_1 penalty would arbitrarily discard highly correlated predictors. By blending Lasso's sparsity with Ridge's proportional shrinkage, ENet retains groups of overlapping signals, providing more robust out-of-sample forecasts.

Model Instantiation and Direct Forecasting Setup. Building upon the theoretical framework established in Eq. (2.15), the Elastic Net is empirically instantiated to predict the asset excess return vector Y_{t+1} directly. To maximize the informational content for this specific task, the expanded predictor vector is formally defined as $X_t = [Z_t^\top, F_t^\top]^\top \in \mathbb{R}^{60}$. This formulation contains exactly 60 variables: the macro-technical set Z_t, containing the 37 Goyal et al. macroeconomic predictors and 20 custom technical features, alongside F_t, which represents the 3 Fama–French factors.

A dedicated regression is fitted independently for each of the five target assets i. Within each rolling iteration, the algorithm minimizes the penalized Mean Squared Error over the active training window $[t - T_{train}, t]$, computing the coefficient vector $\hat{\Lambda}_i$ as follows:

$$\hat{\Lambda}_i = \arg\min_{\Lambda} \left(\frac{1}{T_{train}} \sum_{\tau=t-T_{train}}^{t-1} \left(Y_{i,\tau+1} - \Lambda^T X_\tau \right)^2 + \alpha \left(\rho \sum_{j=1}^{P} |\Lambda_j| + \frac{1-\rho}{2} \sum_{j=1}^{P} \Lambda_j^2 \right) \right) \tag{4.7}$$

where α is the global penalty scaling parameter (following the notation established in Eq. (2.15) As dictated by the mechanics of L_1 and L_2 regularization, both the feature matrix X_τ and the target returns Y_τ are rigorously standardized prior to optimization.

Our ENet specification minimizes the standard MSE rather than a Huber loss (first term of Eq. 4.7), ensuring direct comparability with the OLS and Lasso benchmarks, which share the same objective function. Expected returns are then generated by multiplying the optimized coefficients $\hat{\Lambda}_i$ by the out-of-sample predictors X_t, and reverting the standardized predictions to the original scale using the asset-specific mean and standard deviation from the training set.

Rolling Window and Grid Search Specification. The predictive architecture relies on a walk-forward optimization framework, structured via a rolling window to swiftly adapt to macroeconomic regime shifts (as illustrated in *Figure 4.1*). Since the Elastic Net is a linear model, a strict 180-month rolling window is used to discard obsolete macroeconomic relationships. Within each quarterly iteration, this 15-year block is split into 120 months for training and 60 months for validation.

The optimization space spans 150 combinations per target asset, utilizing an aggressive two-dimensional grid search. For the L_1-ratio (ρ), the search space is constrained to $\rho \in \{0.1, 0.5, 0.9\}$, adopting the proven asset pricing specification of Gu et al. (2020). For the overall penalty (α), we deploy a fixed, log-linear grid of 50 values spanning from 0.001 (10^{-3}) to 31.622 ($10^{1.5}$). Rather than a linear grid, this log-space structure enforces a *coarse-to-fine* exploration: it spans a wide range of regularization regimes while densifying evaluation points at lower magnitudes, where model sensitivity is empirically highest[7]. This is confirmed by the algorithm's execution: although the grid permits penalizations up to 31.62, the optimized α systematically converged to much smaller values, with a median of 0.301 for most assets and 0.373 for REITs[8]. This range aligns with the optimal penalization interval (0.24–0.29) identified in the OLS-Lasso Step B, confirming a consistent structural noise level across the expanded dataset.

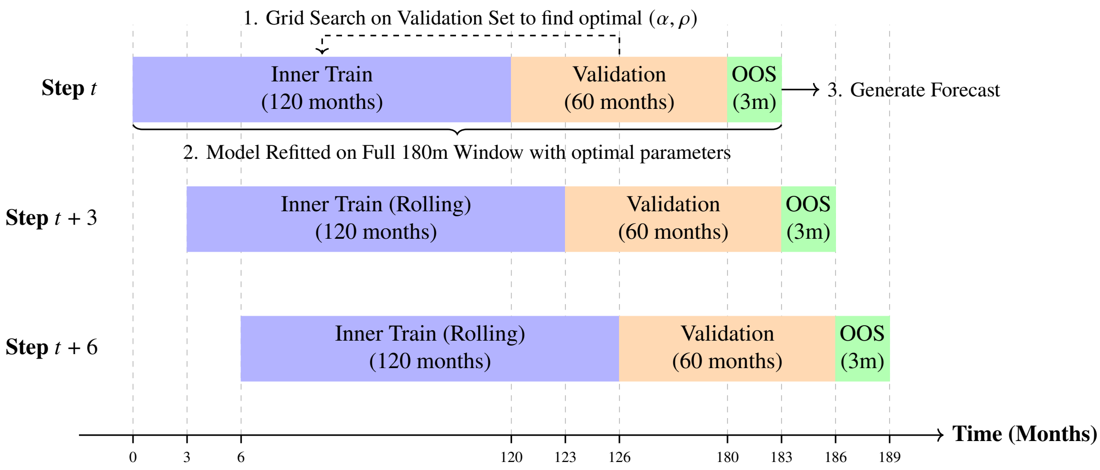

Figure 4.1: Rolling Walk-Forward Validation for the Elastic Net. At each quarterly step, hyperparameters are tuned on a 120-month training and 60-month validation window, then the optimal model is refitted on the full 180-month block to generate the 3-month OOS forecast. *Source:* Author's elaboration.

[7]The validation MSE typically exhibits a U-shaped form with respect to α. A log-linear grid, also adopted by Gu et al. (2020), concentrates evaluation points near the minimum, which lies at lower penalty magnitudes, ensuring precise identification of the optimal shrinkage without wasting resources on high-bias flat regions.

[8]See *Figure A.2* in *Appendix C.3* for the full walk-forward evolution of both α^* and ρ^*.

Empirical Calibration and Asset-Specific Blending. At each rolling iteration, the (α, ρ) pair minimizing the validation MSE is selected, and the model is refitted on the full 180-month window before out-of-sample forecasting. Log diagnostics show that this dynamic calibration captures the diverse statistical properties of the target universe. For equities and risk-on assets (e.g., S&P 500, Commodities, Gold), the optimizer favours Lasso-like sparsity, selecting $\rho = 0.9$ in approximately 58% to 67% of windows. For safe-haven fixed-income assets like 7-10 Year Treasuries, the optimizer instead detects a different signal structure, shifting toward Ridge-like dense shrinkage ($\rho \leq 0.5$ in 62.3% of iterations) to retain a broader set of active predictors. This adaptability empirically validates the L_2 penalty: a pure Lasso would be suboptimal for multi-asset direct forecasting.

4.2.2 Neural Networks Implementation

While the Elastic Net handles high dimensionality, its linearity limits its ability to capture complex, regime-dependent interactions. For instance, a momentum signal's predictive power may change drastically during an interest rate shock. To map these non-linearities, the pipeline introduces Artificial Neural Networks (NNs), directly motivated by Gu et al. (2020), whose *horse race* showed that NNs systematically outperform both linear models and tree-based algorithms in out-of-sample R^2 and portfolio performance.

Architecture and Dimensionality Setup. As shown by Gu et al. (2020), overly deep networks are counterproductive in the low signal-to-noise environment of financial markets, as excess capacity tends to fit noise rather than genuine signals. Their results show that forecasting performance peaks at three hidden layers (*NN-3*), while deeper architectures (*NN-4*, *NN-5*) rapidly overfit and degrade out-of-sample predictability. Our model adopts the exact NN-3 topology, illustrated in *Figure 4.2*. Since our dataset contains 60 dense macro-technical predictors rather than the hundreds of sparse firm-level characteristics used by Gu et al., directly adopting their network size would over-parameterize the model and cause severe overfitting.

To align the model's learning capacity with our restricted predictor space, the network's architecture is structurally compressed. Specifically, we halve the original $[32, 16, 8]$ neuron configuration proposed by Gu et al. (2020) for their NN-3 implementation, adopting a narrower $[16, 8, 4]$ bottleneck design that forces dimensionality reduction and prevents the memorization of noise. In the architecture, non-linearities are introduced through the Rectified Linear Unit (ReLU) activation function, which is applied after each hidden layer alongside Batch

Normalization (BN)[9] to ensure a stable and efficient optimization.

Consistent with our theoretical framework, the pipeline utilizes a *Single-Task* approach, training five independent networks per evaluation step. A shared *Multi-Task* architecture was tested but discarded: the dominant volatility of equities overshadowed the subtler signals of lower-variance assets such as fixed income, distorting gradient updates and yielding unreliable forecasts.

Before training, the target vector Y is standardized to keep the optimization numerically stable and avoid gradient explosions during backpropagation. Standardising all assets on a common scale also ensures that the network is subject to a uniform level of risk exposure, regardless of the magnitude of individual returns. Once the model generates its OOS predictions, these are rescaled back to the original distribution before being passed to the portfolio optimizer.

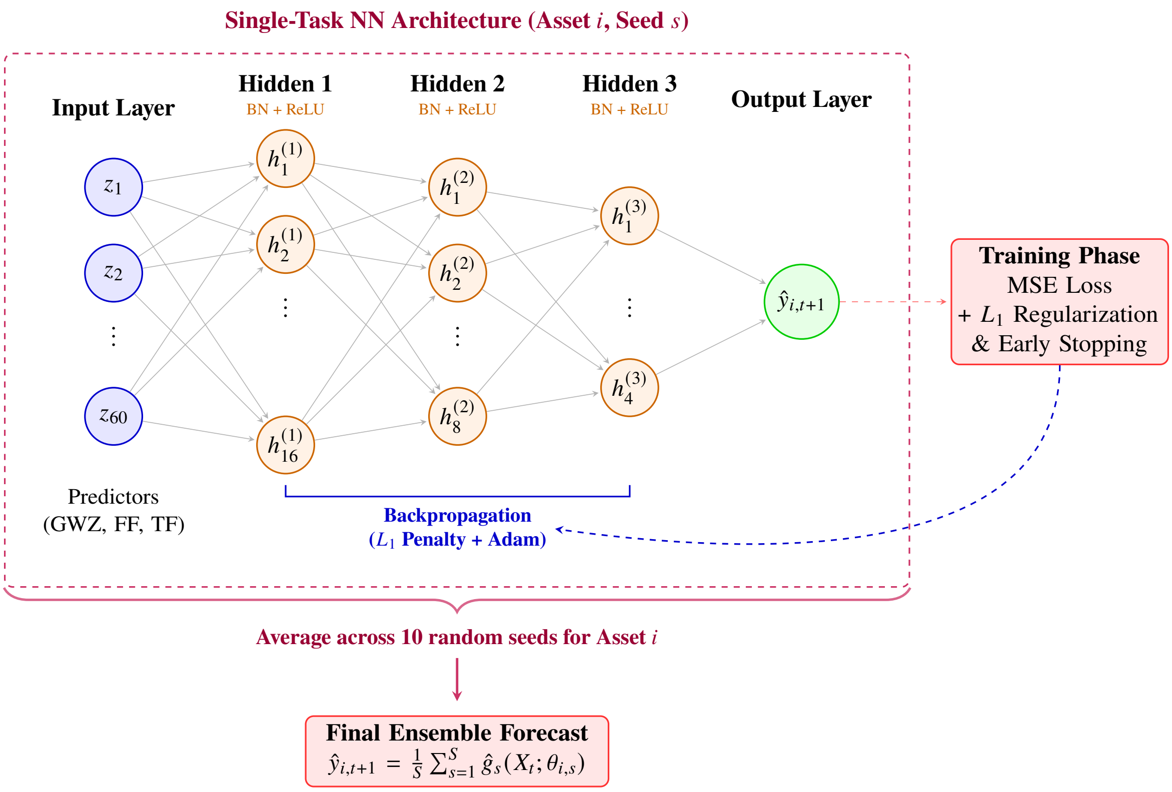

Figure 4.2: Single-Task Neural Network architecture for a generic asset i. The model utilizes an MSE Loss function with L_1 regularization, Batch Normalization (BN), and ReLU activation. The final prediction is the result of an ensemble average across 10 independent training seeds to ensure numerical stability. *Source:* Author's elaboration.

[9]Practically, Batch Normalization is implemented using mini-batches of 16 observations (see *Appendix C.1, Algorithm 3*). In this single-asset context, it provides the stability needed for successful convergence, effectively countering the high level of noise in individual stock returns.

Expanding Window and Hyperparameter Grid Search. Unlike the constant 180-month rolling window utilized for the ENet, NN-3 requires more data to properly initialize and tune their weights. Therefore, the backtesting engine transitions to an *expanding window* framework, showed in *Figure 4.3*). At each quarterly evaluation step t, the algorithm utilizes all available historical data from the beginning of the sample ($t = 0$) up to $t - 60$ as the active training set. This means the inner training window starts at 120 months and continuously expands, while the most recent 60 months $[t - 60, t]$ are strictly reserved as the validation set.

The architecture applies an L_1 weight penalty to the network connections in order to control the model's complexity. The optimal penalty is selected dynamically at each step via a grid search across five orders of magnitude: $\alpha \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, values that correspond to the specifications used in Gu et al. (2020, Appendix A.5, p. 13). The empirical analysis shows that the validation set predominantly selects the maximum available penalty ($\alpha = 10^{-1}$) across all five assets and for the entire OOS period[10], confirming the high noise level of the macroeconomic dataset. However, the use of more restrictive parameters (for example, $\alpha \geq 0.2$) leads to severe underfitting; indeed, it can be seen that excessive regularization drastically reduces the network's weights to zero, compromising its predictive ability and drastically reducing out-of-sample performance. Finally, the architecture intentionally bypasses the use of Dropout[11], relying instead on BN applied before each ReLU activation to mitigate internal covariate shifts and gently regularize the forward pass.

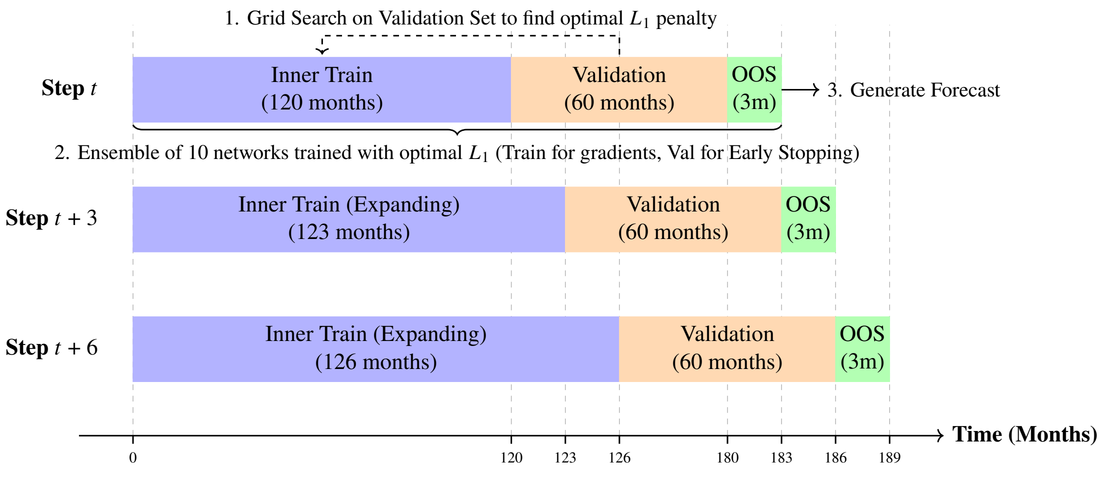

Figure 4.3: Expanding Walk-Forward Validation for Neural Networks. At each quarterly step, hyperparameters are tuned on an expanding training window (from 120 months) and a rolling 60-month validation window. An ensemble of 10 networks is then trained on the same split with Early Stopping to generate the 3-month OOS forecast. *Source:* Author's elaboration.

[10] See the evolution of optimal L_1 penalty $\alpha^\star$ per asset in *Figure A.3* in *Appendix C.3*

[11] Dropout randomly deactivates a fraction of neurons during training to prevent co-adaptations. Gu et al. (2020) exclude it from their architecture, relying solely on L_1 shrinkage and Early Stopping for regularization.

Optimization Routine and Early Stopping. In the training phase, the standard Mean Squared Error (MSE) is minimized. The Huber Loss was omitted because the targets are diversified macroeconomic indices rather than individual stocks. Unlike single equities, these aggregated indices exhibit significantly lower maximum drawdowns, meaning their extreme values often capture important market regimes rather than idiosyncratic noise. In this context, such observations provide crucial information and should not be down-weighted.

The optimization is powered by the Adam algorithm (Adaptive Moment Estimation)[12], instantiated with an initial learning rate of 10^{-3} and a batch size of 16 observations[13]. The training loop allows for a maximum of 100 epochs, but crucially relies on an aggressive *Early Stopping* protocol. After each epoch, the MSE is evaluated on the 60-month validation set and if the validation error fails to improve for exactly 5 consecutive epochs (the *patience* parameter), the optimization is immediately halted, and the weights from the best-performing epoch are restored. Empirically, this mechanism proves highly restrictive: across the 143 walk-forward steps, optimization converges within a narrow range of 8 to 13 average epochs. This rapid convergence suggests that the network identifies structural predictive signals early in training, with Early Stopping preventing the model from entering the noise-memorization phase and effectively substituting heavy L_2 penalization.

Ensemble Generation. Since the loss landscape of a neural network is non-convex, Adam's convergence path is highly sensitive to the random initialization of weights. To neutralize this structural variance, the pipeline employs an *Ensemble Learning* routine. For each asset, once the optimal L_1 penalty is identified, the algorithm does not rely on a single model. Instead, it trains $S = 10$ identical networks in parallel using 10 different random seeds for parameter initialization, following the methodology of Gu et al. (2020). Using this technique, the final OOS expected return for asset i at time $t + 1$ is the simple average of the 10 individual forecasts. Averaging across the ensemble stabilizes predictive performance over the full 35-year out-of-sample period.

[12] Introduced by Kingma and Ba (2014), Adam computes adaptive learning rates for individual parameters using estimates of first and second moments of the gradients, making it particularly efficient for financial forecasting as noted by Gu et al. (2020, Appendix A.3, p. 10).

[13] The batch size defines how many observations are processed before each parameter update. Gu et al. (2020, Appendix A.5, p. 13) use a batch size of 10,000, suited to their large equity panel. Our training window of 120 months is much smaller, so a batch size of 16 is used instead. This guarantees multiple weight updates per epoch and introduces enough stochastic noise to help Adam escape local minima, acting as an additional source of implicit regularization.

4.3 Portfolio Allocation Framework: MVO Implementation

Generating accurate expected returns $\hat{\mu}$ is only the first step of the pipeline. While the literature (e.g., Gu et al., 2020) typically evaluates predictive signals through rank-based long-short portfolios, this approach is unfeasible for a universe of five asset classes. Our forecasts are therefore integrated into a Mean-Variance Optimization (MVO) engine that exploits the full magnitude of the predictions. The portfolio is rebalanced quarterly across the entire 1989–2024 out-of-sample period using the forecasts generated by each model at each step.

4.3.1 The Mean-Variance Engine and Covariance Estimation

Algorithmic Implementation and Dimensionality. The allocation is driven by maximizing a standard mean-variance utility function with a constant relative risk aversion parameter of $\gamma = 5.0$[14]. This moderately high value acts as a natural regularizer: by heavily penalizing risk, it anchors the optimizer toward the stable covariance structure rather than the more volatile return forecasts, reducing portfolio turnover. The constrained quadratic programming problem is solved at each quarterly rebalancing date t using the Sequential Least Squares Programming (SLSQP) algorithm. The small dimensionality of our investment universe ensures mathematical tractability. In typical stock-level frameworks, the number of assets exceeds the number of observations ($N > T$), producing a singular covariance matrix that cannot be inverted. In our case, however, the optimizer targets only $N = 5$ asset classes over a $T = 60$ month rolling window, so observations strictly outnumber assets ($T \gg N$) and the sample covariance matrix is naturally full-rank. This guarantees that the matrix inversion required by the optimizer is feasible even without advanced regularization.

Ledoit-Wolf Covariance Regularization. Despite this structural advantage, MVO algorithms inherently act as "estimation-error maximizers" (Michaud, 1989), remaining sensitive to the residual noise of historical samples, as we reported in *Section 2.1.2*. Using only the sample covariance matrix would inject instability into the allocation weights, so the same shrinkage estimator is applied uniformly across all models to isolate the predictive contribution of each architecture. Specifically, the pipeline implements the shrinkage estimator of Ledoit and Wolf (2004), which blends the noisy sample covariance with a structured target matrix to produce a well-conditioned estimate. Preliminary backtests confirmed that Ledoit-Wolf

[14]Empirical backtesting showed that $\gamma = 5.0$ best balances signal exploitation and allocation stability. Lower values produced excessive turnover, while higher values suppressed the impact of return forecasts. This slightly higher risk-aversion calibration also helps reduce the "error-maximization" tendency described by Michaud (1989).

shrinkage consistently provided an additional layer of robustness over the standard sample covariance, even if marginal. Eigenvalue diagnostics confirm this choice: across the 143 rebalancing dates, the condition number peaks at just 12.9, well below singularity thresholds[15], ensuring that the optimizer's outputs reflect genuine predictive signals rather than numerical artifacts.

4.3.2 Real-World Constraints and Turnover Management

Even with robust inputs, the unconstrained maximization of mean-variance utility frequently leads to allocations that are theoretically optimal but practically uninvestable. To assess the true economic viability of the forecasting models, the optimization routine is evaluated under two distinct structural setups:

- **Long-Only Setup (LO):** The optimizer is subject only to a strict long-only constraint ($w_i \geq 0$) and a fully-invested budget constraint ($\sum w_i = 1$). The empirical results reveal the limitations of this theoretical approach: the models exhibit extreme concentration (frequently allocating over 85% of capital to a single asset) and explosive trading volumes. Across all predictive models, average quarterly turnover in the unconstrained setup ranged from 60.2% (OLS-Lasso) to 116.9% (OLS-Base), and in a real trading environment, transaction costs would quickly erode any gains produced by such frequent reallocation.

- **Realistic Setup (Constrained):** As anticipated, a constrained setup was implemented to bring the strategy closer to real-world conditions. First of all, each asset class is capped at a maximum weight of 60%, ensuring basic diversification and preventing extreme concentration. More importantly, is implemented an *adaptive hard turnover constraint* that limits how much the portfolio can change at each quarterly rebalancing. Under normal conditions, the absolute shift in weights is capped at 20% per quarter, but when the rolling 3-month volatility signals an extreme market regime, the turnover cap automatically relaxes to 40%. Such episodes occurred in 5.6% of the sample periods, roughly 8 rebalancing dates, coinciding with major crises such as 2008 and 2020. This prevents the portfolio from remaining stuck in suboptimal allocations at precisely the time when it is essential to rebalance quickly. Overall, these constraints bring mean

[15]The condition number of a matrix Σ is defined as $\text{cond}(\Sigma) = \lambda_{\max}/\lambda_{\min}$, where $\lambda_{\max}$ and $\lambda_{\min}$ are its largest and smallest eigenvalues. A high number means the matrix is unstable and creates extreme, unreliable portfolio weights. Ledoit-Wolf shrinkage solves this by moving extreme eigenvalues closer to the mean. Our diagnostics confirm high stability across all periods: the condition number peaked at only 12.9 (during 2014), and the minimum eigenvalues remained perfectly safe even during stress phases like 2018.

quarterly turnover down to 18%–21% across all models, making the strategies practical and low-cost to implement.

The two optimization setups, Long-Only (LO) and Realistic (RE), provide a controlled framework to separately assess the raw predictive power of each model and its practical deployability under real-world frictions. The following chapter evaluates the out-of-sample performance of each model, assessing their ability to generate risk-adjusted returns across the full 1989–2024 evaluation period.

Chapter 5

Empirical Results and Performance Analysis

This chapter presents the empirical findings of the forecasting models, evaluating their effectiveness using a twofold approach. First, the models are assessed on their predictive accuracy using statistical metrics to forecast asset returns out-of-sample. Subsequently, the analysis shifts to the economic value when integrated into a MVO framework, assessing their practical viability and transaction costs in real-world portfolio management.

5.1 Predictive Power and Forecast Diagnostics

5.1.1 Out-of-Sample R^2 and Directional Accuracy

The statistical predictive accuracy of the four models is evaluated over the out-of-sample period (May 1989 – November 2024, $T = 143$) through two complementary metrics: the R^2_{OOS} statistic and the directional accuracy (hit rate), which evaluates sign prediction.

Statistical Predictive Performance: $\mathbf{R^2_{OOS}}$ results. The results, reported in *Table 5.1* and visualized in *Figure 5.1*, reveal three distinct patterns:

- **OLS overfitting and the role of regularization.** The unpenalized OLS-Base collapses to a mean $R^2_{OOS} = -87.34\%$ despite being restricted to 37 GWZ predictors, well below the 120-observation training window, with particularly extreme values for S&P 500 (-151.09%) and REITs (-187.09%). As documented in *Section 4.1.2*, the collinearity of macroeconomic indicators is sufficient to destabilize the coefficient estimates even

Table 5.1: Out-of-Sample R^2 (%) by Model and Asset Class

Asset	**OLS-Base**	**OLS-Lasso**	**ENet**	**NN-3**
S&P 500	−151.09	−1.30	**+0.46**	**+1.47**
Commodities	−48.20	**+0.39**	**+0.17**	**+0.52**
Gold	−15.72	−3.02	**+0.43**	−2.57
REITs	−187.09	−1.42	**+13.04**	**+6.00**
Treasury 7–10y	−34.61	−3.36	−0.38	−0.23
Mean	−87.34	−1.74	**+2.75**	+1.04

Note: $R^2_{OOS} > 0$ indicates the model beats the historical mean benchmark (Eq. (2.4)). Positive values in bold. OLS-Base: 120m rolling window, 37 predictors, OLS-Lasso: 180m (120+60) rolling window, 57 pred.; ENet (rolling), NN-3 (expanding): 180m (120+60) window, 60 pred. $T = 143$ (May 1989 – Nov. 2024).
Source: Author's calculations.

in this restricted setting, a failure fully anticipated by Goyal and Welch (2008). When the Baseline model is extended to the entire feature space, comprising 57 predictors, the performance deteriorates significantly; in fact, the average R^2_{OOS} drops to −289.9%: without regularization, the additional features become counterproductive. OLS-Lasso resolves this issue by utilising the L_1 penalty to operate safely on the full set of features, as explained in detail in *Section 4.1.3*: the optimised penalty aggressively shrinks the active predictor set to a small fraction of the available variables per window, neutralising multicollinearity noise before it reaches the optimiser. The result is a significant improvement to an average R^2_{OOS} of −1.74%, an improvement of over 85 percentage points compared to OLS-Base. However, even this regularised linear model fails to achieve a positive aggregate; in fact, we see that only Commodities reach +0.39%. This suggests that the limitation lies not only in overfitting but in the linear architecture itself. It should also be noted that a positive R^2_{OOS} is not always synonymous with economically useful forecasts: we will show that several models produce forecasts that are systematically biased or of unrealistic magnitude, thus making them useless in practice despite excellent predictive power on paper.

- **The ML advantage.** Both Machine Learning models achieve a positive aggregate R^2_{OOS}: ENet at +2.75% and NN-3 at +1.04%, marking a qualitative improvement relative to all linear benchmarks. However, ENet's aggregate is largely driven by REITs (+13.04%); excluding that asset class, its mean R^2_{OOS} would fall to approximately +0.17%, barely above the zero threshold. By contrast, NN-3 delivers a more balanced distribution: it achieves positive values for three out of five assets (S&P 500, Commodities, and REITs), reflecting its expanding window architecture: unlike ENet's strict 180-month rolling window, the neural network accumulates the full available history, reducing estimator

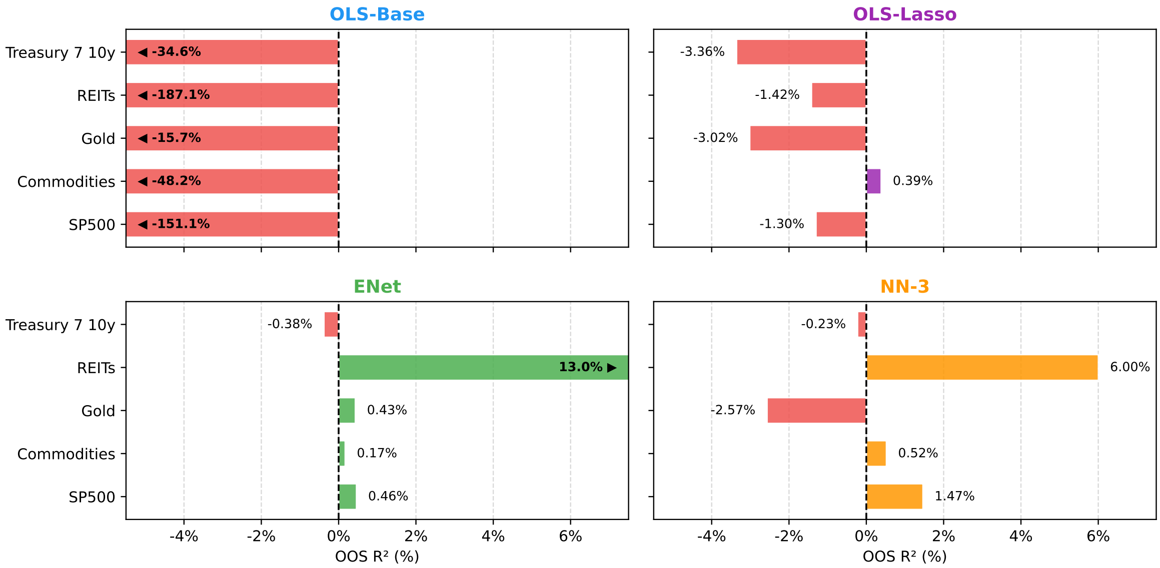

Figure 5.1: Out-of-Sample R^2 (%) by model and asset class (arrows indicate truncated values). OOS period: May 1989–Nov 2024 ($T = 143$). *Source:* Author's elaboration.

variance as the OOS period grows, an advantage documented by Gu et al. (2020) in low signal-to-noise environments. However, as the robustness analysis in *Section 5.3* will show, ML models trained on even longer windows do indeed achieve higher values of R^2_{OOS}; nevertheless, these marginal improvements in statistical accuracy rarely translate into better portfolio performance in terms of the Sharpe ratio or drawdown.

- **Asset-specific patterns.** The cross-asset dispersion in R^2_{OOS} is large, and no model dominates uniformly across the full universe. REITs are the most predictable asset class under both ML models (+13.04% for ENet, +6.00% for NN-3), consistent with Plazzi et al. (2010), who document that real estate expected returns co-move with macroeconomic variables such as those in our GWZ set (term spread, default spread, and short rate), confirming that these signals carry genuine information about real estate discount rates. The S&P 500 ranks second under NN-3 (+1.47%), as expected given the GWZ predictors were originally constructed for U.S. equity premia (Goyal and Welch, 2008; Goyal et al., 2023). Gold and Treasuries remain largely unpredictable: three out of four models yield negative R^2_{OOS} for Gold, reflecting its asymmetric and regime-dependent dynamics, while Treasury values hover near zero for ML models (−0.38%, −0.23%). Notably, no model dominates uniformly: ENet leads on REITs (+13.04% vs. +6.00% for NN-3), while NN-3 outperforms on Commodities (+0.52% vs. +0.17%) and Treasuries (−0.23% vs. −0.38%). This non-dominated structure confirms that return predictability is weak and asset-specific (Goyal et al., 2023).

Directional Accuracy. While the R^2_{OOS} measures the accuracy of return magnitude, it does not directly assess whether a model correctly predicts the *direction* of returns, which is without doubt a relevant dimension for a tactical asset allocator who must decide whether to overweight or underweight each asset class each period. To evaluate this, *Table 5.2* reports the hit rate for each model and asset class, tested against the null hypothesis of no directional predictability using the non-parametric test of Pesaran and Timmermann (1992). The test checks whether the share of correctly predicted return signs is higher than what pure chance would produce. Under the null hypothesis of no predictability, the standardized test statistic follows an asymptotic $\mathcal{N}(0, 1)$ distribution[1].

Table 5.2: Directional Accuracy (Hit Rate) by Model and Asset Class

Asset	**OLS-Base**	**OLS-Lasso**	**ENet**	**NN-3**
S&P 500	52.4%	62.9%***	64.3%***	65.0%***
Commodities	53.8%	51.0%	50.3%	50.3%
Gold	53.8%	51.7%	52.4%	53.1%
REITs	58.0%**	58.7%**	58.7%**	60.8%***
Treasury 7–10y	54.5%	52.4%	53.1%	54.5%

Note: One-sided binomial test, H_1: hit rate > 50% (Pesaran and Timmermann, 1992). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. $T = 143$. *Source:* Author's calculations.

Directional accuracy results align well with the R^2_{OOS} findings. For the S&P 500, OLS-Lasso, ENet, and NN-3 reach hit rates between 62.9% and 65.0%, all significant at the 1% level. In other words, these models correctly predict the sign of the equity premium in roughly two out of three months, a concrete edge for tactical asset allocation. REITs equally show statistically significant directional accuracy across all four models (58.0%–60.8%), consistent with the high R^2_{OOS} values documented above. For the remaining three asset classes, Commodities, Gold, and Treasuries, no model achieves a statistically significant hit rate, confirming that any marginal predictive signal is insufficient to identify the direction of returns. Overall, the hit rate analysis supports a cautiously positive reading of the ML results. Genuine directional predictability is concentrated in the two most cyclically sensitive assets (equities and REITs) where macroeconomic signals are known to track risk premium fluctuations. Conversely, safe-haven assets like Gold and Treasuries remain unpredictable because they are less tied to the economic cycle.

[1]Formally, the test statistic is $S_n = (P - P^*)/\sqrt{\text{Var}(P) - \text{Var}(P^*)}$, where P is the observed hit rate and $P^* = \hat{P}_y \hat{P}_x + (1 - \hat{P}_y)(1 - \hat{P}_x)$ is the expected hit rate under independence, with $\hat{P}_y$ and $\hat{P}_x$ the sample proportions of positive realizations and positive forecasts, respectively. Under H_0, $\sqrt{n}\, S_n \xrightarrow{d} \mathcal{N}(0, 1)$ (Pesaran and Timmermann, 1992, p. 462). In our setting, since the historical mean benchmark generates forecasts with no sign bias ($\hat{P}_x \approx 0.5$), the test reduces to a one-sided binomial test under H_0: HR = 0.5.

5.1.2 Forecast Error Analysis

The R^2_{OOS} metric provides an aggregate summary of predictive accuracy but does not reveal whether forecasts are well-calibrated or systematically biased. This section examines the structure of forecast errors to assess the practical quality of the predictions.

Prediction Calibration Across Models. Plotting realized against predicted returns for each model, by grouping all the observations for each month by concatenating the forecasts for the five asset classes over the entire out-of-sample period, *Figure 5.2* reveals a marked divergence in calibration quality. The ideal predictor lies along the 45° line (slope = 1). OLS-Base produces very weak forecasts (slope = 0.10), consistent with a model that generates almost constant forecasts that match the sign of actual returns purely by chance (Pearson's correlation coefficient $r = 0.11$). OLS-Lasso improves marginally (slope = 0.25), but predictions remain compressed around zero, confirming that L_1 regularization alone does not resolve the calibration problem. ENet (slope = 0.80) and NN-3 (slope = 1.04) are substantially better calibrated, with Pearson correlations of 0.19 and 0.13, respectively. Among all models, NN-3 achieves the best-calibrated forecasts in terms of slope, while ENet records the highest Pearson correlation, reflecting more consistent directional accuracy.

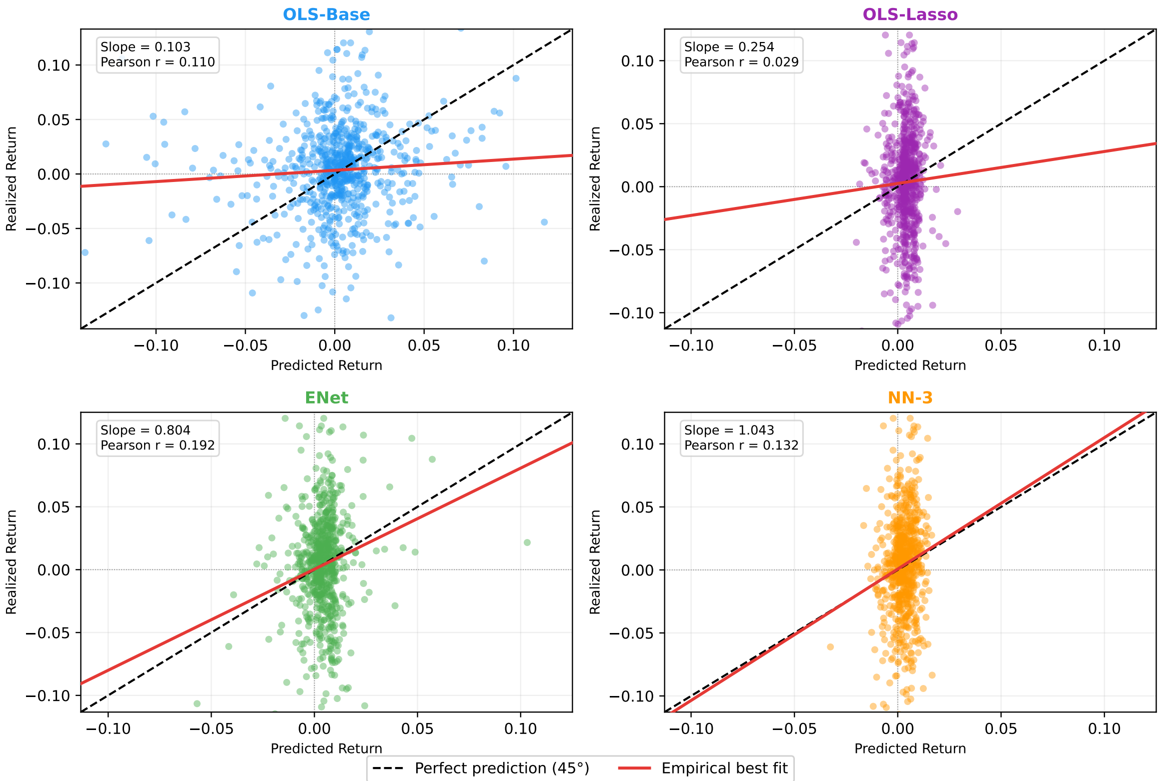

Figure 5.2: Predicted vs. realized returns, all assets pooled. Red line = empirical best fit; dashed = 45° benchmark. OOS Period: May 1989–Nov 2024. *Source:* Author's elaboration.

This calibration advantage is confirmed in the time-series dimension for S&P 500 and REITs, the two assets with positive R^2_{OOS} under ML models. In *Figure 5.3*, a consistent pattern emerges: while OLS-Base generates predictions of comparable scale to realized returns, it does so at the cost of erratic timing and frequent sign errors. Conversely, regularized models (OLS-Lasso, ENet, and NN-3) produce forecasts that are plausible in terms of direction but understated in magnitude. This happens because returns have a large unpredictable component that these methods cannot capture. Since the models only extract the predictable signal, their forecasts cover a much narrower range than realized returns. This explains why these methods are more successful at predicting the direction of returns rather than their exact magnitude: a model can capture the correct sign even when it cannot match the full scale of the movement.

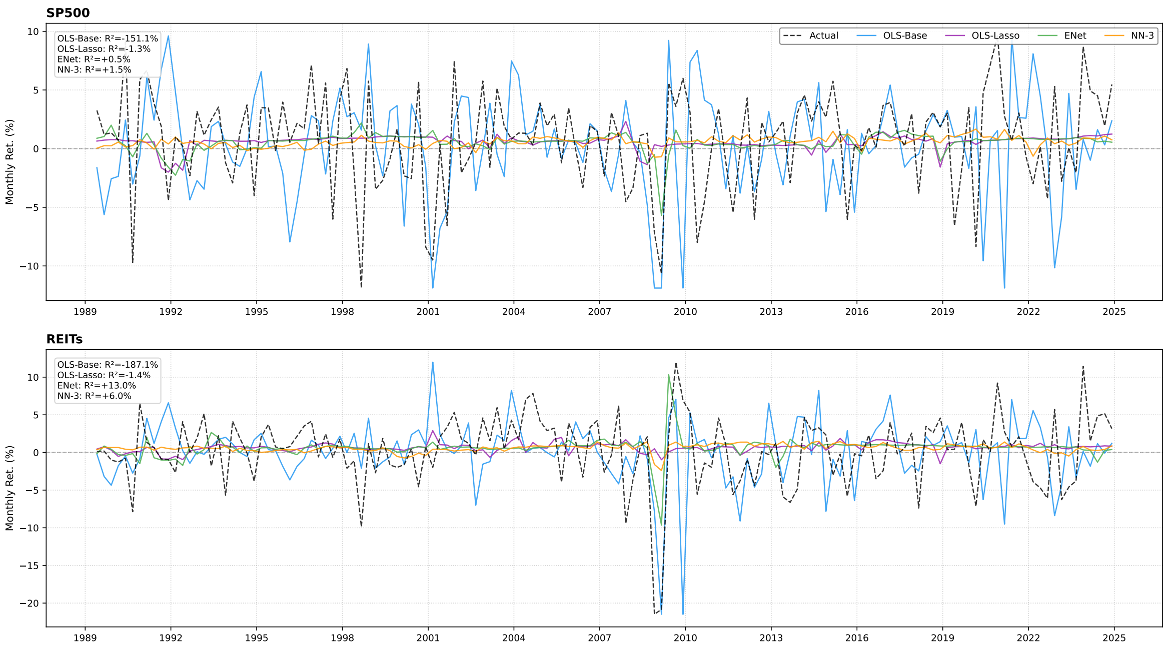

Figure 5.3: Predicted vs. realized monthly excess returns (%) for S&P 500 and REITs. OOS Period: May 1989–Nov 2024. All Assets in *Appendix D.1.1*. *Source:* Author's elaboration.

Systematic Errors by Asset Class. The forecast error distribution shown in *Figure 5.4* reveals two evident patterns across models and asset classes. The first one is that S&P 500 is underpredicted by every model: OLS-Base shows the largest bias ($\mu = -0.80\%$), while regularized and ML models keep it in the range -0.16% to -0.22%. The second one is that REITs go the other way, being systematically overestimated by all penalized and ML models ($\mu \approx +0.39\%$ to $+0.46\%$), whereas OLS-Base is nearly unbiased ($\mu = -0.02\%$). This pattern suggests that regularization introduces a tendency to overestimate REITs risk premia.

A milder but consistent overestimation also appears for Commodities across all models (μ between $+0.13\%$ and $+0.26\%$). Gold shows the most pronounced negative asymmetry under NN-3 ($\mu = -0.53\%$), consistent with its regime-dependent dynamics discussed earlier; OLS-Lasso exhibits a similar negative bias ($\mu = -0.36\%$), second in magnitude among all models. Treasury errors are small in magnitude across all models, though NN-3 exhibits a mild negative bias ($\mu = -0.24\%$).

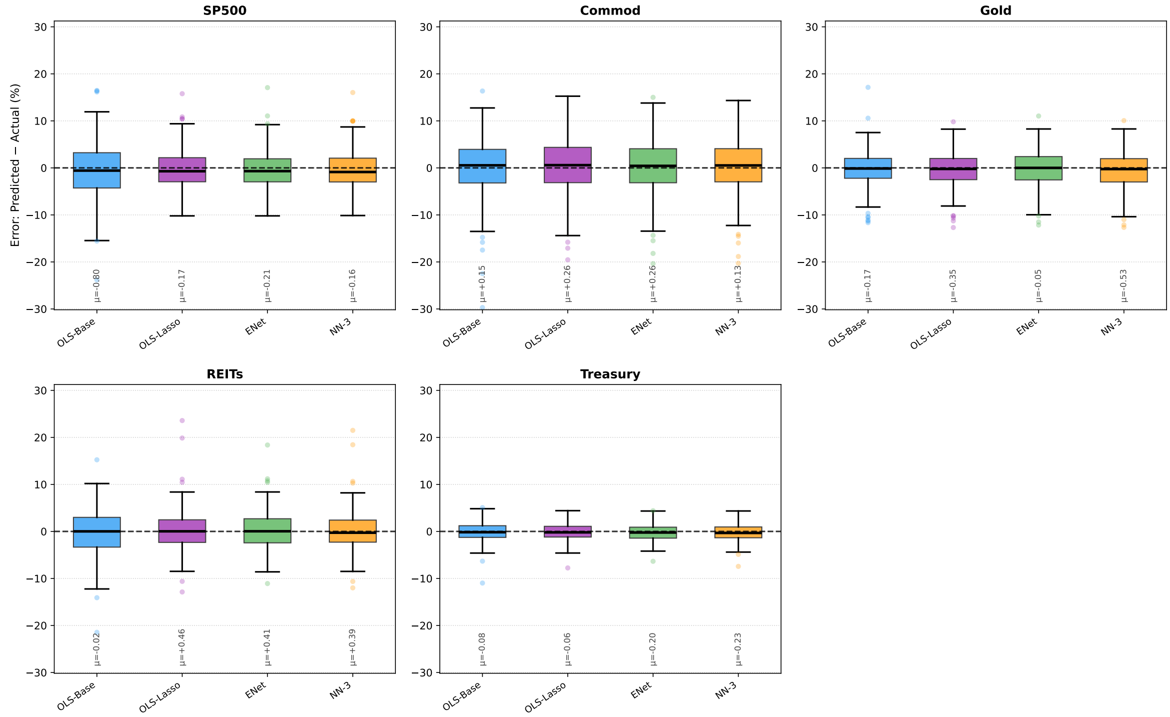

Figure 5.4: Forecast error distribution (Predicted − Actual, %) by asset and model. Dashed line = zero bias; μ = sample mean. OOS Period: May 1989–Nov 2024. Tabulated mean, median, and dispersion statistics in *Appendix D.1.3*. *Source:* Author's elaboration.

Taken together, these results reinforce the conclusion drawn from the R^2_{OOS} analysis: ML models produce forecasts that are better calibrated and less biased than linear alternatives, but the underlying predictive signal remains weak, in line with the theoretical prior that return predictability in financial markets is fragile and episodic (Goyal and Welch, 2008).

5.1.3 Statistical Significance: Diebold–Mariano Test

A positive R^2_{OOS} does not in itself guarantee that a model's superiority over the historical mean is statistically reliable rather than a product of sampling variation. To assess this, *Table 5.3* reports the Diebold–Mariano (DM) test statistic for each model and asset class against the historical mean benchmark[2].

[2]The DM test evaluates whether the model's MSE is significantly lower than the benchmark's via a one-sided HAC-robust t-ratio (Diebold and Mariano, 1995). Note that standard DM statistics do not explicitly account for parameter estimation error (West, 1996) or nested model comparisons (Clark and McCracken, 2001). Given the intractability of finite-sample corrections for ML architectures, we report unadjusted statistics following standard practice (Gu et al., 2020). See *Appendix D.2* for a formal econometric discussion.

Table 5.3: Diebold–Mariano Test Statistics: Each Model vs. Historical Mean Benchmark

Asset	OLS-Base	OLS-Lasso	ENet	NN-3
S&P 500	−2.588	−0.842	+0.094	+0.603
Commodities	−1.355	+0.260	+0.048	+0.259
Gold	−1.208	−1.312	+0.105	−0.959
REITs	−1.539	−0.555	+0.840	+0.964
Treasury 7–10y	−1.452	−2.576	−0.079	−0.098

Note: One-sided DM test, H_1: model MSE < historical mean MSE. HAC Newey–West correction ($\ell = 3$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Positive values indicate the model outperforms the benchmark. No entry reaches conventional significance thresholds. *Source:* Author's calculations.

The results are clear: no model achieves statistical significance against the historical mean for any asset class. While ENet and NN-3 post positive DM statistics for S&P 500 (+0.09, +0.60) and REITs (+0.84, +0.96), consistent with their positive R^2_{OOS} values, the statistics are well below the conventional significance thresholds. This is not surprising given the well-documented difficulty of beating the historical mean out-of-sample (Goyal and Welch, 2008), and reflects the low signal-to-noise ratio inherent in quarterly return prediction at the asset class level. Overall, the evidence from ML models confirms that the predictive benefits of machine learning models are real, but modest in scale. Diebold–Mariano pairwise comparisons confirm that OLS-Base is outperformed by all models ($p < 0.01$ for S&P 500; $p < 0.10$ for REITs), while OLS-Lasso and ML models remain statistically indistinguishable[3].

5.1.4 Feature Importance and Economic Drivers

The positive out-of-sample R^2 values documented in this Section establish that the models' forecasts contain genuine predictive content beyond the historical mean; the analysis below identifies *which* predictors are responsible for that content and whether their selection is economically interpretable. Because the four models rely on fundamentally different statistical architectures, a universal importance measure is not applicable. For the linear models, importance reflects the magnitude of estimated coefficients or their implied t-statistics, quantities directly interpretable as linear predictive weights. For the neural network, we instead use a gradient-based saliency score, which captures the local sensitivity of the forecast to each input and is therefore not comparable in scale to coefficient-based metrics.[4] Figure 5.5 reports the top-15 predictors for each model.

[3] Pairwise DM test results for S&P 500 and REITs are reported in *Tables A.6* and *A.7* in *Appendix D.2.2*.

[4] *OLS-Base*: mean absolute t-statistic of the predictor's loading on the Goyal–Welch factors, $|\bar{t_k}|$, across the OOS window. *OLS-Lasso*: mean absolute implied coefficient $|B \cdot C|$, where B maps factors to returns and C maps standardised predictors to factors via penalised Lasso. *Elastic Net*: mean absolute fitted coefficient $|\hat{\beta}_k|$ averaged over OOS windows. *NN-3*: gradient × input saliency $\overline{|\partial \hat{y}_t / \partial z_{k,t} \cdot z_{k,t}|}$, following Gu et al. (2020).

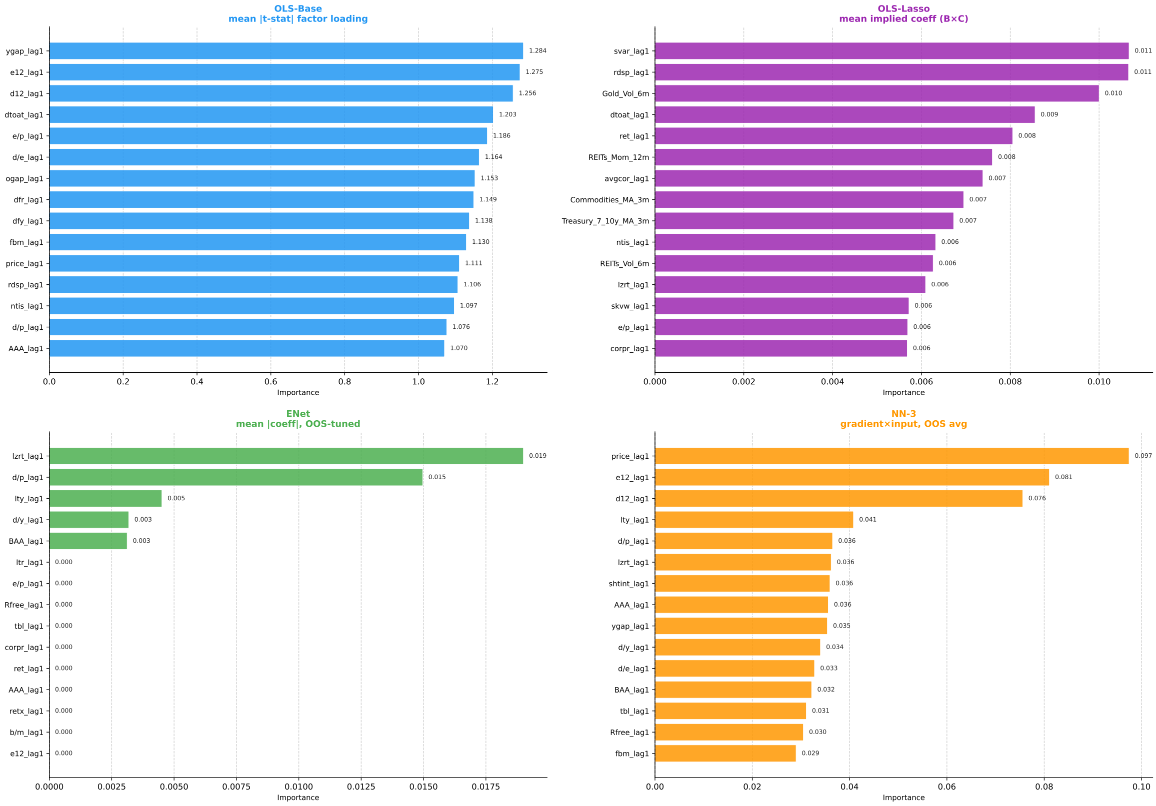

Figure 5.5: Out-of-sample predictor importance: top 15 predictors per model. Heterogeneous x-axis scales across panels; see *footnote 4* for definitions. *Source:* Author's elaboration

Clear differences emerge across model architectures in which predictors each model relies.

OLS-Base draws only from the GWZ set with valuation ratios (`e12`, `d12`, `d/p`, `e/p`), the stock-bond yield gap (`ygap`), and interest rate spreads (`lty`, `lzrt`) dominating, in line with their role as slow-moving equilibrium (Goyal and Welch, 2008; Campbell and Shiller, 1988).

OLS-Lasso is the only model that incorporates custom technical features with macroeconomic predictors. In particular, it include cross-asset momentum and volatility signals (`Gold_Vol_6m`, `REITs_Mom_12m`, `Commod_MA_3m`) together with GWZ indicators (`svar`, `rdsp`). This demonstrates how LASSO selects predictive variables based only on their marginal contribution to the accuracy of the forecast, without giving preference to fundamental or technical variables (Neely et al., 2014).

Elastic Net enforces the strongest parsimony: only 5 of the 57 candidate predictors receive non-zero weight in aggregate (`lzrt`, `d/p`, `lty`, `d/y`, `BAA`), all from the GWZ family and all related to interest-rate levels or valuation spreads. We observed that for several asset classes, particularly Gold and Commodities, whose excess returns are less tightly linked to equity-market fundamentals, the model selects *zero* active predictors, as evidence of the weak signal from our predictors for these assets.

NN-3 relies primarily on GWZ valuation ratios (`price`, `e12`, `d12`), but uses them more flexibly and spreads importance more widely across the predictor set. For some asset classes, no single predictor stands out as clearly important, suggesting that the network builds its forecasts by combining weak signals from many inputs at once rather than leaning on a few dominant variables.

We note that the Fama-French factors (*Mkt-RF*, *SMB*, *HML*) do not appear in any of the top 15 positions; we recall that in the OLS-Base model they are treated as first-stage factors rather than predictors, whereas in the ML models their lagged values are used directly as predictors.

5.2 Economic Value and Portfolio Performance

5.2.1 Portfolio Performance: Long-Only vs. Realistic Setup

The performance results achieved by the MVO portfolios are summarized in *Table 5.4*, which reports the full set of out-of-sample (OOS) performance metrics for the *Long-Only* (unconstrained) and *Realistic* (constrained) setups, respectively. *Figures 5.6* and 5.7 display the corresponding cumulative wealth indices (NAV), offering a visual picture of how each strategy evolves over time.

Table 5.4: Out-of-Sample Portfolio Performance Metrics – Long-Only and Realistic Setups.

Strategy	**Ret**	**Vol**	**Sharpe**	**Sortino**	**MDD**	**VaR$_{5\%}$**	**CVaR$_{5\%}$**	**Turn**
Panel A: Long-Only (Unconstrained)								
Hist-Avg	7.54	13.40	0.562	0.733	−37.46	−5.78	−8.14	107.82
OLS-Base	9.59	13.10	0.732	1.096	−23.59	−5.00	−7.48	467.55
OLS-Lasso	6.67	11.33	0.588	0.782	−30.11	−5.27	−7.02	240.98
ENet	10.80	13.57	0.796	1.204	−25.32	−4.43	−7.31	333.91
NN-3	5.96	13.16	0.453	0.504	−52.75	−4.01	−8.30	333.41
Panel B: Realistic (Constrained)								
Hist-Avg	6.83	11.81	0.579	0.710	−37.37	−4.57	−7.33	55.19
OLS-Base	4.43	10.23	0.433	0.565	−32.49	−4.14	−6.56	83.92
OLS-Lasso	4.85	11.69	0.415	0.463	−57.45	−4.72	−8.11	73.26
ENet	6.25	11.44	0.546	0.668	−27.14	−4.50	−7.31	76.96
NN-3	5.01	12.10	0.414	0.463	−54.63	−4.73	−8.41	79.75
Panel C: Benchmarks								
1/N	5.08	9.04	0.562	0.682	−36.44	−3.24	−5.71	0.00
Buy&Hold	6.73	12.71	0.530	0.614	−51.60	−5.17	−8.59	0.00
60/40	6.34	9.19	0.690	0.969	−29.70	−3.87	−5.46	0.00

Note: Returns, volatility and Turnover are annualised (%); MDD, VaR, and CVaR are expressed as percentages.
Source: Author's calculations.

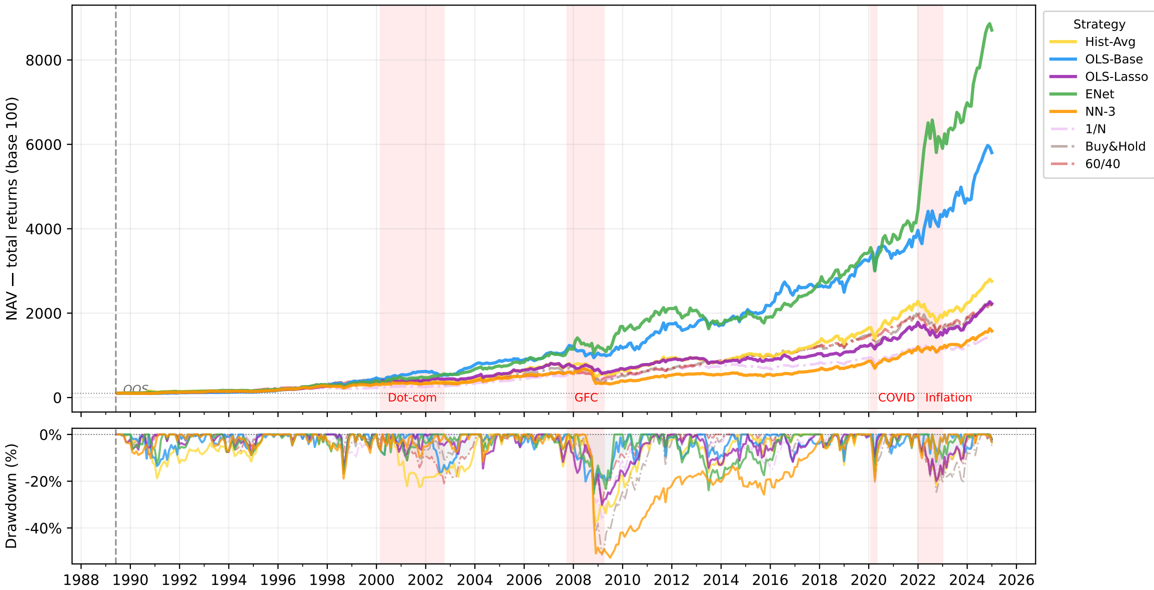

Figure 5.6: Cumulative NAV (base = 100, top panel) and drawdown (bottom panel) for the Long-Only setup. OOS period: May 1989 – Nov 2024. Models: Hist-Avg (gold), OLS-Base (blue), OLS-Lasso (purple), ENet (green), NN-3 (orange). Benchmarks: $1/N$ (light pink), Buy&Hold (gray dotted), 60/40 (pink dash-dot). Shaded areas mark Dot-com (2000–2002), GFC (2007–2009), COVID (2020), and Inflation shock (2022). *Source:* Author's elaboration.

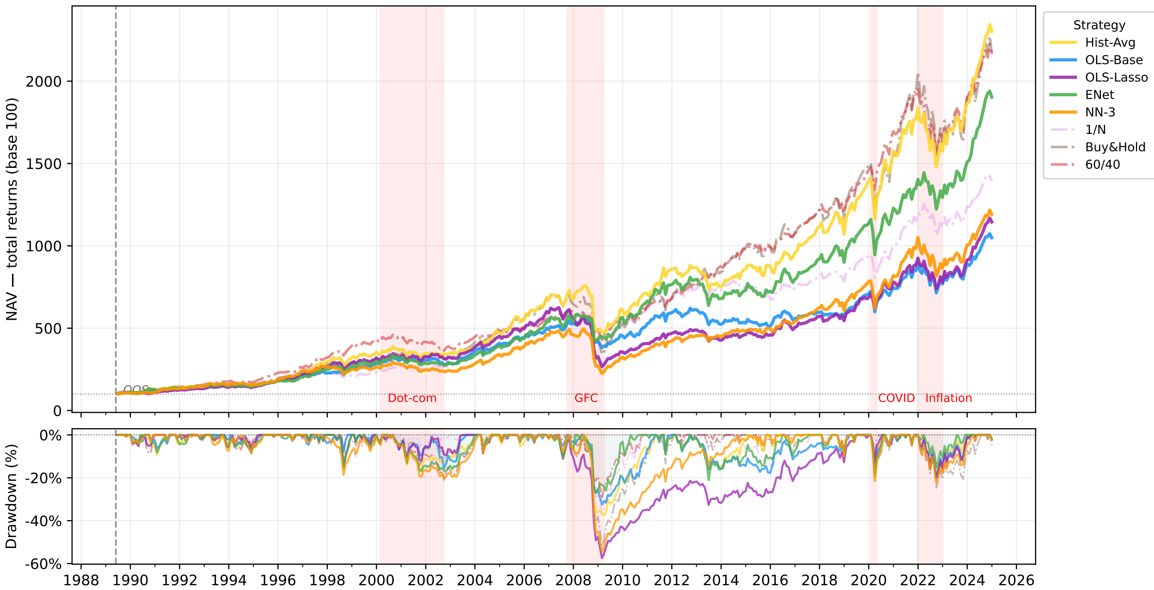

Figure 5.7: Cumulative NAV (base = 100, top panel) and drawdown (bottom panel) for the Realistic setup. Notation as in *Figure 5.6*. *Source:* Author's elaboration.

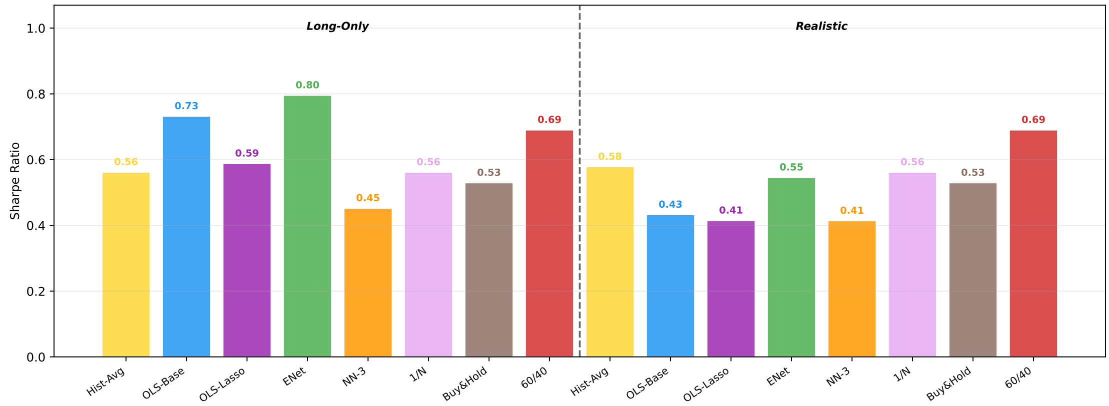

Figure 5.8: Annualised Sharpe Ratio (excess returns) for all strategies across the Long-Only and Realistic setups. *Source:* Author's elaboration.

Strategy performance: Long-Only setup. *Figure 5.8* summarises the Sharpe ratios across all strategies and setups. In the Long-Only setting, ENet is the top performer both in terms of returns (10.80% annualised) and risk-adjusted performance (SR = 0.796), as confirmed by the NAV trajectory in *Figure 5.6*, which shows a clear separation from the other strategies over the full out-of-sample period. However, the Sharpe gap over OLS-Base is modest (0.796 vs 0.732), suggesting that while ENet generates substantially higher gross returns, it does not achieve proportionally lower volatility. OLS-Lasso and Hist-Avg cluster around SR $\approx$ 0.56–0.59, with returns of 6.67% and 7.54% respectively. NN-3 lags all models: its SR = 0.453 and annualised return of 5.96% are accompanied by the worst tail risk profile in the unconstrained setup, with a CVaR of −8.30% and a maximum drawdown of −52.75%, both reflecting the sharp NAV collapses visible during crisis periods in *Figure 5.6*.

Strategy performance: Realistic setup and benchmark comparison. Once turnover constraints are applied, risk-adjusted performance deteriorates across all active strategies (*Figure 5.7*). ENet remains the best-performing model (SR = 0.546, annualised return 6.25%), but the performance gap relative to OLS-Lasso and NN-3 narrows considerably, with both converging to SR $\approx$ 0.41. Notably, NN-3 experiences the smallest Sharpe deterioration across configurations (from 0.453 to 0.414, a drop of only 0.039 points), yet its tail risk remains elevated: its CVaR of −8.41% and MDD of −54.63% are the worst among all models in the Realistic setup, confirming that the NAV path of NN-3 is characterised by severe drawdowns concentrated in stress periods. The passive benchmarks provide a helpful reference: the 60/40 portfolio records SR = 0.690 with a CVaR of only −5.46% and a VaR of −3.87%, substantially better tail protection than any active strategy. In the Realistic setup, no ML model surpasses the 60/40 rule on a Sharpe basis, and even the 1/N portfolio (SR = 0.562) outperforms

OLS-Base, OLS-Lasso, and NN-3 and ENet. The primary driver of this deterioration is gross portfolio turnover, which is analysed in detail in the following paragraph.

The role of turnover. The performance degradation from Long-Only to Realistic is directly proportional to gross portfolio turnover. *Table 5.5* reports annualised turnover for the active strategies in the unconstrained setup.

Table 5.5: Annualised Gross Turnover (%) – Long-Only and Realistic Setups.

Strategy	**Long-Only (%)**	**Realistic (%)**
OLS-Base	467.55	83.92
OLS-Lasso	240.98	73.26
ENet	333.91	76.96
NN-3	333.41	79.75

Note: The Realistic column reports the residual turnover after the quarterly-rebalancing and max-turnover constraints are applied. *Source:* Author's calculations.

OLS-Base produces the highest unconstrained turnover by a wide margin (467% annualised), which makes its Long-Only returns essentially unreachable in practice once transaction costs are applied. ENet and NN-3 share very similar gross turnovers ($\approx$ 333%), yet ENet delivers a far more resilient portfolio in the constrained setting; this difference is due to the stabilising effect of L_1/L_2 regularization on return forecasts, which reduces abrupt changes in portfolio weights from one period to the next. OLS-Lasso has the lowest turnover among the ML models (241%), consistent with its sparse forecast structure, but the instability of its predictions still generates significant performance drag once realistic trading frictions are introduced. A deeper analysis of turnover dynamics and portfolio weight evolution is provided in *Section 5.2.2*.

5.2.2 Portfolio Weight Dynamics and Allocation Patterns

Examining how the MVO optimizer allocates capital across asset classes is informative: it reveals whether the model concentrates risk in a few assets or achieves genuine diversification, and whether the resulting allocations are economically interpretable. *Figure 5.9* shows the quarterly evolution of portfolio weights for each model in the Long-Only (left) and Realistic (right) setups. The contrast between the two columns is immediate: unconstrained portfolios display extreme weight swings, with allocations jumping from near-zero to near-100% within a single quarter, while the Realistic constraints produce smoother and more stable weight paths. This instability must not be overlooked: rebalancing 300–400% of portfolio value annually generates transaction costs large enough to eliminate the gross excess return entirely, even under conservative cost assumptions[5]. An explicit simulation is beyond the scope of this thesis, but the evidence from the Realistic setup, where a 20% quarterly turnover cap already depresses Sharpe ratios by 0.2–0.3 points, strongly suggests that a full cost model would place all ML strategies well below the 60/40 benchmark on a net basis.

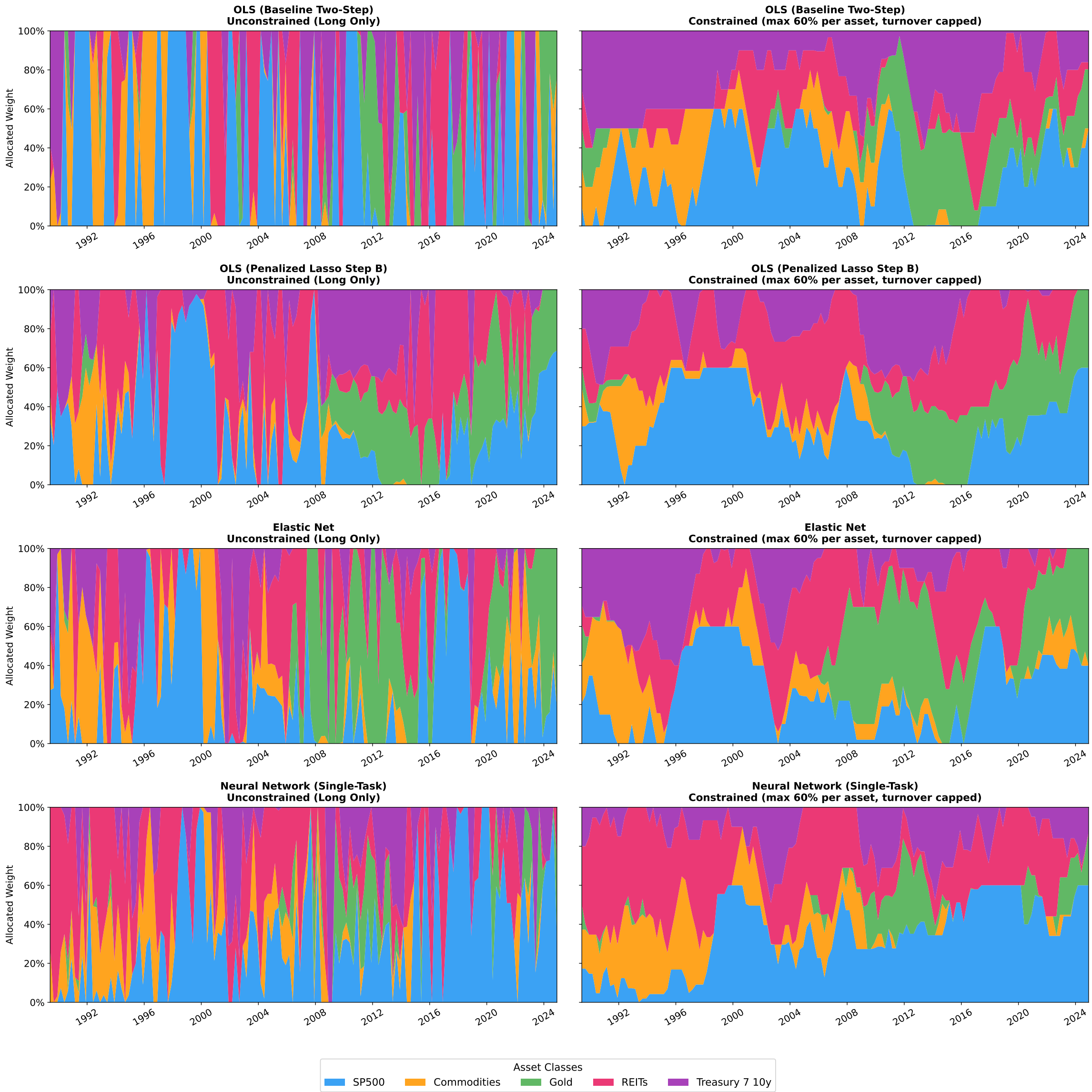

Figure 5.9: Quarterly asset allocation weights evolution – Long-Only (left) and Realistic (right) setup for each model. OOS Period: May 1989 – Nov 2024. *Source:* Author's elaboration.

Constrained weight dynamics and Mean allocation analysis. The Realistic weight plots in *Figure 5.9* reveal both common patterns and model-specific dynamics. In the early 1990s, OLS-Base starts with an extremely high Treasury allocation (purple near 80–100%), which gradually gives way to equities and Gold through the late 1990s. Around 2000, OLS-Lasso

[5] At $c = 0.10\%$ per transaction (institutional one-way cost), a 400% annual turnover implies a drag of roughly 80 bps per year, alone exceeding the excess return of OLS-Lasso in the Long-Only setup.

undergoes a sharp shift: the large SP500 block that dominated the late 1990s collapses and is replaced by a growing REITs component that persists for most of the subsequent period. The 2007–2009 GFC is the only episode where all four models move in the same direction, with Treasury weights expanding across the board. After 2012, the models diverge: OLS-Base progressively rebuilds its Treasury position, while ENet and NN-3 maintain more stable multi-asset allocations. The 2020 shock produces no visible reallocation in any model, as quarterly rebalancing means the optimizer only observes the drawdown *after* the trough, an inherent limitation that will be discussed in *Section 5.2.3*.

Table 5.6 shows the average portfolio weights over time. ENet builds the most balanced portfolio, allocating 20% in Gold, 27% in the S&P 500, and 23–26% in REITs. This wide diversification is likely a direct consequence of the model's regularization, which prevents extreme positions and distributes weights across uncorrelated assets. Meanwhile, OLS-Base holds nearly 29% in Treasury bonds as part of the Realistic strategy, and this conservative allocation protects against large declines but has a negative impact on returns when interest rates rise, as was seen in 2022. OLS-Lasso puts more than 30% of its assets into REITs in both configurations, and this excessive concentration makes the portfolio too exposed to the highly volatile property market. Finally, NN-3 generates a realistic, growth-oriented portfolio that takes into account traditional market risk premiums, allocating 65–67% to the S&P 500 and REITs, whilst keeping gold below 7%.

Table 5.6: Time-averaged portfolio weights (%) – Long-Only vs. Realistic Setups.

	Long-Only					***Realistic***				
Model	**SP500**	**Comm.**	**Gold**	**REITs**	**Tsy**	**SP500**	**Comm.**	**Gold**	**REITs**	**Tsy**
OLS-Base	37.2	13.3	11.4	18.8	19.3	27.4	12.2	14.6	17.0	28.7
OLS-Lasso	30.2	5.7	14.3	32.0	17.6	31.5	6.4	14.9	30.4	16.8
ENet	27.3	15.3	20.6	22.8	14.0	27.0	11.0	20.0	25.5	16.5
NN-3	35.5	12.1	7.1	30.2	15.0	34.2	12.5	6.2	33.1	14.0

Source: Author's calculations.

MVO and the Ledoit-Wolf caveat. The covariance diagnostics confirm that the estimator works as expected throughout the sample: the condition number averages 6.6 and never exceeds 12.9. Notably, the worst-conditioned matrices appear in the low-volatility period of 2014–2015, not during market stress episodes, which confirms that the covariance matrix is not the primary source of portfolio instability. The root cause lies instead in the return forecasts themselves: their high variability from quarter to quarter pushes the optimizer toward extreme corner solutions, regardless of how well-conditioned the covariance matrix is. This is a structural limitation of the predict-then-optimise pipeline: a stable covariance estimate alone cannot produce stable weights when the expected-return inputs are noisy.

5.2.3 Tail Risk and Crisis Performance

Standard metrics such as volatility and the Sharpe ratio do not account for what is known in finance as tail risk. Maximum drawdown (MDD) quantifies this risk by measuring the largest decline a portfolio has recorded from its highest point to its lowest. This indicator shows significant differences among all models, as illustrated in *Figure 5.10*. In the Long-Only setup, ENet records the smallest MDD (−25.3%), while NN-3 suffers the deepest decline (−52.7%), comparable to the passive Buy&Hold (−51.6%) and far worse than the 60/40 benchmark (−29.7%). The pattern persists in the Realistic setup: ENet's MDD increase only to −27.1%, while NN-3 remains at −54.6% and OLS-Lasso deteriorates dramatically to −57.4%.

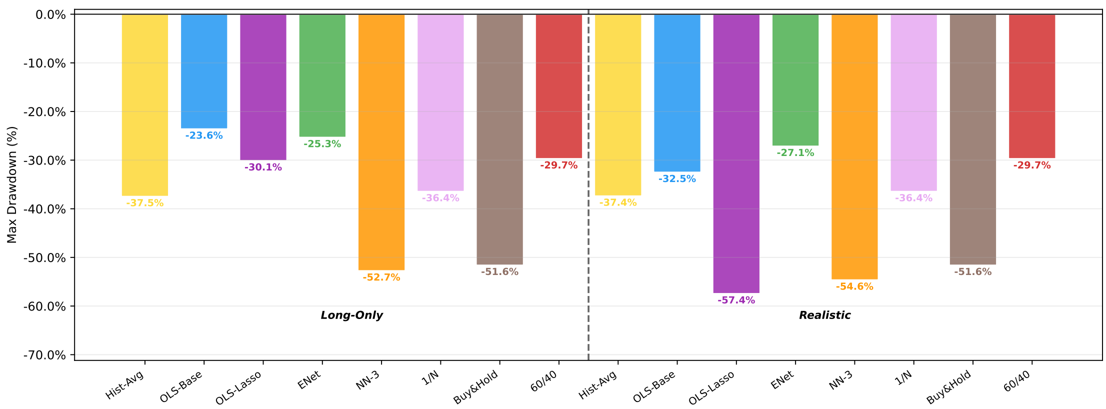

Figure 5.10: Maximum drawdown (%) – Long-Only (left) and Realistic (right) setup for all strategies and passive benchmarks. *Source:* Author's elaboration.

The differences in worst-case losses come directly from how each model allocates its capital. ENet's heavy reliance on Gold proved highly beneficial over time. During historical market crashes, this asset balanced the severe losses suffered by equities and REITs. Similarly, OLS-Base managed to limit its MDD thanks to its large safety cushion in Treasury bonds, although this cautious approach proved counterproductive during the interest rate hikes of 2022. On the other hand, even though NN-3 follows a highly realistic investment approach, its strong concentration in risky assets caused severe maximum drawdowns. Because stocks and REITs often fall together during panics, this lack of protection significantly damaged the model's overall performance. Finally, OLS-Lasso suffered a massive collapse (−57.4%) in the Realistic setup. This huge drop happened precisely because its heavy bet on the property market failed terribly during major crises.

Crisis sub-period analysis. To understand tail risk, it is important to analyse how each strategy performs during the major crises faced in the out-of-sample period. *Figure 5.11* and *Table 5.7* report the annualised Sharpe Ratio for the Long-Only setup during the four major market dislocations: the Dot-Com Bubble burst (Mar. 2000 – Oct. 2002), the Global Financial Crisis (Dec. 2007 – Jun. 2009), the Covid-19 crash (Feb. – Apr. 2020), and the 2022 rate-hike sell-off (Jan. – Dec. 2022).

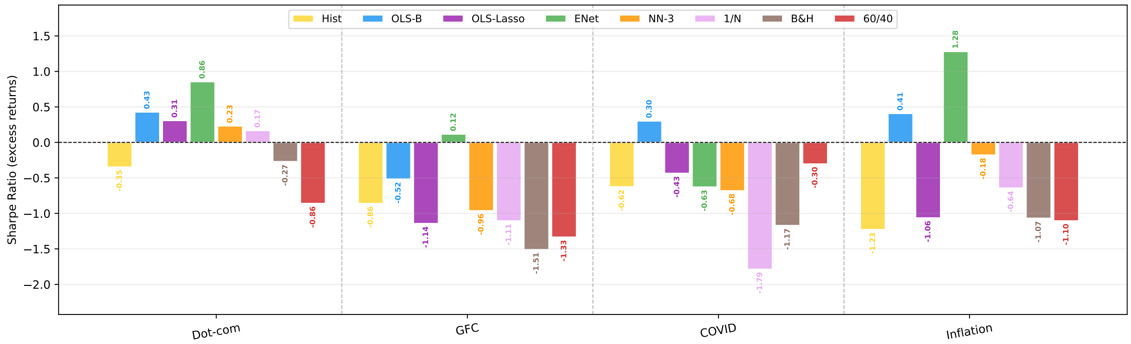

Figure 5.11: Sharpe Ratio (on excess returns) during the four crisis sub-periods for all strategies and passive benchmarks. *Source:* Author's elaboration.

Table 5.7: Sharpe Ratio by crisis sub-period – Long-Only Setup.

Strategy	**Dot-com**	**GFC**	**COVID**	**Inflation**
Hist-Avg	−0.35	−0.86	−0.62	−1.23
OLS-Base	0.43	−0.52	**0.30**	0.41
OLS-Lasso	0.31	−1.14	−0.44	−1.06
ENet	**0.86**	**0.12**	−0.63	**1.28**
NN-3	0.23	−0.96	−0.68	−0.18
1/N	0.17	−1.11	−1.79	−0.64
Buy&Hold	−0.27	−1.51	−1.17	−1.07
60/40	−0.86	−1.33	−0.30	−1.11

Source: Author's calculations.

As we can see, ENet is the only model that has a positive Sharpe Ratio in three of the four crisis windows. Its near-zero GFC result ($SR = +0.12$) stands in sharp contrast to every other strategy and passive benchmark, including the supposed safe-haven 60/40 ($SR = -1.33$). During the inflation shock, the model performed very well ($SR = 1.28$). This success comes from holding plenty of Treasury bonds and Gold, which both protect the portfolio when interest rates are unpredictable. OLS-Base achieved positive Sharpe Ratios in three periods, but it failed during the GFC (−0.52). In that specific crash, holding nearly 29% in Treasuries was simply not enough to cover the heavy stock market losses. Finally, NN-3 shows a negative Sharpe Ratio in almost every crisis. This confirms that, because the portfolio is heavily concentrated in risky assets, any market crisis results in a significant loss for the portfolio.

The quarterly-rebalancing constraint. It is important to remember that the 3-month rebalancing frequency introduces an additional source of tail-risk exposure: the optimizer observes each shock only *ex post*. During the COVID crash of March 2020, the entire drawdown was absorbed at the pre-crisis allocation; by the May rebalancing date, markets had partially recovered and the window for defensive repositioning had closed. This is an inherent trade-off of low-frequency rebalancing: it reduces transaction costs, but prevents timely de-risking during fast-moving crises.

5.2.4 Alpha Generation and Performance Significance

To assess whether the documented excess performance reflects genuine alpha or merely compensated market exposure, we estimate a single-factor CAPM regression for each strategy in both setups[6].

Table 5.8: Jensen's alpha and market beta – Long-Only vs. Realistic Setups.

	Long-Only				***Realistic***			
Strategy	α **(ann.)**	β	R^2	**Sig.**	α **(ann.)**	β	R^2	**Sig.**
Hist-Avg	3.24%	0.491	0.312	*	2.56%	0.489	0.398	
OLS-Base	5.75%	0.439	0.261	***	0.34%	0.468	0.485	
OLS-Lasso	2.57%	0.469	0.398	*	−0.17%	0.574	0.560	
ENet	7.78%	0.346	0.151	***	2.22%	0.460	0.376	
NN-3	0.89%	0.579	0.450		−0.43%	0.622	0.614	
1/N	1.51%	0.406	0.466		1.51%	0.406	0.466	
Buy&Hold	0.71%	0.683	0.669		0.71%	0.683	0.669	
60/40	1.32%	0.570	0.892	**	1.32%	0.570	0.892	**

Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$, $T = 427$ observations. *Source:* Author's calculations.

In the Long-Only setup, ENet and OLS-Base produce statistically significant extra returns at the 1% level ($\alpha = 7.78\%$ and 5.75%, respectively). They also take on low market risk ($\beta \approx 0.35$–0.44), which suggests that they have a real ability to predict the future. On the other hand, NN-3 generates a tiny extra return (0.89%) that is statistically meaningless. This confirms that its performance comes simply from following the overall market ($\beta = 0.58$) rather than from actual predictive ability.

The Realistic setup, however, tells a strikingly different story: every ML alpha collapses to near zero and loses statistical significance entirely. ENet's alpha contracts from 7.78% to 2.22% (without any significance), OLS-Base from 5.75% to 0.34%, and OLS-Lasso turns

[6]The model is $r^e_{p,t} = \alpha + \beta\,(R_{m,t} - r_{f,t}) + \varepsilon_t$, estimated by OLS on monthly excess returns ($T = 427$ observations). The market excess return is proxied by the S&P 500 excess return; $H_0: \alpha = 0$ is tested with a standard t-test.

marginally negative (−0.17%), just the same as NN-3 (−0.43%). This compression is the clearest robustness check available: Long-Only alphas are largely an artefact of the cost-free, unconstrained rebalancing that setup implicitly allows. Once realistic constraints are imposed, every ML alpha collapses to near zero, while the passive 60/40 benchmark retains a statistically significant alpha of 1.32% ($p < 0.05$). Notably, the 60/40 itself retains significance precisely because it is a simple, cost-efficient rule with low turnover, not because it gives more accurate forecasts. That ML models cannot outperform it net of frictions suggests that their predictive edge is largely consumed by rebalancing costs rather than being absent altogether.

5.3 Robustness Checks: Sensitivity to Window Lengths

To assess whether the findings in *Section 5* are artefacts of the primary architectural choice, all four models are evaluated across eight alternative configurations, varying training and validation window lengths (120+60, 180+60, 180+120, 240+120 months) and the estimation scheme for linear models (Expanding, `E`, versus Rolling, `R`), while NN-3 always uses an expanding window[7]. The primary specification, 120+60 months with Rolling OLS/Lasso/ENet and Expanding NN-3 (hereafter RRRE), is the configuration discussed throughout this chapter. *Table 5.9* summarises the eight configurations.

Table 5.9: Overview of the eight backtest configurations.

Setup	**Train (m)**	**Val (m)**	**OLS/Lasso/EN**	**OOS Length (m)**	**OOS Period**
120+60_RRRE⋆	120	60	Rolling	427	1989–2024
120+60_EEEE	120	60	Expanding	427	1989–2024
180+60_RRRE	180	60	Rolling	367	1994–2024
180+60_EEEE	180	60	Expanding	367	1994–2024
180+120_RRRE	180	120	Rolling	307	1999–2024
180+120_EEEE	180	120	Expanding	307	1999–2024
240+120_RRRE	240	120	Rolling	247	2004–2024
240+120_EEEE	240	120	Expanding	247	2004–2024

Note: OLS/Lasso/EN window type: E = Expanding, R = Rolling; NN-3 always E. *Source:* Author's elaboration.
⋆ Primary specification used in *Section 5*

5.3.1 Impact of Training Windows on Predictive Accuracy

Window length and estimation type are implementation choices that, in principle, can materially affect out-of-sample accuracy: a longer window provides more data but may include stale observations, while an expanding window accumulates history at the cost of reduced

[7] A full summary of all tested configurations and their hyperparameter settings is reported in *Appendix D.4*.

adaptability. To quantify these effects, *Table 5.10* reports the aggregate mean R^2_{OOS} (%) across all five assets for each of the eight configurations.

Table 5.10: Aggregate mean R^2_{OOS} (%) across asset classes, by model and configuration.

Setup	OLS-Base	OLS-Lasso	ENet	NN-3
120+60_RRRE⋆	−87.34	−1.74	+2.75	+1.04
120+60_EEEE	−3.99	**+0.31**	+2.15	+1.04
180+60_RRRE	−18.93	−1.72	**+4.66**	+1.24
180+60_EEEE	−4.28	+0.20	+2.35	**+1.24**
180+120_RRRE	−24.21	−1.36	+1.72	−0.21
180+120_EEEE	−6.57	−0.92	−0.89	−0.21
240+120_RRRE	−11.62	−0.28	+2.46	+0.08
240+120_EEEE	−9.61	−1.54	−0.84	+0.08

Source: Author's calculations.

- **OLS-Base:** The Baseline model shows structural weaknesses in all configurations. In particular, the rolling window approach leads to extremely poor out-of-sample R^2 values (−87.34%). On the other hand, the expanding window approach provides a less unstable output (minimum −3.99%) and fails to produce any positive predictive value.
- **OLS-Lasso:** Regularization ensures greater stability compared to the baseline model, yielding R^2_{OOS} values ranging from −1.74% to +0.31%. This is the only model in which increasing window sizes consistently yields higher results than rolling windows, confirming the ability of L_1 regularization to remove noise.
- **ENet:** Elastic Net proves to be the most robust model of all, with a positive average R^2_{OOS} value in six out of eight cases. The best result (+4.66%) is achieved with 180+60_RRRE, rather than with the chosen primary specification, demonstrating that adding just 60 months to the rolling training period increases Elastic Net's predictive power. Elastic Net's predictions deteriorate when used with validation periods longer than 120 months.
- **NN-3:** The Neural Network yields positive R^2_{OOS} values in six out of eight cases, ranging from −0.21% to +1.24%, demonstrating strong consistency and stability across specifications. Since NN-3 always uses an expanding window, the RRRE and EEEE variants produce identical results. In line with Gu et al. (2020), predictive accuracy is not monotonic in sample size, as distant observations add noise rather than signal in low signal-to-noise environments.

Overall, predictive power is strongest when models adapt quickly to recent conditions rather than relying on stale data. The 120+60_RRRE setup strikes the best balance between sufficient training history and responsiveness to recent trends. As the *Section 5.3.2* will confirm, it is also the specification that best translates forecasts into actual performance, alternative windows that improve R^2_{OOS} often produce unrealistic allocations with poor returns.

5.3.2 Stability of Portfolio Metrics across Specifications

To evaluate the impact of changes in window size and type on the performance of the different models, *Table 5.11* reports Sharpe Ratio for both the Long-Only (LO) and Realistic (RE) setups and the annualised turnover for the Long-Only setup across all configurations.

Table 5.11: Sharpe Ratio and annualised Long-Only turnover (%) across configurations.

	OLS-Base			**OLS-Lasso**			**ENet**			**NN-3**		
Setup	SR_{LO}	Turn	SR_{RE}	SR_{LO}	Turn	SR_{RE}	SR_{LO}	Turn	SR_{RE}	SR_{LO}	Turn	SR_{RE}
120+60_RRRE*	**0.73**	468	0.43	0.59	241	0.42	**0.80**	334	**0.55**	0.45	333	0.41
120+60_EEEE	0.51	426	0.43	**0.65**	316	**0.53**	0.45	427	0.44	0.45	366	0.39
180+60_RRRE	0.67	482	**0.60**	0.61	239	0.42	0.52	359	0.48	0.46	338	0.41
180+60_EEEE	0.56	399	0.53	0.69	271	0.58	0.47	400	0.45	0.46	338	0.41
180+120_RRRE	0.66	500	0.51	0.37	175	0.35	0.32	347	0.43	**0.53**	354	0.42
180+120_EEEE	0.54	373	0.55	0.46	220	0.42	0.44	357	0.41	0.63	316	0.51
240+120_RRRE	0.50	412	0.51	0.56	182	0.40	0.44	358	0.47	0.57	327	**0.42**
240+120_EEEE	0.62	342	0.40	0.50	222	0.41	0.43	393	0.36	0.57	327	0.42

Note: SR_{LO} = Long-Only; SR_{RE} = Realistic; Turn = annualised LO turnover (%). Bold = best SR per Setup.
Source: Author's calculations.

Robustness of model performance. Evaluating the predictive and economic performance of the models across these different specifications delivers three consistent findings:

First, Elastic Net is the best-performing model under constraints, where its Realistic Sharpe Ratio ranges from 0.36 to 0.55 across all scenarios and outperforms or matches NN-3 in six out of eight comparisons. It is evident from the primary setting (SR_{RE} = 0.546) that this is the highest Sharpe ratio achieved by ENet under constraints, and this justifies the choice to adopt smaller windows (120+60) because in most cases an increase in predictive power did not translate into improved performance.

Second, the use of rolling versus expanding produces a significant trade-off for both linear models (OLS-Base and ENet), in fact the transition from 120+60_RRRE to 120+60_EEEE results in a notable statistical improvement for both models and particularly for OLS-Base, as its R^2_{OOS} goes from −87% to −4%, while its Long-Only Sharpe ratio goes from 0.732 to 0.511, and that of ENet drops from 0.796 to 0.446.The expanding window forces both models to fit stale macroeconomic relationships from the 1970s and 1980s alongside current signals. The resulting expected-return forecasts become statistically less wrong on average but economically misaligned with current conditions, driving the MVO optimizer to allocations that erode gross returns despite lower measured forecast error. This validates the rationale for the rolling window developed in *Section 4*.

Third, longer validation windows reduce predictive accuracy without offering any portfolio gains. Specifically, the 180+120 and 240+120 specifications sacrifice OOS sample length (307 and 247 months respectively) to extend hyperparameter tuning. For ENet and OLS-Lasso, this trade-off is generally unfavourable: both R^2_{OOS} and Realistic Sharpe Ratios decline without compensating improvements. For NN-3, the effect is marginal. Ultimately, these configurations just confirm that the baseline 120+60 architecture does not suffer from underfitting due to insufficient tuning data.

Jensen's alpha. As we have seen previously, Jensen's α proves to be highly informative; we will therefore analyse the *alphas* summarised in *Table 5.12* for the Long-Only strategy. The 1% significance achieved by ENet (7.78%) and OLS-Base (5.75%) in the primary specification is not replicated in any alternative: OLS-Base retains 5% significance only in the 180+60_RRRE and 180+120_RRRE configurations, while ENet's alpha falls to insignificant levels in all other setups. This confirms the conclusion of *Section 5.2.4* about the performance significance of the models: the abnormal returns documented in the Long-Only setting partly reflect the OOS sample length advantage of the 120+60 rolling architecture and should not be interpreted as a robust, window-invariant feature of the pipeline. A further caveat applies to the configurations with the shortest OOS span: the 240+120 specification covers only 247 months (June 2004 – November 2024), a period dominated by the post-GFC bull market in equities; alpha estimates obtained on this window are therefore subject to a structural timing bias and likely overstate the strategies' unconditional performance.

Table 5.12: Jensen's α (% ann.) and significance – Long-Only setup.

Setup	**OLS-Base**	**OLS-Lasso**	**ENet**	**NN-3**
120+60_RRRE*	5.75%***	2.57%*	7.78%***	0.89%
120+60_EEEE	3.04%	3.05%**	2.21%	1.04%
180+60_RRRE	4.72%**	2.75%*	2.15%	0.76%
180+60_EEEE	2.75%	2.95%**	1.70%	0.76%
180+120_RRRE	5.70%**	0.71%	1.05%	2.95%
180+120_EEEE	3.44%	0.85%	2.14%	3.55%**
240+120_RRRE	0.91%	1.16%	1.27%	1.20%
240+120_EEEE	2.49%	−0.28%	0.84%	1.20%

Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. CAPM single-factor regression on monthly excess Long-Only returns. *Source:* Author's calculations.

Chapter 6

Conclusions

6.1 Summary of Main Findings

This thesis evaluates whether Machine Learning models for expected returns yield more efficient multi-asset portfolios than classical and passive benchmarks. Based on a 35-year out-of-sample analysis (May 1989–November 2024) across five asset classes, the findings are structured around the four research questions outlined in *Section 1.2*.

RQ1 — Methodological comparison: factor model and regularization. The OLS baseline, which forecasts Fama–French factor premia using the 37 Goyal–Welch–Zafirov macroeconomic predictors and then maps them to asset returns, collapses to a mean R^2_{OOS} of -87.3% (*Table 5.1*). This failure is not due to the factor structure being wrong, the Oracle test cited in *Section 4.1.1* confirms that the loadings $\hat{B}$ are valid and that realized factors explain up to 99% of S&P 500 return variation when used with perfect foresight. The real bottleneck lies in Step B, where the standard OLS model try to predict three factor premia using a 120-month window. Because the 37 predictors are highly correlated, the resulting estimates become highly unstable, confirming that multicollinearity severely impairs the model's accuracy even in this restricted setup, consistent with Goyal and Welch (2008). When the predictor space is expanded to the full 57-variable set without regularization, the average R^2_{OOS} collapses further to -289.9%, as documented in *Section 4.1.2*. By selecting just 4.2 variables on average, OLS-Lasso mitigates multicollinearity and recovers mean R^2_{OOS} to -1.7%. However, the aggregate remains negative, with only Commodities yielding a positive result (+0.39%).

This answer is clear: a structured factor approach requires regularization to be numerically stable, but linear regularization alone is insufficient to generate out-of-sample predictability.

RQ2 — The value added of Machine Learning. Both ML models improve meaningfully over all linear benchmarks. The Elastic Net achieves a mean R^2_{OOS} of +2.75% and NN-3 of +1.04% (*Table 5.1*), marking the first positive aggregate values across the full model comparison. Reinforcing this result, the directional accuracy analysis in *Table 5.2* shows that all regularized and ML models correctly predict the sign of the S&P 500 equity premium roughly two-thirds of the time (highly significant hit rates between 62.9% and 65.0%), while maintaining similarly significant accuracy for REITs (58.0%–60.8%). ENet is the more robust forecaster, posting positive R^2_{OOS} in six of eight robustness configurations (*Section 5.3*), and generating the most diversified and crisis-resilient portfolio allocation. NN-3, by contrast, produces allocations that are more concentrated in equities and REITs, which leads to larger drawdowns during stress periods but also to smoother performance between crises; in the Realistic setup, its portfolio weights are arguably more investable than ENet's. However, the Diebold–Mariano test (*Table 5.3*) confirms that the statistical superiority of ML models over the historical mean is not significant at conventional thresholds for any individual asset. Ultimately, addressing whether a *data-driven* approach based on *Elastic Net* and *Neural Networks* can outperform the classical linear baseline by capturing more complex relationships, our findings indicate that it can.

Nevertheless, while these models successfully extract directional accuracy, their statistical predictive advantage remains modest in scale, a result consistent with the seminal findings of Gu et al. (2020) and the broader skepticism regarding equity premium predictability expressed by Goyal et al. (2023).

RQ3 — Drivers of predictability and economic value. The feature importance analysis in *Section 5.1.4* identifies a consistent set of dominant predictors across models: valuation ratios (`e12`, `d12`, `d/p`) and interest rate variables (`lty`, `lzrt`, `BAA`) account for most of the predictive signal, reflecting their established role as slow-moving proxies for time-varying risk premia (Campbell and Shiller, 1988; Goyal and Welch, 2008). Predictability varies significantly across assets: while REITs and the S&P 500 are the easiest to forecast, Gold and Treasury bonds remain nearly impossible to predict for all models. For Gold, Commodities, and REITs the Elastic Net selects zero active predictors for several rolling windows (*Figure A.6* in *Appendix D.3*), which suggests that the GWZ predictor set, originally calibrated on the U.S. equity premium, carries little relevant information for these asset classes.

In terms of economic value, ENet, the best one in the Long-Only setup, achieves an annualized return of 10.80%, a Sharpe ratio of 0.796, and a Jensen's alpha of 7.78% significant at the 1%

level (*Tables 5.4* and *5.8*). However, this alpha is not robust across alternative configurations (*Table 5.12*), and disappears entirely when transaction costs and real market frictions are taken into account.

The results are consistent with those reported by Goyal and Welch (2008) and Goyal et al. (2023), who found that the use of macroeconomic variables is not sufficiently accurate to predict future market trends, and that the benefits observed in-sample or under non-real-world conditions rarely guarantee robust OOS results, both statistically and economically.

RQ4 — Stability and the Realistic setup. The gap between the Long-Only and Realistic setups is one of the most important empirical findings of this thesis. In the unconstrained setting, all active models rebalance between 241% and 468% of portfolio value annually (*Table 5.5*). Under even conservative transaction cost assumptions, this level of turnover would eliminate any gross excess return. Once a 20% quarterly turnover cap and a 60% per-asset weight limit are imposed, the Sharpe ratios of all active strategies fall dramatically, and no ML model surpasses the 60/40 benchmark (SR = 0.690). Every model's Jensen alpha becomes statistically indistinguishable from zero in the Realistic setup (*Table 5.8*), and the 60/40 portfolio, by contrast, retains its significance ($\alpha = 1.32\%$, $p < 0.05$), demonstrating its superiority over all the active strategies tested. Crucially, the failure of active strategies is tied to the MVO process. While ML and OLS-Lasso models often capture the correct return direction, their forecasts are significantly shrunk toward zero and fail to reflect the full magnitude of market movements; this lack of signal intensity prevents the optimizer from constructing meaningful portfolio structures under such restrictive conditions.

Regarding the fourth research question, the results indicate that the MVO process is the main bottleneck. Although ML models show a certain degree of stability compared to the OLS-Base method, their portfolio weights are, in practical terms, not investable. Ultimately, the MVO's inability to best utilize the more accurate forecasts makes the 60/40 strategy outperform all active strategies.

6.2 Limitations and Critical Issues

A critical assessment of this study requires us to acknowledge its inherent structural weaknesses. The results obtained out-of-sample are, in fact, constrained by the specific parameters of the simulation and could deteriorate rapidly outside those limits.

The investment universe is restricted to five assets, ensuring a stable covariance matrix and a tractable optimization. However, these findings may not generalize to larger portfolios where estimation error grows. In high-dimensional settings, Ledoit–Wolf shrinkage would require more advanced support, such as factor-implied or sparse precision estimators.

The quarterly rebalancing frequency is a deliberate choice that reduces transaction costs and matches the signal horizon of monthly macroeconomic predictors. However, it introduces a structural lag in crisis response: during the COVID-19 crash of February–March 2020, the full drawdown was absorbed at the pre-crisis allocation, and by the time of the next rebalancing in May, markets had already partially recovered. The crisis performance results reported in *Section 5.2.3* are therefore conditioned on this choice, and a monthly rebalancing rule could yield materially different tail risk profiles.

Transaction costs are considered only from a qualitative perspective, and the only practical example is provided in the *Footnote 5* of *Section 5.2.2*. The model does not explicitly account for either bid-ask spreads or transaction costs for each trade. Given the level of turnover reported in *Table 5.5*, a quantitative analysis of transaction costs will drastically reduce the net return, bringing it well below the benchmarks, if not close to zero.

The set of GWZ predictor variables was specifically designed and empirically tested to forecast stock returns in the United States (Goyal and Welch, 2008; Goyal et al., 2023). Consequently, even from a theoretical standpoint, these predictor variables seem to explain rather weakly the risks associated with the commodities market, gold, and Treasury securities. Empirically, we confirm the existence of this misalignment, as these financial instruments remain difficult to predict using this set. Similarly, our technical indicators also fail to provide any significant predictive power due to their unique nature. One of the main structural weaknesses of using both sets of predictors, however, concerns the use of one-month lags.

When point forecasts are used directly as inputs for portfolio construction, the estimation error is significantly amplified: the mean-variance methodology does not account for the estimation risk intrinsic to statistical forecasting and, according to *Section 2.1.2*, acts as an "error maximizer" (Michaud, 1989). Positive estimation errors are exploited through an excessive allocation of weights to assets with average returns higher than actual returns. In fact, the problem manifests itself in highly unstable asset allocations, where even small model estimation errors generate extreme volatility in the weights.

Finally, the architectural choices for the neural network, the compressed $[16, 8, 4]$ topology and the Single-Task setup, were motivated by the size of the predictor space and validated empirically during implementation: both the wider $[32, 16, 8]$ architecture and a Multi-Task configuration were tested and produced materially worse out-of-sample results. Similarly, the hyperparameter grids for OLS-Lasso and ENet were designed with a coarse-to-fine log-linear structure specifically to avoid cherry-picking, covering a wide range of penalty magnitudes before concentrating around the empirically dominant region. However, a systematic model selection study using held-out data across all architectures simultaneously would be required to establish whether the chosen specifications are genuinely optimal or merely better than the specific alternatives tested.

6.3 Suggestions for Future Research

To address the current framework's limitations and fully utilize the potential of Machine Learning in portfolio management, future research should focus on four main aspects.

Faced with the various performance metrics evaluated in this study, the fundamental question remains whether these statistical forecasts can truly add tangible value to an investor's strategy. To rigorously confirm the stability of this predictive edge, future research should employ *Block* and *Circular Bootstrap* procedures. By preserving the autocorrelation structure of financial returns under a scenario of zero predictability, these techniques would provide a robust assessment of whether the positive R^2_{OOS} values of ENet and NN-3 reflect genuine forecasting skill or mere sampling variation. This deeper validation is especially relevant given the sensitivity of Jensen's alpha to the choice of the evaluation window (*Table 5.12*).

A second area to explore would be refining the portfolio optimization phase. The current fixed quarterly rebalancing schedule could be adjusted to monthly intervals to increase adaptability in light of changing conditions, or, better yet, an adaptive strategy could be implemented. Within an adaptive system, trading would occur when necessary, for example, when asset weightings shift outside certain ranges or during periods of high volatility. Furthermore, feeding raw data on expected returns directly into the optimizer can lead to amplified estimation errors. A more robust solution would be the Black-Litterman framework (Black and Litterman, 1992), which combines the model's return forecasts, treated as investor *views*, with the equilibrium expected returns implied by the market portfolio under the CAPM. The degree of confidence placed in each view can be calibrated dynamically, for instance using a rolling R^2_{OOS} or a Diebold–Mariano statistic, so that the optimizer relies more heavily on model signals when their recent predictive accuracy is higher.

A third aspect, and probably the most important one, is the expansion of the set of predictors.

As argued in *Section 6.2*, the GWZ set is largely uninformative for Gold, Commodities, and Treasury bonds. Future work could find or construct asset-class-specific predictors, such as commodity inventory data, global trade indicators, or real yield curve slopes for fixed income, and test whether they improve R^2_{OOS} for the three currently unpredictable asset classes. A broader extension would also include international asset classes (e.g., MSCI World ex-U.S. equities, Emerging Market equities and bonds) to test whether the predictability patterns documented here are specific to the U.S. market or reflect more general macroeconomic dynamics.

Finally, regarding model architecture, future research could explore networks designed to analyze multiple past months simultaneously. Overcoming the current one-month lag limitation would allow the models to better capture long-term market trends. Additionally, the rolling hyperparameter distributions (*Appendix C.3*) confirm that optimal model settings shift substantially over time, suggesting that models capable of dynamically adapting to new market regimes would be highly beneficial. Furthermore, the Multi-Task learning approach could be successfully revisited. By adjusting the learning process to prevent highly volatile assets (like Equities) from overshadowing the others, a Multi-Task network could efficiently uncover shared patterns across correlated assets, such as the S&P 500 and REITs.

Bibliography

Black, F. and R. Litterman (1992). "Global portfolio optimization". In: *Financial Analysts Journal* 48.5, pp. 28–43. URL: https://www.jstor.org/stable/4479577.

Campbell, J. Y. and R. J. Shiller (1988). "The dividend-price ratio and expectations of future dividends and discount factors". In: *The Review of Financial Studies* 1.3, pp. 195–228. URL: https://doi.org/10.1093/rfs/1.3.195.

Chopra, V. K. and W. T. Ziemba (1993). "The effect of errors in means, variances, and covariances on optimal portfolio choice". In: *Journal of Portfolio Management* 19.2, pp. 6–11. URL: https://people.duke.edu/~charvey/Teaching/BA453_2006/Chopra_The_effect_of_1993.pdf.

Clark, T. E. and M. W. McCracken (2001). "Tests of equal forecast accuracy and encompassing for nested models". In: *Journal of Econometrics* 105.1, pp. 85–110. URL: https://doi.org/10.1016/S0304-4076(01)00071-9.

Cochrane, J. H. (2005). *Asset Pricing*. Revised. Princeton University Press.

Cochrane, J. H. (2011). "Presidential address: Discount rates". In: *The Journal of Finance* 66.4, pp. 1047–1108. URL: https://doi.org/10.1111/j.1540-6261.2011.01671.x.

Diebold, F. X. and R. S. Mariano (1995). "Comparing Predictive Accuracy". In: *Journal of Business & Economic Statistics* 13.3, pp. 253–263. URL: https://www.jstor.org/stable/1392185.

Fama, E. F. (1970). "Efficient Capital Markets: A Review of Theory and Empirical Work". In: *The Journal of Finance* 25.2, pp. 383–417. URL: https://doi.org/10.2307/2325486.

Fama, E. F. and K. R. French (1992). "The cross-section of expected stock returns". In: *The Journal of Finance* 47.2, pp. 427–465. URL: https://www.jstor.org/stable/2329112.

Fama, E. F. and K. R. French (1993). "Common risk factors in the returns on stocks and bonds". In: *Journal of Financial Economics* 33.1, pp. 3–56. URL: https://doi.org/10.1016/0304-405X(93)90023-5.

Fama, E. F. and K. R. French (2015). “A five-factor asset pricing model”. In: *Journal of Financial Economics* 116.1, pp. 1–22. URL: https://dx.doi.org/10.2139/ssrn.2287202.

French, K. R. (2024). *Data Library*. Last Accessed: 2026-04-13. URL: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Goyal, A. and I. Welch (2008). “A comprehensive look at the empirical performance of equity premium prediction”. In: *The Review of Financial Studies* 21.4, pp. 1455–1508. URL: https://doi.org/10.1093/rfs/hhm014.

Goyal, A. et al. (2023). *A Comprehensive 2022 Look at the Empirical Performance of Equity Premium Prediction*. Working Paper 21-85. Swiss Finance Institute Research Paper. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3929119.

Goyal, A. et al. (2024). *Comprehensive 2022 Look – Data and Code*. Last Accessed: 2026-04-13. URL: https://docs.google.com/spreadsheets/d/10_nkOkJPvq4eZgNl-1ys63PzhbnM3S2y/edit?gid=1060486139.

Gu, S. et al. (2020). “Empirical asset pricing via machine learning”. In: *The Review of Financial Studies* 33.5, pp. 2223–2273. URL: https://doi.org/10.1093/rfs/hhaa009.

Harvey, C. R. et al. (2016). “. . . and the cross-section of expected returns”. In: *The Review of Financial Studies* 29.1, pp. 5–68. URL: https://doi.org/10.1093/rfs/hhv059.

Hastie, T. et al. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd. New York, NY: Springer Science & Business Media.

Kingma, D. P. and J. Ba (2014). “Adam: A method for stochastic optimization”. In: *arXiv preprint arXiv:1412.6980*. URL: https://doi.org/10.48550/arXiv.1412.6980.

Ledoit, O. and M. Wolf (2004). “Honey, I shrunk the sample covariance matrix”. In: *The Journal of Portfolio Management* 30.4, pp. 110–119. URL: https://ssrn.com/abstract=433840.

Lintner, J. (1965). “The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets”. In: *The Review of Economics and Statistics* 47.1, pp. 13–37. URL: https://www.jstor.org/stable/1924119.

Markowitz, H. (1952). “Portfolio selection”. In: *The Journal of Finance* 7.1, pp. 77–91. URL: https://doi.org/10.2307/2975974.

Michaud, R. O. (1989). “The Markowitz optimization enigma: Is ’optimized’ optimal?” In: *Financial Analysts Journal* 45.1, pp. 31–42. URL: https://doi.org/10.2469/faj.v45.n1.31.

Moskowitz, T. J. et al. (2012). "Time series momentum". In: *Journal of Financial Economics* 104.2, pp. 228–250. URL: https://doi.org/10.1016/j.jfineco.2011.11.003.

Neely, C. J. et al. (2014). "Forecasting the equity risk premium: The role of technical indicators". In: *Management Science* 60.7, pp. 1772–1791. URL: https://www.jstor.org/stable/42919633.

Pesaran, M. H. and A. Timmermann (1992). "A Simple Nonparametric Test of Predictive Performance". In: *Journal of Business & Economic Statistics* 10.4, pp. 461–465. URL: https://www.jstor.org/stable/1391822.

Plazzi, A. et al. (2010). "Expected Returns and the Expected Growth in Rents of Commercial Real Estate". In: *The Review of Financial Studies* 23.9, pp. 3469–3519. URL: https://ssl.lu.usi.ch/entityws/Allegati/pdf_pub4890.pdf.

Ross, S. A. (1976). "The arbitrage theory of capital asset pricing". In: *Journal of Economic Theory* 13.3, pp. 341–360. URL: https://doi.org/10.1016/0022-0531(76)90046-6.

Sharpe, W. F. (1964). "Capital asset prices: A theory of market equilibrium under conditions of risk". In: *The Journal of Finance* 19.3, pp. 425–442. URL: https://www.jstor.org/stable/2977928.

Shiller, R. J. (1981). "Do stock prices move too much to be justified by subsequent changes in dividends?" In: *The American Economic Review* 71.3, pp. 421–436. URL: https://www.jstor.org/stable/1802789.

Tobin, J. (1958). "Liquidity preference as behavior towards risk". In: *The Review of Economic Studies* 25.2, pp. 65–86. URL: https://doi.org/10.2307/2296205.

West, K. D. (1996). "Asymptotic inference about predictive ability". In: *Econometrica* 64.5, pp. 1067–1084. URL: https://doi.org/10.2307/2171838.

Zou, H. and T. Hastie (2005). "Regularization and Variable Selection via the Elastic Net". In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 67.2, pp. 301–320. URL: https://www.jstor.org/stable/3647580.

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A Goyal–Welch–Zafirov Predictor Definitions

The 37 monthly predictors used in this study are listed below. Variables marked † originate from Goyal and Welch (2008); those marked ‡ were introduced in Goyal et al. (2023), where full definitions and original references are provided. All series are lagged one month ($t-1$) to prevent look-ahead bias.

Table A.1: List and Description of 37 GWZ Predictors

Code	Name	Description
Panel A: Raw Price, Return, and Earnings Series†		
`price`	Price Index Level	S&P 500 price index value (Shiller website; S&P Corp. post-1987).
`d12`	12-Month Dividends	12-month rolling sum of S&P 500 dividends.
`e12`	12-Month Earnings	12-month rolling sum of S&P 500 earnings; post-1987 interpolated from quarterly data.
`ret`	Market Return (gross)	CRSP total S&P 500 return including dividends.
`retx`	Market Return (ex-div.)	CRSP S&P 500 capital gain return, excluding dividends.
`Rfree`	Risk-Free Return	Monthly Treasury-bill return.
Panel B: Valuation Ratios†		
`d/p`	Dividend–Price Ratio	$\log D_{12,t} - \log P_t$ (Campbell and Shiller 1988, as in Goyal and Welch 2008).
`d/y`	Dividend Yield	$\log D_{12,t} - \log P_{t-1}$ (Campbell and Shiller 1988, as in Goyal and Welch 2008).
`e/p`	Earnings–Price Ratio	$\log E_{12,t} - \log P_t$ (Campbell and Shiller 1988, as in Goyal and Welch 2008).
`d/e`	Dividend–Payout Ratio	$\log D_{12,t} - \log E_{12,t}$ (Lamont 1998, as in Goyal and Welch 2008).
`b/m`	Book-to-Market Ratio	Book-to-market of the DJIA from Value Line; lagged by fiscal year (Kothari and Shanken 1997, as in Goyal and Welch 2008).
Panel C: Interest Rates, Spreads, and Bond Returns†		
`tbl`	Treasury Bill Rate	3-month T-Bill secondary market rate; NBER pre-1934, FRED post-1934 (Campbell 1987, as in Goyal and Welch 2008).
`lty`	Long-Term Govt. Yield	Yield on long-term U.S. government bonds from Ibbotson *SBBI* (Fama and French 1989, as in Goyal and Welch 2008).
`ltr`	Long-Term Govt. Return	Total return on long-term U.S. government bonds from Ibbotson *SBBI* (Fama and French 1989, as in Goyal and Welch 2008).

Table A.1 (continued)

Code	Name	Description
corpr	Corporate Bond Return	Total return on long-term corporate bonds from Ibbotson (Fama and French 1989, as in Goyal and Welch 2008).
AAA	AAA Bond Yield	Yield on AAA-rated corporate bonds from FRED (Fama and French 1989, as in Goyal and Welch 2008).
BAA	BAA Bond Yield	Yield on BAA-rated corporate bonds from FRED (Fama and French 1989, as in Goyal and Welch 2008).
tms	Term Spread	$lty_t - tbl_t$ (Fama and French 1989, as in Goyal and Welch 2008).
dfy	Default Yield Spread	$BAA_t - AAA_t$ (Fama and French 1989, as in Goyal and Welch 2008).
dfr	Default Return Spread	$corpr_t - ltr_t$ (Fama and French 1989, as in Goyal and Welch 2008).
Panel D: Macroeconomic and Market Activity Indicators†		
infl	Inflation Rate	Monthly CPI-U change (BLS); extra lag applied due to publication delay (Fama and Schwert 1977, as in Goyal and Welch 2008).
ntis	Net Equity Expansion	12-month net equity issues by NYSE firms scaled by total market cap (CRSP) (Boudoukh et al. 2007, as in Goyal and Welch 2008).
svar	Stock Variance	Sum of squared daily S&P 500 returns per month (Guo 2006, as in Goyal and Welch 2008).
Panel E: New Variables from Goyal et al. (2023)‡		
lzrt	Log Zero-Return Illiquidity	Log fraction of trading days with zero equity return; adjusted for 1997 and 2001 tick-size changes (Chen, Eaton, and Paye 2018, as in Goyal et al. 2023).
ogap	Output Gap	Deviation of log industrial production from a linear–quadratic trend; full-sample version only, real-time recomputation required for OOS use (Cooper and Priestley 2009, as in Goyal et al. 2023).
wtexas	Oil Price Change	Monthly % change in WTI crude oil price (Driesprong, Jacobsen, and Maat 2008, as in Goyal et al. 2023).
ndrbl	New Orders/Shipments	Ratio of durable goods new orders to shipments (Census Bureau) (Jones and Tuzel 2013, as in Goyal et al. 2023).
skvw	Avg. Stock Skewness	Cross-sectional average of individual stock return skewness within each month (Jondeau, Zhang, and Zhu 2019, as in Goyal et al. 2023).
tail	Tail Risk	Time-varying tail risk estimated from the cross-section of individual stock returns (Kelly and Jiang 2014, as in Goyal et al. 2023).

Table A.1 (continued)

Code	Name	Description
`fbm`	B/M Cross-Section Factor	PLS-based latent factor extracted from the cross-section of book-to-market ratios; full-sample version only, real-time recomputation required for OOS use (Kelly and Pruitt 2013, as in Goyal et al. 2023).
`dtoy`	Distance to 52-Week High	$(H_t^{52w} - P_t)/H_t^{52w}$; DJIA proximity to its 52-week maximum (Li and Yu 2012, as in Goyal et al. 2023).
`dtoat`	Distance to All-Time High	$(H_t^{\infty} - P_t)/H_t^{\infty}$; DJIA proximity to its lifetime historical maximum (Li and Yu 2012, as in Goyal et al. 2023).
`ygap`	Stock–Bond Yield Gap	Dividend-price ratio minus 10-year Treasury yield (Maio 2013, as in Goyal et al. 2023).
`rdsp`	Stock Return Dispersion	Cross-sectional std. dev. of returns on 100 size and book-to-market sorted portfolios (Maio 2016, as in Goyal et al. 2023).
`tchi`	Technical Indicators	First principal component of 14 moving-average and momentum signals on the S&P 500; full-sample version only, real-time recomputation required for OOS use (Neely, Rapach, Tu, and Zhou 2014, as in Goyal et al. 2023).
`avgcor`	Avg. Stock Correlation	Average pairwise daily return correlation of individual stocks (Pollet and Wilson 2010, as in Goyal et al. 2023).
`shtint`	Short Interest	Aggregate NYSE short interest ratio; real-time recomputation required for OOS use (Rapach, Ringgenberg, and Zhou 2016, as in Goyal et al. 2023).

Source: Author's elaboration based on Goyal and Welch (2008) and Goyal et al. (2023)

B Supplementary OLS Analyses: Dimensionality and Perfect Foresight

B.1 Curse of Dimensionality: OLS-Base vs. Unrestricted OLS

This section provides empirical support for the predictor-set restriction applied to the OLS-Base model (37 GWZ variables) compared to the unrestricted OLS-Full specification (57 variables), motivated by the dimensionality concerns discussed in *Section 4.1.2*. Diagnostic out-of-sample metrics are reported in *Table A.2*.

Table A.2: OOS metrics: OLS-GWZ (37 predictors) vs. OLS-Full (57 predictors).

	RMSE		$\mathbf{R^2_{OOS}}$ (%)	
Asset	*GWZOnly*	*GWZ+ Custom*	*GWZOnly*	*GWZ+ Custom*
S&P 500	0.0668	0.0889	−151.09	−343.91
Commodities	0.0760	0.1162	−48.20	−246.92
Gold	0.0446	0.0603	−15.72	−111.84
REITs	0.0799	0.1280	−187.09	−636.36
Treasury 7–10y	0.0223	0.0279	−34.61	−110.50
Mean	0.0579	0.0842	−87.34	−289.91

Note: OOS period: May 1989–November 2024 ($T = 143$). Both specifications use a 120-month rolling training window and identical walk-forward architecture. The unrestricted OLS-Full includes the 37 GWZ predictors plus 20 custom technical features. Diebold–Mariano pairwise tests (HAC, $\ell = 3$) yield DM statistics between −0.98 and −1.37 (all $p > 0.17$), confirming that the performance gap is not driven by sampling variation. *Source:* Author's calculations.

B.2 OLS-Oracle: Perfect Foresight Benchmark

As introduced in *Section 4.1.1*, *Table A.3* reports the complete out-of-sample metrics for the "Oracle" benchmark. This idealized perfect-foresight specification uses actual future values (F_{t+1}) to isolate the predictive bottleneck of the macroeconomic forecasts.

Table A.3: OOS metrics of the "Oracle" Model (Perfect Foresight).

Asset	**RMSE**	$\mathbf{R^2_{OOS}}$ (%)
S&P 500	0.0031	99.47
Commodities	0.0598	8.01
Gold	0.0428	−6.92
REITs	0.0369	38.90
Treasury 7–10y	0.0189	3.22

Note: OOS evaluation starts from index 180 (May 1989). The estimation is based on a fixed 120-month rolling train window. Instead of standard predictors, the model uses the actual F_{t+1} as predictions. *Source:* Author's calculations.

C Model Architecture and Implementation

C.1 Algorithm implementation

The complete Python implementation (Jupyter Notebook, HTML preview, and charts) developed for this thesis is available on *GitHub*.

GitHub Repository URL: `https://github.com/botta-andrea/Bachelor_Thesis_USI-Asset_Allocation_via_ML_Factor-Models`

C.1.1 OLS Baseline and OLS-Lasso

Inputs. $Y \in \mathbb{R}^{T\times N}$, $F \in \mathbb{R}^{T\times K}$, $Z \in \mathbb{R}^{T\times P}$; $T_{\text{train}} = 120\,\text{m}$ (Baseline) or $180\,\text{m}$ (Lasso, $T_{\text{val}} = 60\,\text{m}$); $\Delta = 3\,\text{m}$.

Algorithm 1 OLS Baseline / OLS-Lasso: Two-Step Rolling Estimation

---

Input: `Y, F, Z,` T_{train}, Δ, `lambda_range` (Lasso only)

Output: $\left\{\hat{\mathbb{E}}_t[Y_{t+1}],\ \hat{\Sigma}_{t+1}\right\}_t$

for $t = t_0,\ t_0+\Delta,\ \ldots,\ T$ **do**

  Set $\mathcal{W} = [t - T_{\text{train}},\ t)$; skip if $|\mathcal{W}| < T_{\text{train}}$.

  # Step A — factor loadings

  $(\hat{\alpha}_t,\ \hat{B}_t) = \arg\min_{\alpha,B} \|Y_{\mathcal{W}} - \mathbb{1}\alpha^\top - F_{\mathcal{W}}B^\top\|_F^2$; $\hat{V}_{\varepsilon,t} = \operatorname{diag}\big(\widehat{\operatorname{Var}}(\hat{\varepsilon}_{\mathcal{W}})\big)$

  # Step B — factor forecast

  if `use_lasso` **then**

    Partition $\mathcal{W}$ into $\mathcal{W}_{\text{in}}$ and $\mathcal{W}_{\text{val}}$; for each factor k: $\alpha_k^* = \arg\min_{\alpha\in\mathcal{A}} \dfrac{1}{T_{\text{val}}} \sum_{\tau\in\mathcal{W}_{\text{val}}} \big(F_{k,\tau} - \hat{\lambda}_k^\top Z_{\tau-1}\big)^2$, with $\hat{\lambda}_k$ fitted via L_1-penalized regression on $\mathcal{W}_{\text{in}}$. Refit on full $\mathcal{W}$ with α_k^*; assemble $\hat{\Lambda}_t \in \mathbb{R}^{K\times P}$.

  else

    $(\hat{c}_t, \hat{C}_t) = \arg\min_{c,C} \|F_{\mathcal{W}} - \mathbb{1}c^\top - Z_{\mathcal{W}}C^\top\|_F^2$

  $\hat{\mathbb{E}}_t[F_{t+1}] = \hat{c}_t + \hat{C}_t Z_t$ (Baseline) / $\hat{c}_t + \hat{\Lambda}_t Z_t$ (Lasso)

  # Steps C & D — return forecast and covariance

  $\hat{\mathbb{E}}_t[Y_{t+1}] = \hat{\alpha}_t + \hat{B}_t\,\hat{\mathbb{E}}_t[F_{t+1}]$; $\hat{\Sigma}_{t+1} = \hat{B}_t\,\hat{V}_{f,t}\,\hat{B}_t^\top + \hat{V}_{\varepsilon,t}$

return $\left\{\hat{\mathbb{E}}_t[Y_{t+1}],\ \hat{\Sigma}_{t+1}\right\}_t$

Predictor Importance (OLS-Lasso): $\mathcal{I}_j = \frac{1}{T_{\text{OOS}}}\sum_t \frac{1}{N}\sum_{i=1}^N |\hat{B}_t\,\hat{\Lambda}_t|_{(i,j)}$.

C.1.2 Elastic Net (Direct Forecasting)

Inputs. $X_t = [Z_t^\top, F_t^\top]^\top \in \mathbb{R}^P$ (P=60: 37 GW + 20 custom + 3 lagged FF); T_{train}=120 m (inner), T_{val}=60 m; α-grid: 50 log-linear on $[10^{-3}, 10^{1.5}]$; $\rho \in \{0.1, 0.5, 0.9\}$; Δ=3.

Key distinction. ENet forecasts returns *directly* from X_t, bypassing factor decomposition. $\hat{\Sigma}_{t+1}$ is estimated via Ledoit–Wolf (see *Section 4.3.1*).

Algorithm 2 Elastic Net: Direct Rolling Estimation

Input: `Y, X,` T_{train}, T_{val}, Δ, `alpha_grid`, `l1_ratio_grid`

Output: $\{\hat{\mathbb{E}}_t[Y_{t+1}]\}_t$

for $t = t_0,\ t_0 + \Delta,\ \dots,\ T$ **do**

  Set $\mathcal{W}_{\text{in}} = [t - T_{\text{train}} - T_{\text{val}},\ t - T_{\text{val}})$, $\mathcal{W}_{\text{val}} = [t - T_{\text{val}},\ t)$; skip if $|\mathcal{W}_{\text{in}}| < T_{\text{train}}$.

  for *each asset* $i = 1, \dots, N$ **do**

    Standardize $X_{\mathcal{W}_{\text{in}}}$ and $Y_{i,\mathcal{W}_{\text{in}}}$ to zero mean, unit variance.

    # Grid search over 150 combinations

    for *each* $(\alpha, \rho) \in \mathcal{A} \times \{0.1, 0.5, 0.9\}$ **do**

      Solve on $\mathcal{W}_{\text{in}}$: $\hat{\Lambda}_i = \arg\min_{\Lambda} \dfrac{1}{T_{\text{train}}} \sum_{\tau \in \mathcal{W}_{\text{in}}} \left(Y_{i,\tau} - \Lambda^\top X_\tau\right)^2 + \alpha\left(\rho \|\Lambda\|_1 + \frac{1-\rho}{2}\|\Lambda\|_2^2\right)$

      Evaluate MSE_{val} on $\mathcal{W}_{\text{val}}$.

    $(\alpha_i^*,\ \rho_i^*) \leftarrow \arg\min\ \text{MSE}_{\text{val}}$

    # Refit on full 180m window, predict and rescale

    Refit ENet(α_i^*, ρ_i^*) on $\mathcal{W}_{\text{in}} \cup \mathcal{W}_{\text{val}}$; $\hat{\mathbb{E}}_t[Y_{i,t+1}] = \hat{\Lambda}_i^\top X_t \cdot \hat{\sigma}_{Y_i} + \hat{\mu}_{Y_i}$

return $\{\hat{\mathbb{E}}_t[Y_{t+1}]\}_t$

Importance: Fitted on $\mathcal{W}_{\text{in}}$ of the last window using median-optimal (α_i^*, ρ_i^*): $\mathcal{I}_j = \frac{1}{N}\sum_{i=1}^N |\hat{\Lambda}_{i,j}|$.

C.1.3 Neural Networks – NN-3 (Single-Task Ensemble)

Inputs. Same $X_t \in \mathbb{R}^{60}$, Y_t as *C.1.2*; expanding window (min T_{train}=120 m); T_{val}=60 m; $\alpha \in \{10^{-5}, \ldots, 10^{-1}\}$; ensemble S=10; Δ=3.

Architecture. Three hidden layers $[16, 8, 4]$; BatchNorm + ReLU; linear output; input dim = 60:

$$h^{(l)} = \text{ReLU}\Big(\text{BN}\big(W^{(l)} h^{(l-1)} + b^{(l)}\big)\Big), \qquad \hat{y}_i = W^{(4)} h^{(3)} + b^{(4)}$$

Algorithm 3 NN-3 Single-Task: Expanding Window + Ensemble

Input: `Y, X`, $T_{\text{train}}, T_{\text{val}}, \Delta$, `L1_grid`, S

Output: $\{\hat{\mathbb{E}}_t[Y_{t+1}]\}_t$

for $t = T_{train} + T_{val},\ t + \Delta,\ \ldots,\ T$ **do**
- Set $\mathcal{W}_{\text{in}} = [0,\ t - T_{\text{val}})$ (expanding); $\mathcal{W}_{\text{val}} = [t - T_{\text{val}},\ t)$ (fixed 60 m).
- Standardize $X_{\mathcal{W}_{\text{in}}}$ (`StandardScaler`).
- **for** *each asset* $i = 1, \ldots, N$ **do**
  - Standardize $Y_{i,\mathcal{W}_{\text{in}}}$ to zero mean, unit variance.
  - ***# L1 grid search with seed = 0***
  - **for** *each* $\alpha \in \{10^{-5}, \ldots, 10^{-1}\}$ **do**
    - Train NN-3(α) via Adam (η=10^{-3}, batch=16, max 100 epochs, patience=5); minimize $\mathcal{L} = \frac{1}{T}\sum_{\tau}\big(Y_{i,\tau} - \hat{g}(X_\tau;\theta)\big)^2 + \alpha\|\theta\|_1$
    - evaluate MSE_{val}.
  - $\alpha_i^* \leftarrow \arg\min_\alpha\ \text{MSE}_{\text{val}}(\alpha)$
  - ***# Ensemble: re-train S networks with seeds*** $0, \ldots, S-1$
  - **for** *seed* $s = 0, 1, \ldots, S-1$ **do**
    - Re-initialize NN-3(α_i^*) with seed s; train with Early Stopping (patience=5, restore best weights); store $\hat{g}_{i,s}(\,\cdot\,;\theta_{i,s})$.
  - $\hat{\mathbb{E}}_t[Y_{i,t+1}] = \frac{1}{S}\sum_{s=1}^{S}\hat{g}_{i,s}(X_t;\theta_{i,s}) \cdot \hat{\sigma}_{Y_i} + \hat{\mu}_{Y_i}$

return $\{\hat{\mathbb{E}}_t[Y_{t+1}]\}_t$

Importance: Gradient × Input averaged over ensemble and OOS steps: $\mathcal{I}_{i,j,t} = \frac{1}{S}\sum_s \left|\frac{\partial \hat{g}_{i,s}}{\partial X_{j,t}}\right| \cdot |X_{j,t}|$; $\mathcal{I}_j = \frac{1}{N \cdot T_{\text{OOS}}}\sum_{i,t}\mathcal{I}_{i,j,t}$.

C.2 Architecture and Hyperparameter Summary

Table A.4: Hyperparameter Specifications for All Methods

Parameter	**OLS-Base**	**OLS-Lasso**	**ENet**	**NN-3**
A. Input				
Predictors P	37	57	60	60
Predictor set	GWZ	GWZ+ custom	GWZ+ custom + FF-3	GWZ+ custom + FF-3
Standardize X / Y	—	✓	✓	✓
Forecasting target	$E[F_{t+1}]$	$E[F_{t+1}]$	$Y_{i,t+1}$	$Y_{i,t+1}$
B. Estimation Window				
Window scheme	Rolling	Rolling	Rolling	Expanding
Inner train	120 m	120 m	120 m	≥120 m
Validation	—	60 m	60 m	60 m (rolling)
Total window	120 m	180 m	180 m	full history
Forecast step Δ	3 m	3 m	3 m	3 m
OOS start	$t_0 = 180$	$t_0 = 180$	$t_0 = 180$	$t_0 = 180$
C. Regularization				
Loss function	MSE	MSE	MSE	MSE
Huber loss	—	—	—	—
Penalty type	None	L_1	$L_1 + L_2$	L_1 (weights)
α grid	—	50, $[10^{-2}, 10^{0}]$	50, $[10^{-3}, 10^{1.5}]$	$\{10^{-5}, \ldots, 10^{-1}\}$
α spacing	—	log-linear	log-linear	log-linear
L_1-ratio ρ	—	1 (fixed)	$\{0.1, 0.5, 0.9\}$	—
Grid combos / asset	—	50	150	5
Tuning criterion	—	MSE_{val}	MSE_{val}	MSE_{val}
D. Architecture & Optimization (NN-3 only)				
Hidden layers	—	—	—	3
Neurons per layer	—	—	—	[16, 8, 4]
Activation	—	—	—	BN + ReLU
Dropout	—	—	—	—
Optimizer	—	—	—	Adam
Learning rate η	—	—	—	10^{-3}
Batch size	—	—	—	16
Max epochs	—	—	—	100
Early stopping patience	—	—	—	5 epochs
E. Ensemble				
Ensemble size S	—	—	—	10 seeds
Aggregation	—	—	—	simple avg

Note: GWZ= 37 Goyal–Welch macroeconomic predictors; custom = 20 technical features; FF-3 = 3 Fama–French factors lagged one period. For OLS-Lasso and ENet, the optimal hyperparameters are identified on the validation set and the model is subsequently refitted on the full combined window before forecasting. For NN-3, the validation set is used solely for L_1 grid search and early stopping; no combined refit. Notation — indicates parameter not applicable. *Source:* Author's elaboration.

C.3 Optimal Hyperparameter Distributions Over Time

Figures A.1–A.3 report the walk-forward evolution of the optimal hyperparameters selected at each OOS rebalancing step ($T_{\mathrm{OOS}} = 143$). *Figure A.1* refers to **OLS-Lasso**: it shows the optimal L_1 regularization strength α^* selected at each step for each of the three Fama–French factors in Step B of the estimation. *Figure A.2* refers to the **Elastic Net**: it reports both the optimal penalty α^* and the L_1-ratio ρ^* per asset, capturing the balance between ridge and lasso shrinkage over time. *Figure A.3* refers to **NN-3**: it displays the optimal L_1 weight-decay penalty α^* per asset, selected via grid search on the validation window at each rebalancing step.

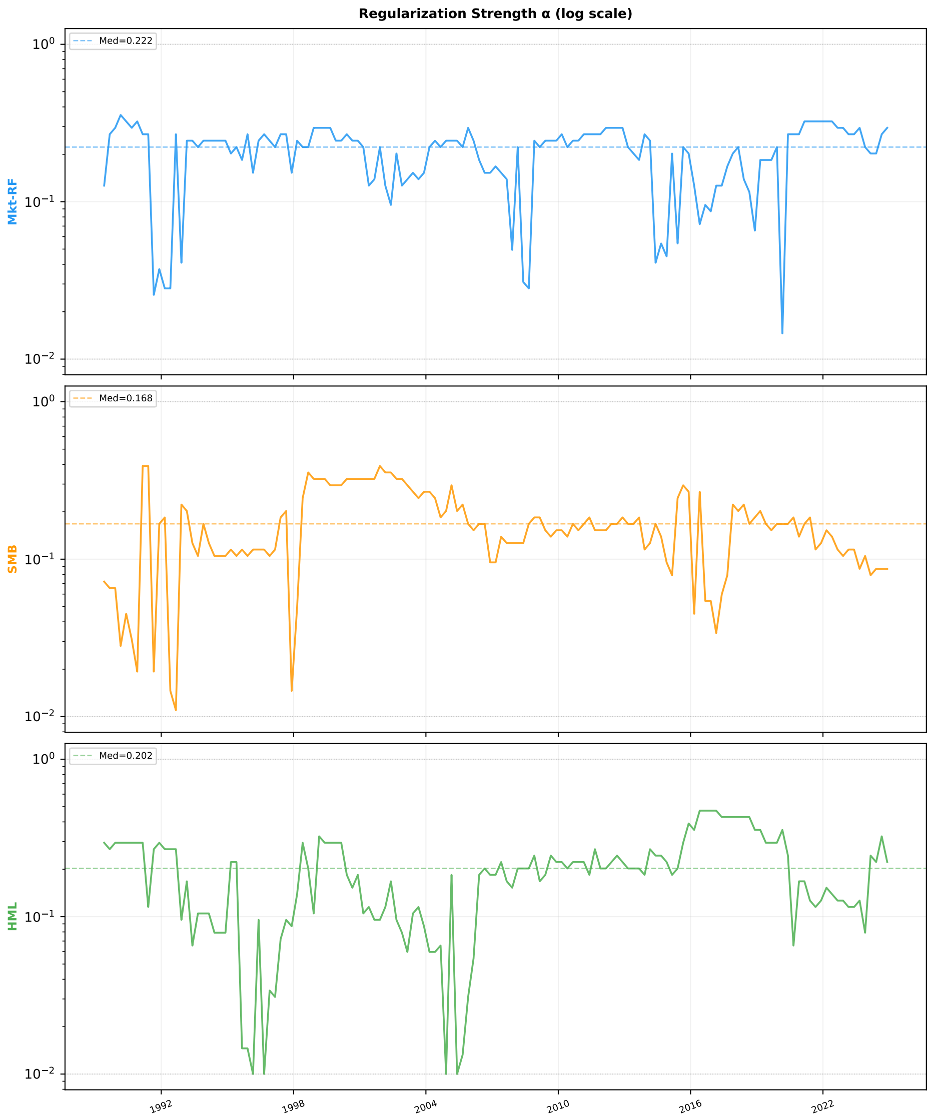

Figure A.1: OLS-Lasso (Step B): optimal α^* over time for each Fama–French factor. Dashed line: time-series median. Dotted lines: grid bounds. *Source:* Author's elaboration.

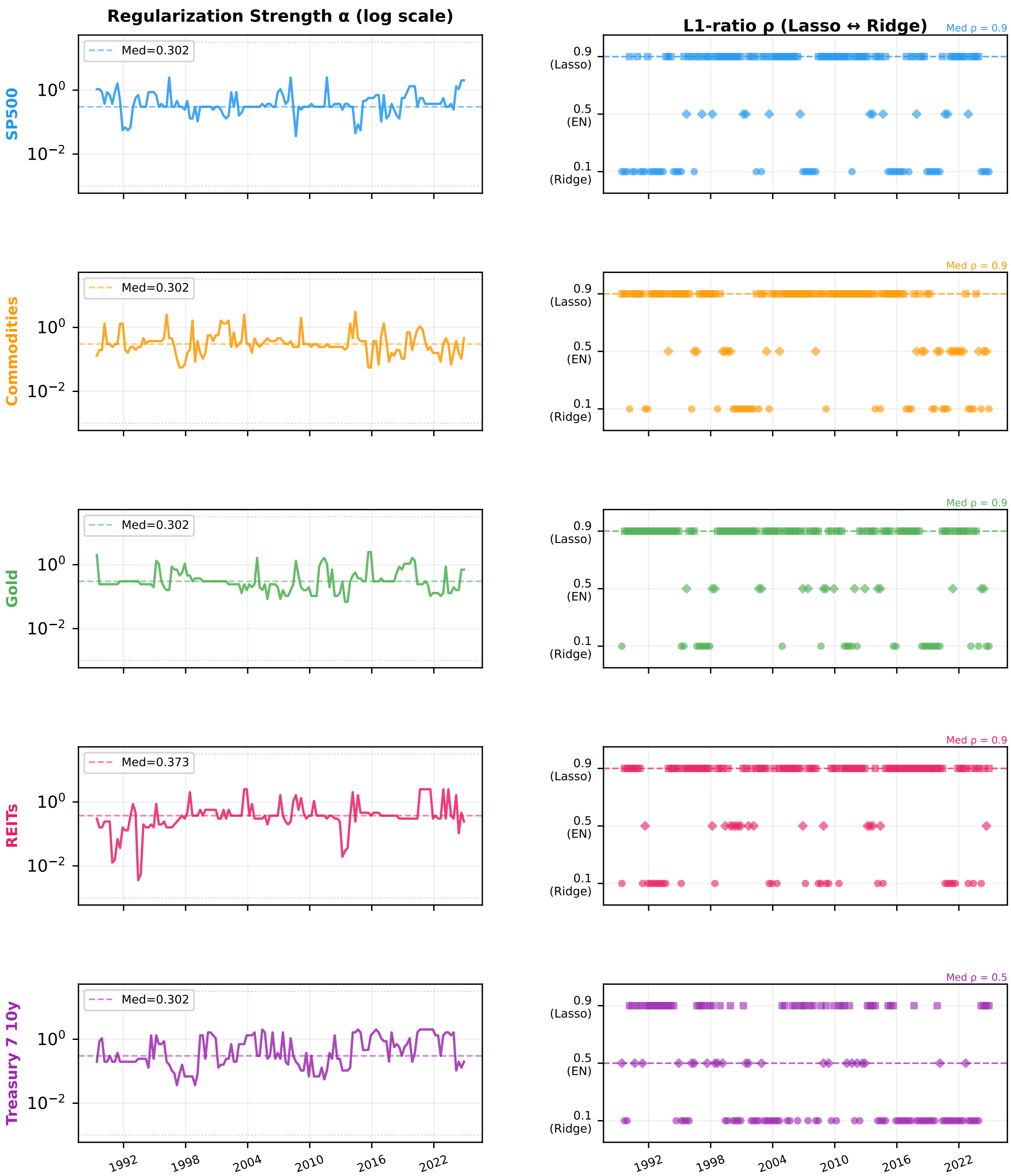

Figure A.2: Elastic Net: optimal regularization strength α^* (left) and L_1-ratio ρ^* (right) per asset over the OOS period. *Source:* Author's elaboration.

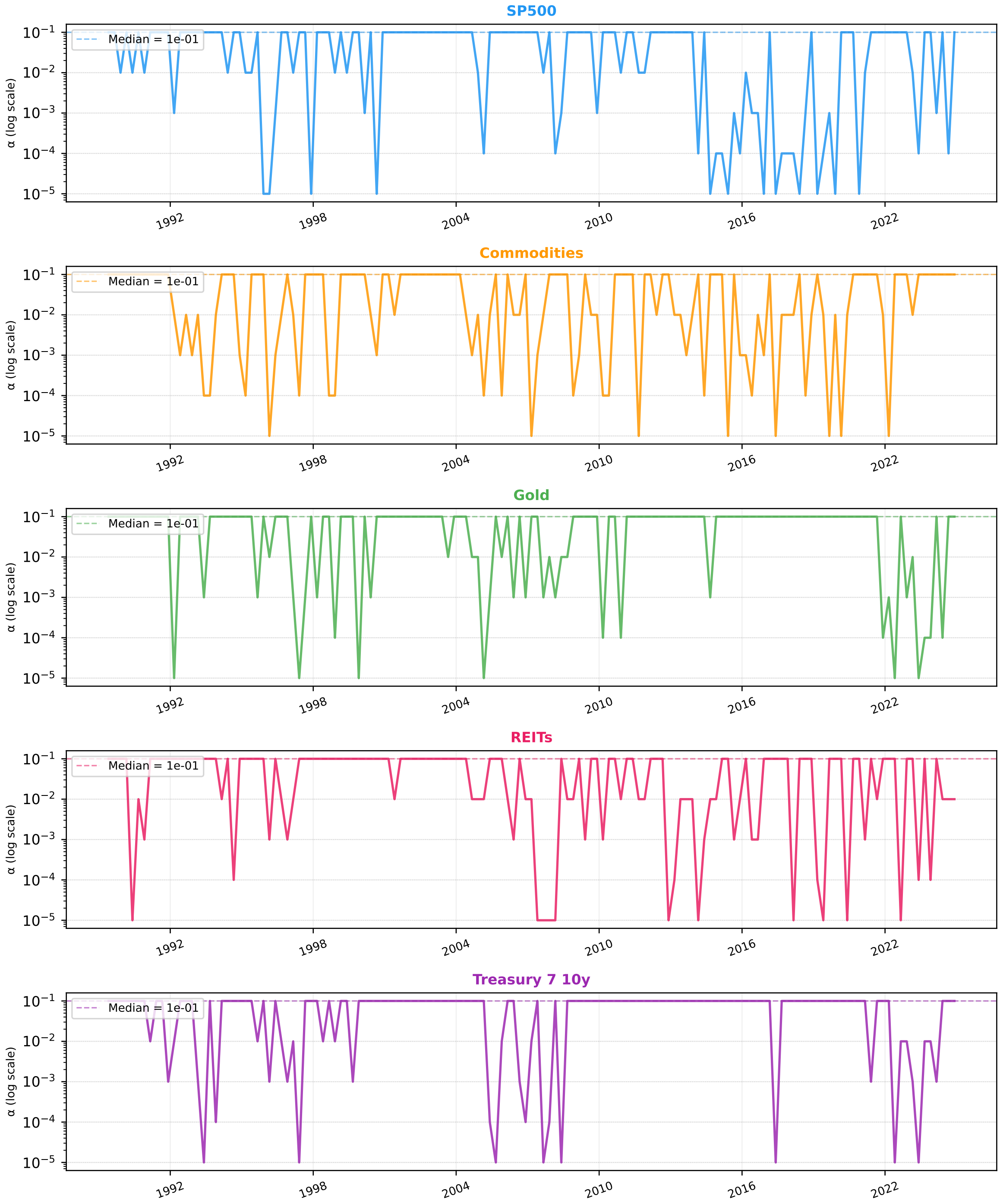

Figure A.3: NN-3: optimal L_1 weight-decay penalty α^* per asset over the OOS period. Dashed line: time-series median. *Source:* Author's elaboration.

D Empirical Results

D.1 Detailed Forecast Diagnostics by Asset Class

D.1.1 Predicted vs. Realized Returns

Expanding upon the time-series analysis presented for the S&P 500 and REITs in *Section 5.1.2* (*Figure 5.3*), here *Figure A.4* illustrate the magnitude and direction of each model's forecasts against the realized returns over the out-of-sample period.

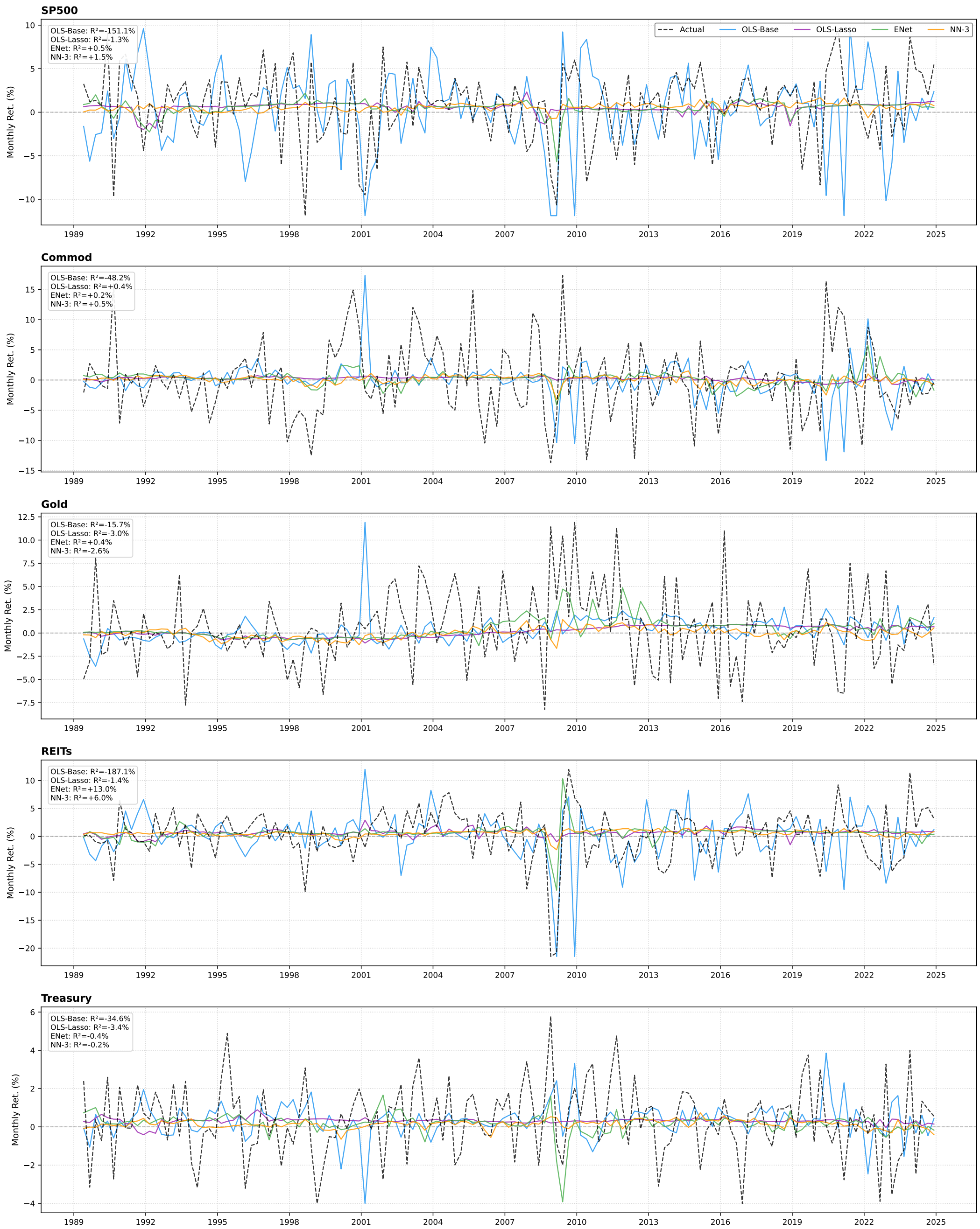

Figure A.4: Predicted vs. realized excess returns for all models and asset classes. Each panel reports the OLS regression line (red) and the 45° line (dashed). *Source:* Author's elaboration.

D.1.2 Regression Slopes by Asset and Model

While *Figure 5.2* in *Section 5.1.2* presented calibration slopes aggregated by model across all asset classes, this subsection disaggregates the results. Here *Figure A.5* detail the specific regression slopes for every individual model-asset pair to assess localized over- or under-dispersion.

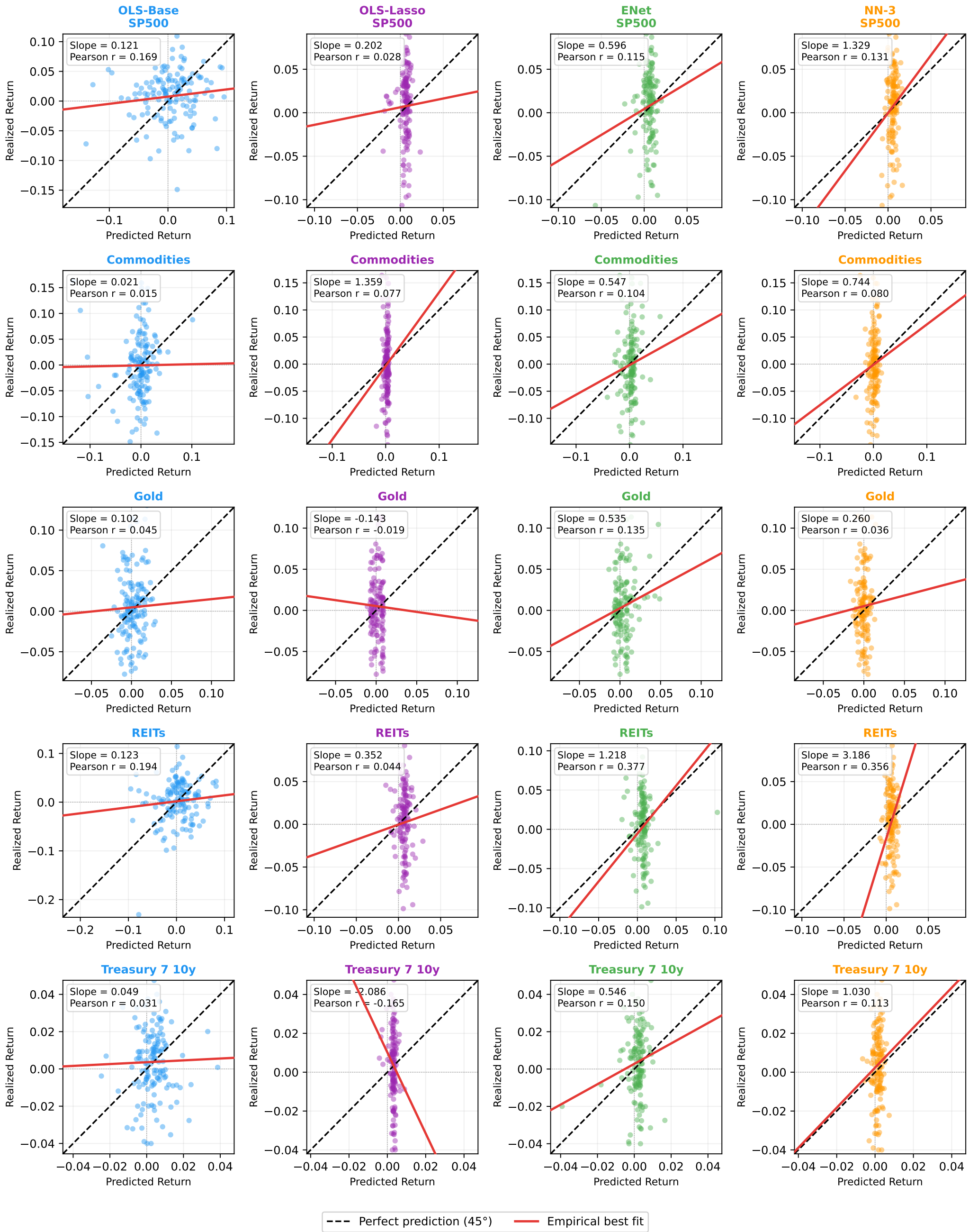

Figure A.5: OLS slope coefficients from regressing realized on predicted returns for each model–asset pair. A slope of 1 indicates perfect calibration; values below 1 signal systematic over-dispersion of forecasts. *Source:* Author's elaboration.

D.1.3 Forecast Error Bias

This section reports the exact numerical data underlying the error distributions visualized in *Figure 5.4* (*Section 5.1.2*). The tabulated metrics provide a granular breakdown of the central tendencies and dispersion of forecast errors.

Table A.5: Mean and median forecast errors by model and asset class (monthly %).

Model	Asset	Mean Err.	Median Err.	Std. Err.	R^2_{OOS}
OLS-Base	S&P 500	−0.802%†	−0.578%	6.659%	−151.09%
	Commodities	+0.148%	+0.545%	7.621%	−48.20%
	Gold	−0.170%	−0.133%	4.468%	−15.72%
	REITs	−0.018%	+0.012%	8.020%	−187.09%
	Treasury	−0.082%	−0.165%	2.234%	−34.61%
OLS-Lasso	S&P 500	−0.170%	−0.714%	4.257%	−1.30%
	Commodities	+0.257%†	+0.582%	6.244%	+0.39%
	Gold	−0.355%†	−0.224%	4.204%	−3.02%
	REITs	+0.460%†	+0.024%	4.744%	−1.42%
	Treasury	−0.064%	−0.200%	1.958%	−3.36%
ENet	S&P 500	−0.215%	−0.690%	4.217%	+0.46%
	Commodities	+0.260%†	+0.431%	6.250%	+0.17%
	Gold	−0.047%	−0.001%	4.147%	+0.43%
	REITs	+0.407%†	+0.035%	4.395%	+13.04%
	Treasury	−0.199%	−0.235%	1.920%	−0.38%
NN-3	S&P 500	−0.162%	−0.875%	4.198%	+1.47%
	Commodities	+0.128%	+0.523%	6.243%	+0.52%
	Gold	−0.532%†	−0.250%	4.175%	−2.57%
	REITs	+0.391%†	−0.276%	4.572%	+6.00%
	Treasury	−0.235%	−0.323%	1.915%	−0.23%

Note: T = 143 OOS observations. † |Mean Error| > 0.25%. Positive values indicate overestimation. *Source:* Author's calculations.

D.2 Diebold–Mariano Test Framework

D.2.1 Test Formulation and Econometric Caveats

The Diebold and Mariano (1995) test evaluates if a model's MSE is significantly lower than a benchmark's. Let $e_{\text{bench},t}$ and $e_{\text{model},t}$ be the forecast errors. The test is defined as:

- **Loss Differential:** $d_t = e^2_{\text{bench},t} - e^2_{\text{model},t}$.
- **Null Hypothesis (H_0):** $E[d_t] \leq 0$ (Model MSE $\geq$ Benchmark MSE).
- **Test Statistic:**

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}_d / P}}$$

  where $\bar{d}$ is the sample mean of d_t, P is the out-of-sample period, and $\hat{V}_d$ is the HAC-robust variance (Newey–West, $\ell = 3$).

Applying this framework to ML models introduces two practical statistical limitations:

- **Estimation Error (West, 1996):** The test assumes perfectly known parameters, ignoring the high uncertainty of estimating ML models on limited data. This underestimates the true forecast error variance, making the test overly optimistic and prone to false positives.
- **Nested Models (Clark and McCracken, 2001):** Because the Historical Average benchmark is effectively just a zero-predictor, "nested" version of our ML models, comparing the two mathematically distorts the test's standard significance thresholds.

Due to the difficulty of adjusting the test for ML models, we follow Gu et al. (2020, Section 1.8 and Table 3) and report the standard DM statistics. However, any significance should be interpreted with caution.

D.2.2 Diebold–Mariano Pairwise Tests

Complementing the predictive performance evaluation discussed in the main text, this subsection provides the complete pairwise matrices for the S&P 500 (*Table A.6*) and REITs (*Table A.7*). These tests rigorously assess the statistical significance of the differences in MSE across all competing models.

Table A.6: Pairwise Diebold–Mariano test statistics for S&P 500 (MSE loss).

	Hist-Avg	OLS-Base	OLS-Lasso	ENet	NN-3
Hist-Avg	—	−2.588	−0.842	+0.094	+0.603
OLS-Base	+2.588***	—	+2.567***	+2.580***	+2.600***
OLS-Lasso	+0.842	−2.567	—	+0.385	+0.985
ENet	−0.094	−2.580	−0.385	—	+0.275
NN-3	−0.603	−2.600	−0.985	−0.275	—

Source: Author's calculations.

Table A.7: Pairwise Diebold–Mariano test statistics for REITs (MSE loss).

	Hist-Avg	OLS-Base	OLS-Lasso	ENet	NN-3
Hist-Avg	—	−1.539	−0.555	+0.840	+0.964
OLS-Base	+1.539*	—	+1.530*	+1.640*	+1.587*
OLS-Lasso	+0.555	−1.530	—	+1.010	+1.476*
ENet	−0.840	−1.640	−1.010	—	−0.727
NN-3	−0.964	−1.587	−1.476	+0.727	—

Note: For each entry (i, j), the DM statistic is the t-ratio of the loss differential $d_t = e^2_{i,t} - e^2_{j,t}$, where i denotes the row model and j the column model. Statistical significance is estimated via HAC-robust OLS (Newey–West, $\ell = 3$). A positive entry indicates that model j (column) has a lower MSE than model i (row), supporting the one-sided alternative $H_1 : \text{MSE}_j < \text{MSE}_i$. *Significance:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. *Source:* Author's calculations.

D.3 Predictor Importance by Asset and Model

Similar to the disaggregation of calibration slopes, this section details the top driving features on an asset-by-asset basis for each model. This expands upon *Figure 5.5* (*Section 5.1.4*), which displayed predictor importance aggregated across all asset classes.

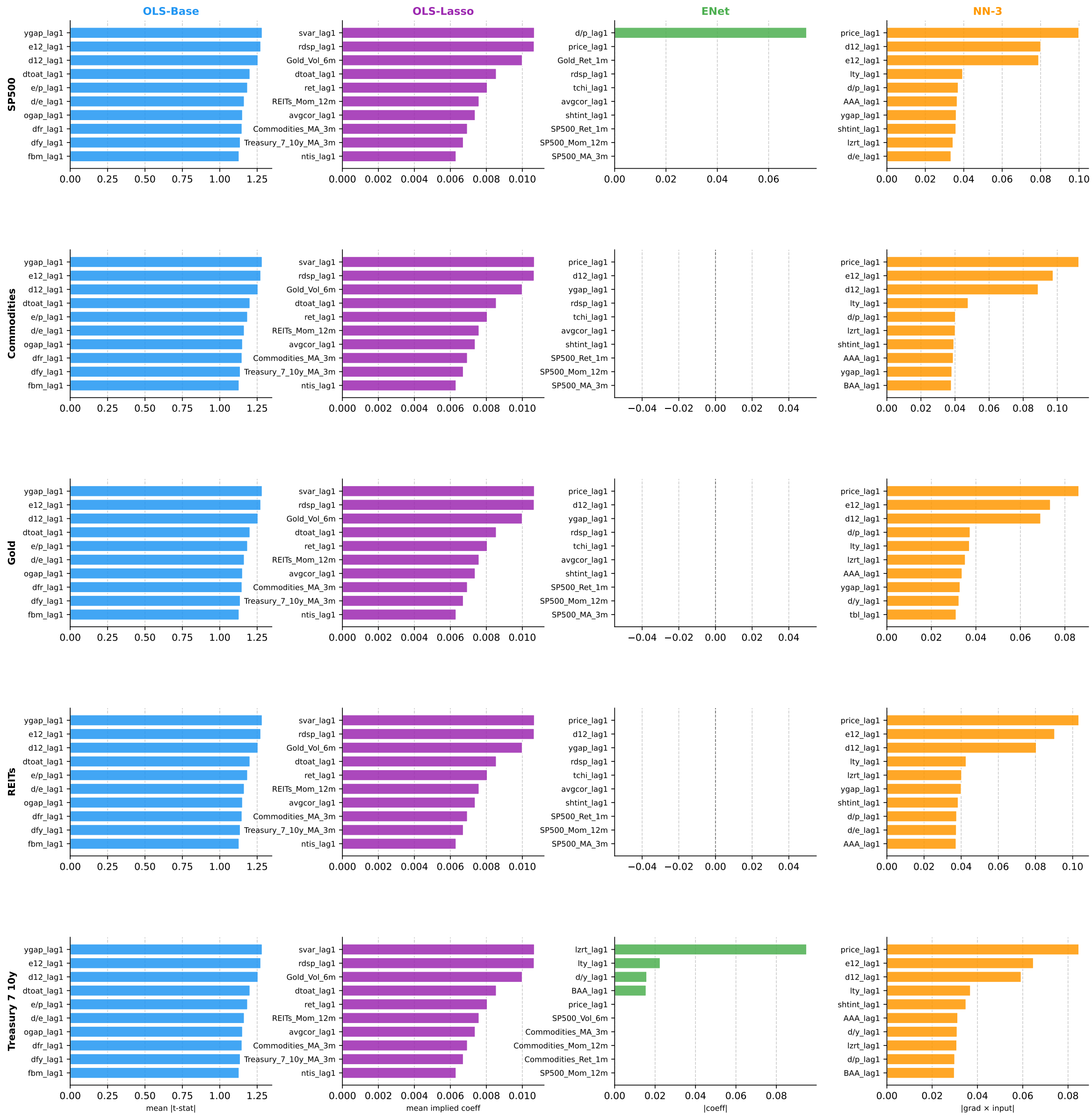

Figure A.6: Top-10 Predictor Importance by Asset Class and Model. Rows correspond to the five asset classes (SP500, Commodities, Gold, REITs, Treasury 7–10y); columns to the four models (OLS-Base, OLS-Lasso, ENet, NN-3). Importance is measured by mean $|t\text{-stat}|$ (OLS-Base), mean implied coefficient (OLS-Lasso), $|\hat{\beta}|$ (ENet), and mean $|\text{grad} \times \text{input}|$ (NN-3), averaged over all walk-forward windows. *Source:* Author's elaboration.

D.4 Robustness Analysis: Summary Table

Expanding on *Section 5.3*, this table details the full empirical metrics across all alternative window configurations.

Table A.8: Robustness Analysis Summary Table. Column labels: *training window – OOS step* (months). *RRRE*: Rolling for OLS-Base, OLS-Lasso, ENet; Expanding for NN-3. *EEEE*: Expanding for all models. Sharpe ratios and Jensen $\hat{\alpha}$ annualised on monthly excess returns. VaR/CVaR at 5% (monthly). Significance: $^{*}p$<0.10, $^{**}p$<0.05, $^{***}p$<0.01.

		120 + 60		**180 + 60**		**180 + 120**		**240 + 120**	
		RRRE	*EEEE*	*RRRE*	*EEEE*	*RRRE*	*EEEE*	*RRRE*	*EEEE*
Panel A: Window Configuration									
Training window (months)		120	120	180	180	180	180	240	240
OOS step / window (months)		60	60	60	60	120	120	120	120
Window type (all models)		Rolling	Expnd.	Rolling	Expnd.	Rolling	Expnd.	Rolling	Expnd.
OOS start (approximate)		06/1989	06/1989	10/1994	10/1994	05/1999	05/1999	06/2004	06/2004
N_{OOS} (months)		427	427	367	367	307	307	247	247
Panel B: OOS Predictive Performance (Pooled Across 5 Assets)									
Aggregate R^2_{OOS} (%)									
	OLS-Base	−87.34	−3.99	−18.93	−4.28	−24.21	−6.57	−11.62	−9.61
	OLS-Lasso	−1.74	0.31	−1.72	0.20	−1.36	−0.92	−0.28	−1.54
	ENet	2.75	2.15	4.66	2.35	1.72	−0.89	2.46	−0.84
	NN-3	1.04	1.04	1.24	1.24	−0.21	−0.21	0.08	0.08
Aggregate Slope (pred. vs. actual, pooled)									
	OLS-Base	0.103	0.418	0.250	0.427	0.250	0.417	0.401	0.406
	OLS-Lasso	0.254	0.951	0.210	0.953	0.348	0.384	1.686	0.219
	ENet	0.804	0.765	0.958	0.786	0.937	0.545	0.922	0.642
	NN-3	1.043	1.043	1.122	1.122	0.750	0.750	1.013	1.013
Average Hit Rate (%, mean across 5 assets; H_0: 50%)									
	OLS-Base	54.5	55.9	55.6	57.1	55.7	56.5	56.1	56.1
	OLS-Lasso	55.3	57.0	55.8	57.7	56.7	56.5	56.4	56.1
	ENet	55.8	58.8	57.2	59.7	57.7	56.1	59.3	55.2
	NN-3	56.7	56.7	57.4	57.4	58.3	58.3	58.3	58.3
Panel C: Portfolio Performance — Long-Only (LO)									
Sharpe Ratio (excess returns, annualised)									
	Hist-Avg	0.562	0.502	0.615	0.615	0.466	0.510	0.593	0.593
	OLS-Base	0.732	0.511	0.671	0.556	0.655	0.537	0.496	0.623
	OLS-Lasso	0.588	0.651	0.613	0.689	0.365	0.459	0.564	0.502
	ENet	0.796	0.446	0.521	0.472	0.317	0.441	0.436	0.434
	NN-3	0.453	0.453	0.460	0.460	0.534	0.632	0.573	0.573
	$1/N$	0.562	0.562	0.565	0.565	0.585	0.585	0.558	0.558
	B&H	0.530	0.530	0.529	0.529	0.483	0.483	0.508	0.508
	60/40	0.690	0.690	0.699	0.699	0.551	0.551	0.719	0.719
Maximum Drawdown (%)									
	Hist-Avg	−37.46	−40.89	−37.46	−37.46	−40.89	−37.46	−37.46	−37.46
	OLS-Base	−23.59	−46.03	−32.83	−33.56	−37.16	−33.56	−52.53	−33.56
	OLS-Lasso	−30.11	−47.18	−30.11	−43.09	−58.92	−38.84	−42.49	−38.84
	ENet	−25.32	−61.49	−43.87	−57.68	−55.35	−42.40	−50.32	−42.40
	NN-3	−52.75	−53.35	−52.75	−52.75	−36.75	−27.84	−27.84	−27.84
	$1/N$	−36.44	−36.44	−36.44	−36.44	−36.44	−36.44	−36.44	−36.44
	B&H	−51.60	−51.60	−51.60	−51.60	−51.60	−51.60	−51.60	−51.60
	60/40	−29.70	−29.70	−29.70	−29.70	−29.70	−29.70	−29.70	−29.70
CVaR 5% (monthly, %)									
	Hist-Avg	−8.14	−8.96	−8.42	−8.42	−9.18	−8.35	−8.58	−8.58
	OLS-Base	−7.48	−8.41	−6.69	−8.01	−7.24	−8.46	−8.18	−8.47
	OLS-Lasso	−7.02	−7.30	−7.28	−7.04	−9.48	−7.60	−7.54	−7.78
	ENet	−7.31	−9.62	−8.44	−9.37	−10.57	−8.27	−9.16	−8.84
	NN-3	−8.30	−9.17	−8.97	−8.97	−9.28	−7.57	−7.83	−7.83
	$1/N$	−5.71	−5.71	−6.08	−6.08	−6.44	−6.44	−6.94	−6.94
	B&H	−8.59	−8.59	−9.11	−9.11	−9.54	−9.54	−10.29	−10.29
	60/40	−5.46	−5.46	−5.54	−5.54	−5.63	−5.63	−5.86	−5.86
VaR 5% (monthly, %)									
	Hist-Avg	−5.78	−6.32	−5.78	−5.78	−6.75	−6.12	−6.12	−6.12

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		120 + 60		180 + 60		180 + 120		240 + 120	
		RRRE	*EEEE*	*RRRE*	*EEEE*	*RRRE*	*EEEE*	*RRRE*	*EEEE*
	OLS-Base	−5.00	−6.04	−5.42	−5.35	−5.96	−5.52	−5.93	−5.87
	OLS-Lasso	−5.27	−4.93	−5.42	−4.58	−5.86	−5.27	−4.82	−5.14
	ENet	−4.43	−5.60	−5.54	−5.50	−6.51	−5.75	−5.53	−5.87
	NN-3	−4.01	−4.71	−4.14	−4.14	−5.75	−4.28	−5.26	−5.26
	$1/N$	−3.24	−3.24	−3.51	−3.51	−3.76	−3.76	−3.98	−3.98
	B&H	−5.17	−5.17	−5.39	−5.39	−5.54	−5.54	−5.83	−5.83
	60/40	−3.87	−3.87	−3.87	−3.87	−3.92	−3.92	−3.99	−3.99
Panel D: Portfolio Performance — Realistic / Turnover-Constrained (RE)									
Sharpe Ratio (excess returns, annualised)									
	Hist-Avg	0.579	0.561	0.606	0.606	0.521	0.539	0.581	0.581
	OLS-Base	0.433	0.433	0.600	0.533	0.513	0.553	0.503	0.491
	OLS-Lasso	0.415	0.534	0.423	0.577	0.349	0.424	0.438	0.424
	ENet	0.546	0.437	0.479	0.449	0.430	0.409	0.505	0.378
	NN-3	0.414	0.392	0.406	0.406	0.415	0.505	0.488	0.488
	$1/N$	0.562	0.562	0.565	0.565	0.585	0.585	0.558	0.558
	B&H	0.530	0.530	0.529	0.529	0.483	0.483	0.508	0.508
	60/40	0.690	0.690	0.699	0.699	0.551	0.551	0.719	0.719
Maximum Drawdown (%)									
	Hist-Avg	−37.37	−34.04	−37.37	−37.37	−34.04	−37.37	−37.37	−37.37
	OLS-Base	−32.49	−52.28	−31.45	−43.65	−35.71	−43.65	−47.30	−43.65
	OLS-Lasso	−57.45	−42.38	−57.45	−34.87	−55.81	−48.72	−49.34	−48.99
	ENet	−27.14	−47.33	−39.99	−46.25	−42.22	−48.35	−39.07	−47.96
	NN-3	−54.63	−56.54	−54.63	−54.63	−54.86	−45.62	−45.62	−45.62
	$1/N$	−36.44	−36.44	−36.44	−36.44	−36.44	−36.44	−36.44	−36.44
	B&H	−51.60	−51.60	−51.60	−51.60	−51.60	−51.60	−51.60	−51.60
	60/40	−29.70	−29.70	−29.70	−29.70	−29.70	−29.70	−29.70	−29.70
CVaR 5% (monthly, %)									
	Hist-Avg	−7.33	−7.45	−7.74	−7.74	−7.96	−7.77	−8.30	−8.30
	OLS-Base	−6.56	−7.53	−5.17	−6.66	−6.05	−7.05	−7.69	−7.53
	OLS-Lasso	−8.11	−7.52	−8.59	−7.17	−8.86	−8.47	−8.31	−9.01
	ENet	−7.31	−8.61	−8.70	−8.72	−9.18	−9.23	−8.76	−9.88
	NN-3	−8.41	−8.94	−9.00	−9.00	−10.04	−8.11	−8.60	−8.60
	$1/N$	−5.71	−5.71	−6.08	−6.08	−6.44	−6.44	−6.94	−6.94
	B&H	−8.59	−8.59	−9.11	−9.11	−9.54	−9.54	−10.29	−10.29
	60/40	−5.46	−5.46	−5.54	−5.54	−5.63	−5.63	−5.86	−5.86
Panel E: Statistical Tests									
Jensen $\hat{\alpha}$ (%, ann., LO; H_0: $\alpha = 0$)									
	OLS-Base	5.75***	3.04	4.72**	2.75	5.70**	3.44	0.91	2.49
	OLS-Lasso	2.57*	3.05**	2.75*	2.95**	0.71	0.85	1.16	−0.28
	ENet	7.78***	2.21	2.15	1.70	1.05	2.14	1.27	0.84
	NN-3	0.89	1.04	0.76	0.76	2.95	3.55**	1.20	1.20
t-stat for Jensen $\hat{\alpha}$ (LO)									
	OLS-Base	3.002	1.455	2.444	1.396	2.494	1.566	0.409	1.088
	OLS-Lasso	1.714	2.067	1.664	2.068	0.327	0.631	0.649	−0.189
	ENet	3.654	1.001	1.088	0.811	0.386	1.006	0.463	0.336
	NN-3	0.535	0.570	0.415	0.415	1.452	2.104	0.661	0.661
Jensen $\hat{\alpha}$ (%, ann., RE; H_0: $\alpha = 0$)									
	OLS-Base	0.34	0.34	2.59*	2.75	2.83	2.83	0.08	0.08
	OLS-Lasso	−0.17	3.05**	−0.26	2.95**	0.06	0.85	−0.39	−0.39
	ENet	2.22	2.21	0.35	1.70	1.02	1.02	0.57	0.57
	NN-3	−0.43	1.04	−0.94	0.76	0.37	0.37	−0.77	−0.77
t-stat for Jensen $\hat{\alpha}$ (RE)									
	OLS-Base	0.269	0.269	1.886*	1.396	1.619	1.619	0.048	0.048
	OLS-Lasso	−0.131	2.067**	−0.178	2.068**	0.031	0.631	−0.211	−0.211
	ENet	1.444	1.001	0.233	0.811	0.564	0.564	0.283	0.283
	NN-3	−0.337	0.570	−0.698	0.415	0.223	0.223	−0.540	−0.540
Diebold–Mariano stat, SP500 (H_0: equal MSE vs. Hist-Avg)									
	OLS-Base	−2.588	−1.624	−1.704	−1.495	−1.636	−1.376	−1.690	−1.611
	OLS-Lasso	−0.842	−0.347	−0.677	−0.382	−0.334	+0.282	+0.533	−0.005
	ENet	+0.094	−0.468	−1.261	−0.563	+1.446*	−0.072	+1.136	+0.660
	NN-3	+0.603	+0.603	+0.792	+0.792	+1.782**	+1.782**	+1.507*	+1.507*

Note: Jensen $\hat{\alpha}$ (%, ann.) is derived from regressing LO/RE excess returns on the FF market factor, using White-corrected t-statistics. For the Diebold–Mariano test (with HLN 1997 correction, SP500 only); significance levels for this test are based on one-tailed p-values. *Source:* Author's calculations.

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